Name:

GCSE COMPUTER SCIENCE

COMP1 Revision Booklet

Your Essential Exam Revision Guide

Welcome to your revision guide. It will help guide you through the key ideas you *really* need to know. You'll find clear explanations, helpful reminders, and practice questions to train your brain step by step.

Use this booklet little and often to organise your thinking, grow your confidence, and walk calmly into your exams knowing you're ready.

Paper 1 Exam Date:
Summer 2027 - check your exam timetable

Contents / Part 1

Select a topic to find diagrams, clear explanations and practice questions.

1.1.1 Architecture of the CPU	**4**
1.1.2 CPU Performance	**21**
1.1.3 Embedded Systems	**33**
1.2.1 Primary Memory	**45**
1.2.2 Secondary Storage	**58**
1.2.3 Units of Data Storage	**71**
1.2.4 Data Representation	**88**
1.2.5 Compression	**111**
1.3.1 Computer Networks and Topologies	**123**
1.3.2 Wired, Wireless Networks and Protocols	**145**

Each topic follows Theory → Key terminology → Worked examples → Exam practice → Answers.

Contents / Part 2

Synoptic practice follows the topic-by-topic revision sections.

1.4.1 Threats to Computer Systems **165**

1.4.2 Preventing Vulnerabilities **181**

1.5.1 Operating Systems **199**

1.5.2 Utility Software **216**

1.6.1 Impacts, Legislation and TIER **233**

1.6.1 Open-source and Proprietary Licences **253**

Synoptic Practice A: questions 269

Synoptic Practice A: answers and misconceptions 278

Synoptic Practice B: questions 280

Synoptic Practice B: answers and misconceptions 290

Use these final practice sets to test knowledge across several topics. Check the answers after completing each set.

01 THEORY

Architecture of the CPU

The Central Processing Unit (CPU) processes instructions and data so that programs can run. It repeatedly fetches, decodes and executes instructions.

The fetch–decode–execute cycle

1 FETCH

Obtain the next instruction from main memory.

2 DECODE

The control unit interprets the instruction.

3 EXECUTE

Carry out the action, such as a calculation.

Why is this a cycle?

Once the instruction is executed, the processor continues with the next instruction. The stages repeat while the program runs.

One useful analogy

Think of a head chef: they collect an order (fetch), interpret what is requested (decode) and prepare the meal (execute). The analogy helps with sequence, but in an exam use the technical wording above.

What OCR expects you to know

- The purpose of the CPU and the three stages of the cycle.
- The functions of the ALU, control unit, cache and registers.
- What the MAR, MDR, Program Counter and Accumulator store - especially the difference between a memory address and data.

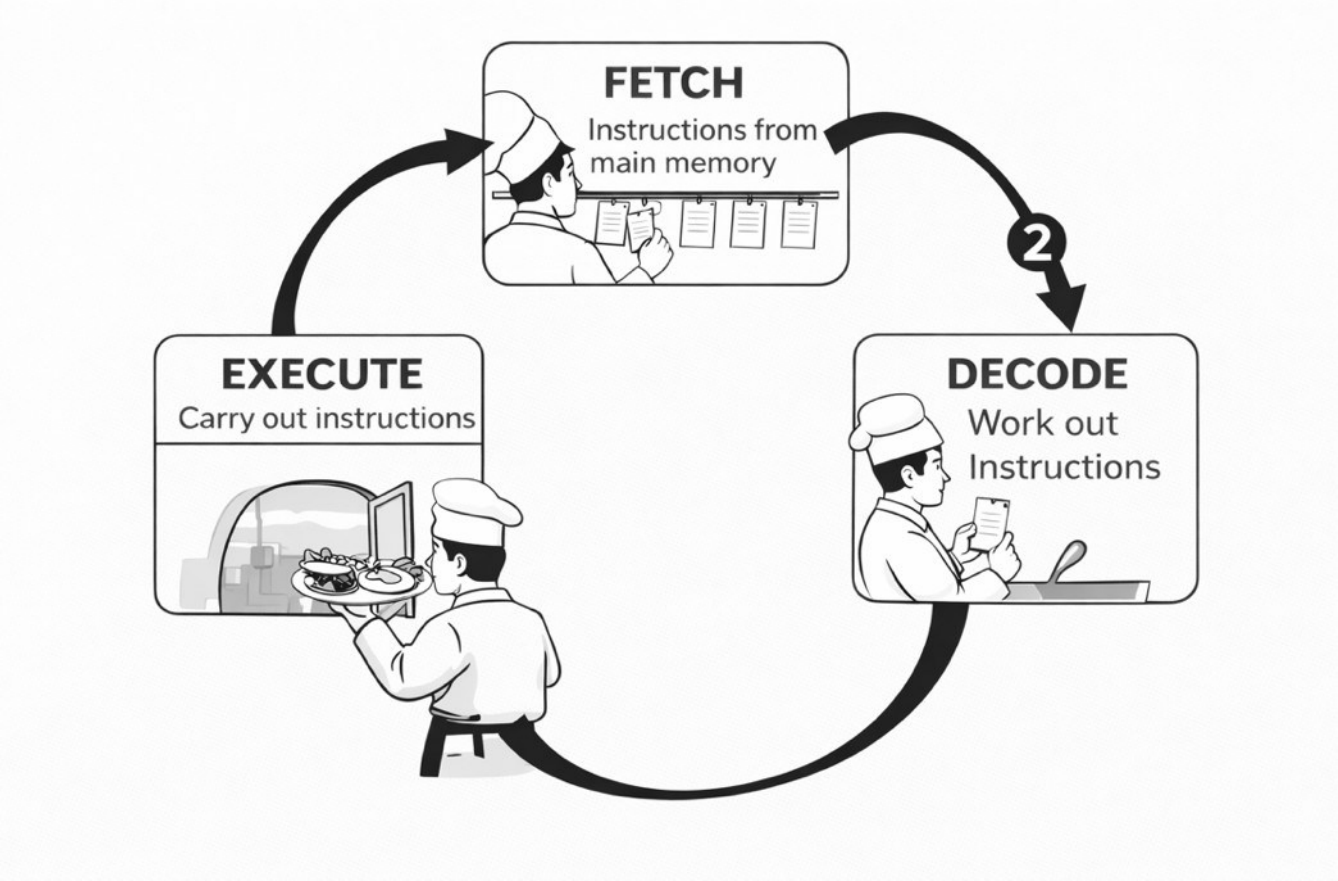
Specification boundary

J277 does not require a step-by-step account of transferring values between different registers at every stage of the cycle.

THEORY / VISUAL ANALOGY

The chef follows an order

Use the chef analogy to remember the stages. The explanation on the right gives accurate GCSE exam wording.



RESTAURANT ANALOGY		CORRECT EXAM WORDING
FETCH	Chef collects the next order.	The CPU fetches the next instruction from memory.
DECODE	Chef reads the order to understand it.	The Control Unit decodes the instruction.
EXECUTE	Chef carries out the order.	The CPU executes the instruction; the ALU may perform a calculation.

Remember: This is a repeating cycle, not a one-off task. Use the kitchen to recall the order; use fetch, decode, execute in your answer.

01 THEORY / FETCH–DECODE–EXECUTE

Three different instructions, three cycles

A chef does not cook the same order forever. They finish one instruction, then pick up the next, different instruction. The CPU follows the same repeating pattern.

INSTRUCTION 01 — **CHOP CARROTS**

FETCH	DECODE	EXECUTE
Chef picks up the order: CHOP CARROTS.	Chef works out what chopping carrots requires.	Chef chops the carrots.

NEXT INSTRUCTION

INSTRUCTION 02 — **STIR THE SOUP**

FETCH	DECODE	EXECUTE
Chef picks up a new order: STIR SOUP.	Chef works out what stirring the soup requires.	Chef stirs the soup.

NEXT INSTRUCTION

INSTRUCTION 03 — **PLATE THE MEAL**

FETCH	DECODE	EXECUTE
Chef picks up the next order: PLATE MEAL.	Chef works out how to plate the meal.	Chef plates the meal.

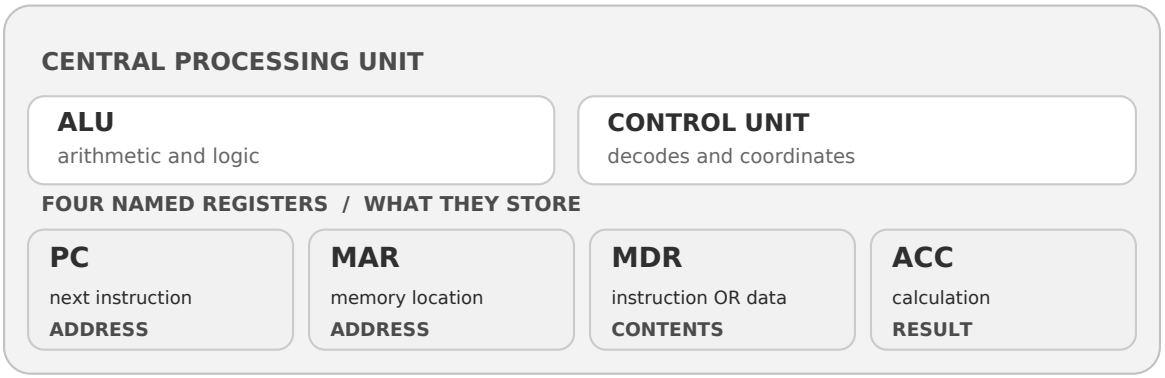
EXAM CONNECTION

For each instruction, the CPU fetches it from memory, the Control Unit decodes it, and the CPU executes it. It then fetches the next instruction. The stages repeat; the instruction changes.

01 THEORY

Components and registers

A CPU includes several parts with different purposes. Each contributes to handling instructions efficiently.



Core components

Arithmetic Logic Unit (ALU)
Performs arithmetic calculations and logical operations.

Control Unit (CU)
Decodes instructions and coordinates CPU activity using control signals.

Cache
Fast memory that holds frequently used data and instructions for quick access.

Registers
Very small, very fast storage locations for values currently needed by the CPU.

Von Neumann registers: address or data?

PC Program Counter
Address of the next instruction.

MAR Memory Address Register
Address of the memory location being accessed.

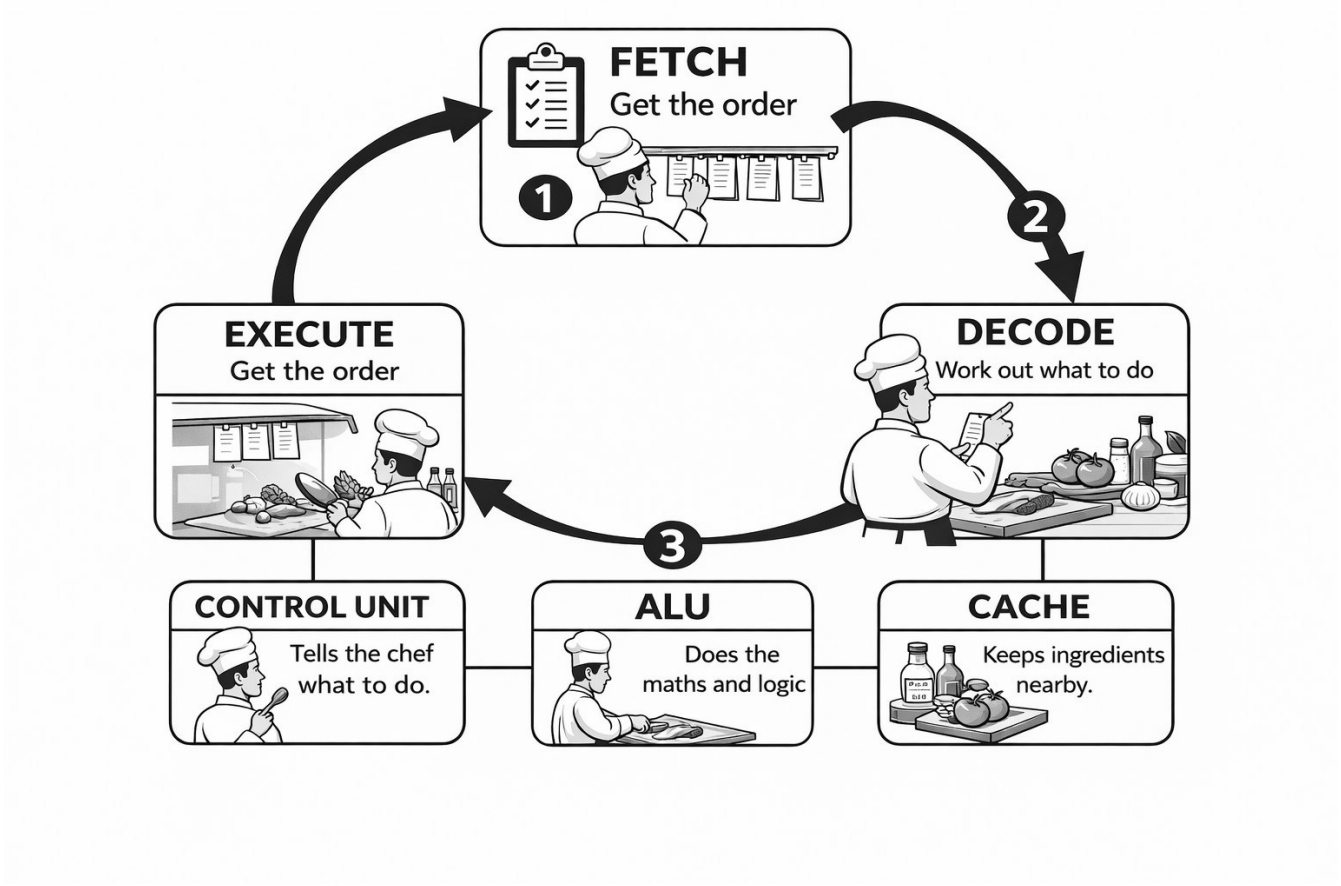
MDR Memory Data Register
Data or instruction being transferred to or from memory.

ACC Accumulator
Intermediate result of arithmetic or logic operations.

Remember: PC / MAR = addresses; MDR = instruction OR data; ACC = result.

01 THEORY / VISUAL ANALOGY The CPU works like a kitchen

The same familiar setting links the fetch–decode–execute cycle to the roles of the CPU components.

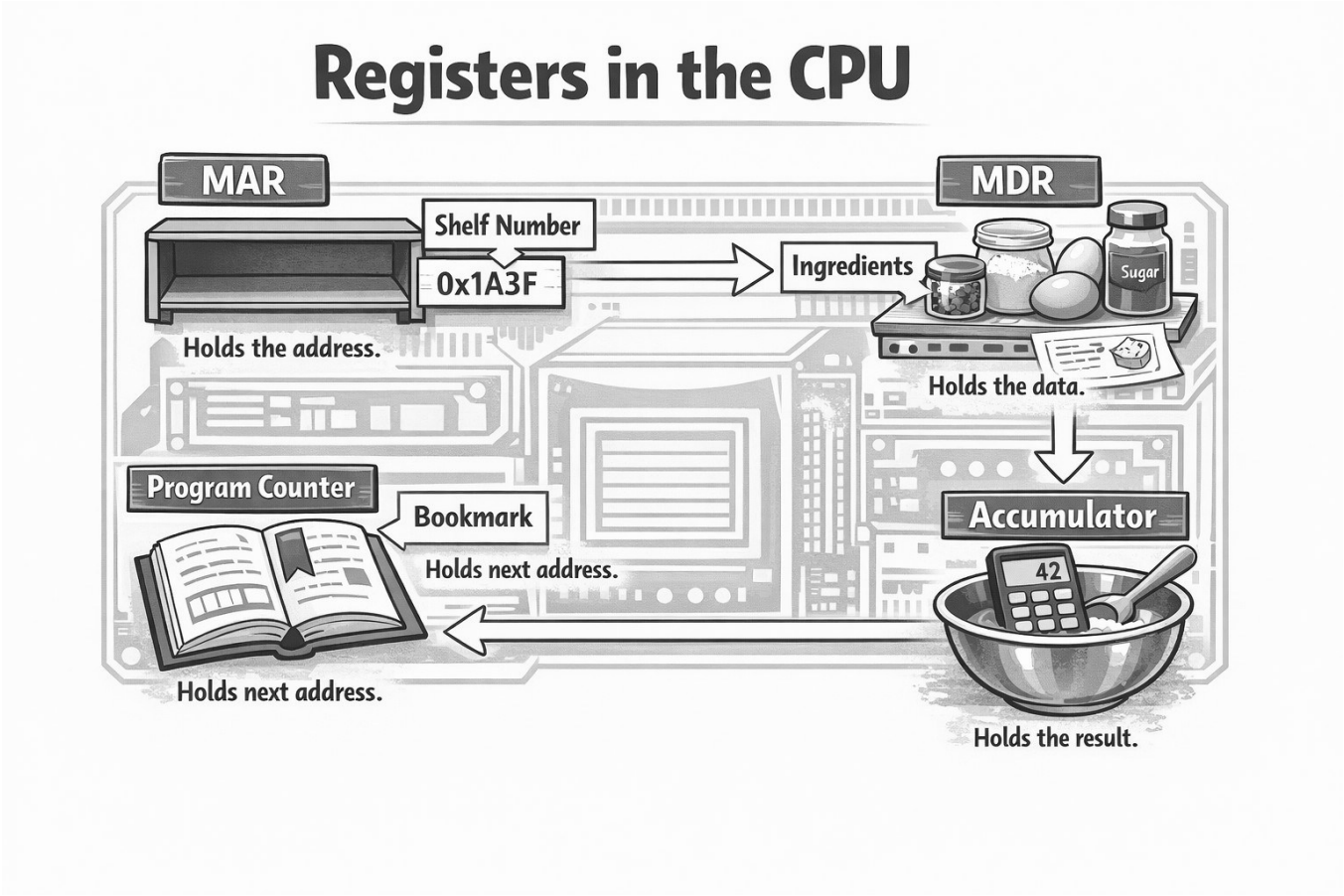


WHAT THE PICTURE REPRESENTS		CORRECT EXAM WORDING
CONTROL UNIT	Head chef directs the work.	Decodes instructions and coordinates operations using control signals.
ALU	Does the maths and logic needed.	Performs arithmetic operations and logical comparisons.
CACHE	Common ingredients kept close by.	Small, fast memory for frequently used data and instructions.

Exam tip: The analogy makes each role memorable; it does not replace the technical terms Control Unit, ALU and cache.

01 THEORY / VISUAL ANALOGY Registers: shelves and ingredients

The kitchen analogy distinguishes a memory address (where something is) from data (what it contains).



VISUAL CLUE		CORRECT EXAM WORDING
MAR	Shelf number identifies a location.	Memory Address Register stores the address of the memory location being accessed.
MDR	Ingredients are the actual contents.	Memory Data Register holds data or an instruction being transferred to or from memory.
PC	Bookmark shows where to read next.	Program Counter stores the address of the next instruction.
ACC	Mixing bowl holds the current result.	Accumulator stores an intermediate arithmetic or logical result.

02 KEY TERMINOLOGY

Words that earn marks

Use these terms accurately when you explain how the CPU works. The yellow highlighting is intentionally placed behind each keyword, as in the RHSC Computing lessons.

CPU
Processes data and executes instructions so that programs run.

Fetch
Retrieves the next instruction from main memory.

Decode
Interprets the instruction to identify the action needed.

Execute
Carries out the operation described by the instruction.

ALU
Performs arithmetic and logical operations.

Control Unit
Decodes instructions and coordinates CPU activities.

Cache
Small, fast memory holding frequently used instructions/data.

Register
Tiny, fast storage location within the processor.

Program Counter
Stores the address of the next instruction.

MAR
Stores the address of the memory location being accessed.

MDR
Stores data/instructions travelling to or from memory.

Accumulator
Stores intermediate results of calculations.

Exam reminder
For register questions, include the word address when describing the PC or MAR, and data/instruction when describing the MDR.

03 WORKED EXAMPLES

See how marks are earned

WORKED EXAMPLE 1

Question: Explain the three stages of the fetch–decode–execute cycle. [3 marks]

FETCH	The CPU retrieves the next instruction from memory.
DECODE	The control unit interprets the instruction.
EXECUTE	The CPU carries out the required instruction, such as an ALU calculation.

Why this earns 3 marks

Each distinct stage has the correct action. The answer uses precise exam language rather than relying on the chef analogy.

WORKED EXAMPLE 2

Question: A program is about to access memory location 120. Which register holds the location address, and which holds the data read from that location? [2 marks]

MAR	Holds the address 120 - the location to access.
MDR	Holds the actual data or instruction read from that location.

The distinction

The MAR stores where; the MDR stores what. You do not need to memorise the sequence of data passing between registers for this specification point.

04 EXAM PRACTICE

Try it yourself · 1 of 5

Answer the questions before looking at the separate mark scheme. Marks are shown beside each question.
Attempt each question before checking the separate mark scheme.

Q1 1 MARK

What is the purpose of the CPU?

Q2 1 MARK

Name one component of the CPU.

Q3 1 MARK

What does the Program Counter store?

Q4 2 MARKS

State two roles of the Control Unit.

04 EXAM PRACTICE

Try it yourself · 2 of 5

Answer the questions before looking at the separate mark scheme. Marks are shown beside each question.
Attempt each question before checking the separate mark scheme.

Q5

2 MARKS

State two types of operation that the ALU performs.

Q6

2 MARKS

Match each CPU component to its function. Connect ALU, Control Unit and Registers to the correct functions.

Choose from:

A holds data and addresses being used

B decodes instructions and controls operations

C performs calculations and logic

ALU	Write A, B or C: _____
Control Unit	Write A, B or C: _____
Registers	Write A, B or C: _____

Q7

3 MARKS

Explain the Fetch–Decode–Execute cycle.

04 EXAM PRACTICE

Try it yourself · 3 of 5

Answer the questions before looking at the separate mark scheme. Marks are shown beside each question.
Attempt each question before checking the separate mark scheme.

Q8 3 MARKS

Explain why registers are used inside the CPU.

Q9 3 MARKS

Explain the role of cache memory in the CPU.

Q10 3 MARKS

Label the ALU, Control Unit and registers on the blank CPU diagram below.



04 EXAM PRACTICE

Try it yourself · 4 of 5

Answer the questions before looking at the separate mark scheme. Marks are shown beside each question.
Attempt each question before checking the separate mark scheme.

Q11 4 MARKS

Describe the role of the ALU and the Control Unit.

Q12 4 MARKS

Describe the role of the MAR and MDR.

04 EXAM PRACTICE

Try it yourself · 5 of 5

Answer the questions before looking at the separate mark scheme. Marks are shown beside each question.
Attempt each question before checking the separate mark scheme.

Q13 4 MARKS

Compare the MAR and MDR.

Q14 6 MARKS

Explain how the CPU processes instructions using the Fetch–Decode–Execute cycle.

05 ANSWERS & MARK SCHEME

Check your work · 1 of 4

Read your answer before self-marking. Award marks for correct computing ideas. These are indicative model answers, not an official OCR mark scheme.

Q1 · 1 mark

The CPU processes data and instructions so that programs can run.

Q2 · 1 mark

Arithmetic Logic Unit (ALU), Control Unit (CU), cache or registers.

Q3 · 1 mark

The memory address of the next instruction to be fetched.

Q4 · 2 marks

It decodes instructions and sends control signals to coordinate CPU operations.

- 1 mark: decodes instructions.
- 1 mark: coordinates/controls execution, e.g. by sending control signals.

Q5 · 2 marks

Arithmetic operations (for example, addition) and logical operations (for example, AND).

- 1 mark: arithmetic.
- 1 mark: logical operations.

05 ANSWERS & MARK SCHEME

Check your work · 2 of 4

Read your answer before self-marking. Award marks for correct computing ideas. These are indicative model answers, not an official OCR mark scheme.

Q6 · 2 marks

ALU → calculations; Control Unit → decodes/coordinates; Registers → small, fast storage for current data and addresses.

Q7 · 3 marks

The CPU fetches the next instruction from memory. The Control Unit decodes it to determine what action is required. The CPU then executes it, possibly using the ALU.

- 1 mark: fetches next instruction from memory.
- 1 mark: decodes/interprets the instruction.
- 1 mark: carries out/executes the instruction.

Q8 · 3 marks

Registers are small, very fast storage locations within the CPU. They hold addresses, data or results currently being used, giving the processor rapid access during instruction processing.

- 1 mark: small storage in/very close to CPU.
- 1 mark: holds data, addresses or results in current use.
- 1 mark: faster access supports efficient processing.

Q9 · 3 marks

Cache is small, fast memory close to or within the CPU. It holds frequently used data and instructions. This reduces time spent accessing the slower main memory.

- 1 mark: small, fast memory.
- 1 mark: frequently used data/instructions.
- 1 mark: quicker access/reduced RAM accesses.

05 ANSWERS & MARK SCHEME

Check your work · 3 of 4

Read your answer before self-marking. Award marks for correct computing ideas. These are indicative model answers, not an official OCR mark scheme.

Q10 · 3 marks

From left to right: ALU; Control Unit; Registers.

Q11 · 4 marks

The ALU performs arithmetic operations such as addition and logical operations such as comparisons. The Control Unit decodes instructions and coordinates how CPU components act by sending control signals.

- 1 mark: ALU arithmetic.
- 1 mark: ALU logic.
- 1 mark: CU decodes.
- 1 mark: CU controls or coordinates instructions/CPU operations.

Q12 · 4 marks

The MAR holds the address of the memory location being accessed. The MDR holds the data or instruction being transferred to or from memory. They are both CPU registers used in memory access.

- 1 mark: MAR identified as a register.
- 1 mark: MAR stores memory address.
- 1 mark: MDR identified as a register.
- 1 mark: MDR stores data/instruction being read or written.

05 ANSWERS & MARK SCHEME

Check your work · 4 of 4

Read your answer before self-marking. Award marks for correct computing ideas. These are indicative model answers, not an official OCR mark scheme.

Q13 · 4 marks

Both are registers used when the CPU accesses memory. The MAR holds a location address - it identifies where the data is stored. The MDR holds the actual data or instruction being read from or written to that location.

- 1 mark: both are registers / involved in memory access.
- 1 mark: MAR stores an address.
- 1 mark: MDR stores data or an instruction.
- 1 mark: explicit comparison of location vs content.

Q14 · 6 marks

The CPU repeats the fetch–decode–execute cycle to run a program. During fetch, it obtains the next instruction from memory. The Control Unit decodes the instruction to determine the required action. During execute, the CPU carries out that action; for example, the ALU might perform an arithmetic or logic operation. The Program Counter identifies the address of the next instruction, while registers provide fast storage for addresses, data and results. The Accumulator may hold the result of an ALU operation.

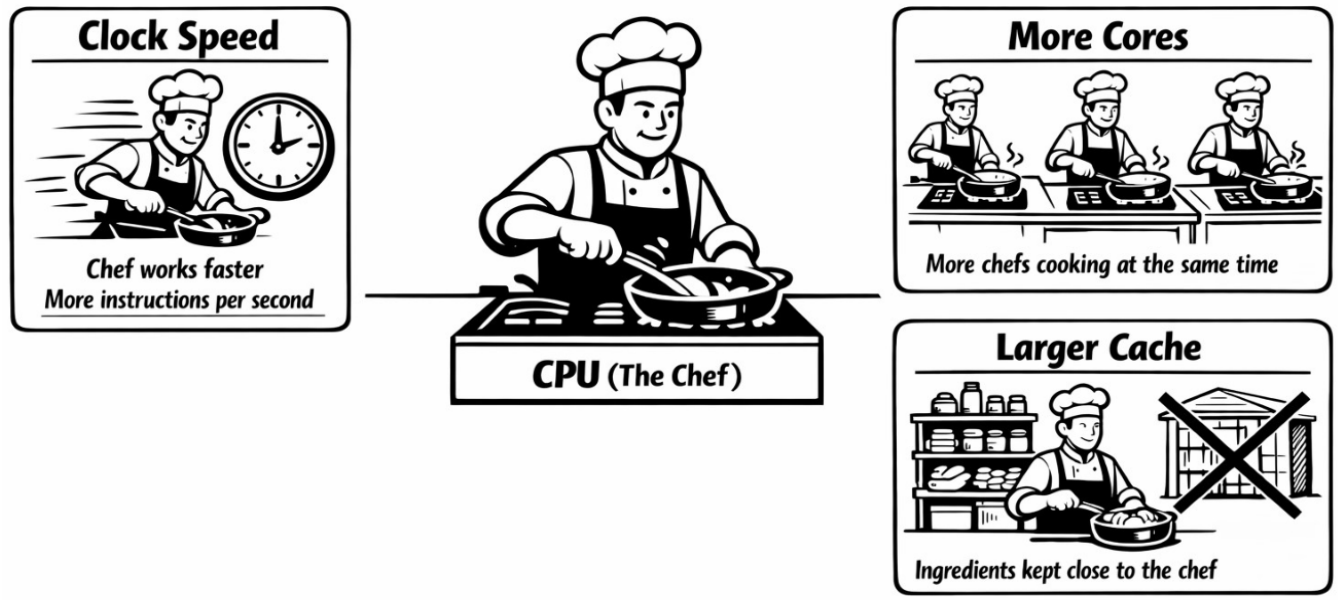
- 1 mark: cycle repeats to process a program.
- 1 mark: fetches instruction from memory.
- 1 mark: decodes/understands instruction.
- 1 mark: executes the required action.
- 1 mark: correct component role (e.g., CU or ALU).
- 1 mark: correct register role (PC, MAR, MDR or Accumulator).

01 THEORY

CPU performance

Three connected ideas: clock speed, cache size and number of cores.

One familiar kitchen, three ways to improve performance



Link the picture to the technical explanation

1 **Clock speed**

The chef works faster → more clock cycles per second.

2 **Number of cores**

More chefs can work at the same time → parallel work.

3 **Cache size**

Ingredients are kept close → less waiting for frequently used data.

Exam reminder

A faster clock, larger cache and more cores may improve performance, but the effect depends on the workload. More cores help most when tasks can run in parallel.

01 THEORY

See the image. Explain the CPU.

The drawings are your memory cues; the right-hand wording earns marks.



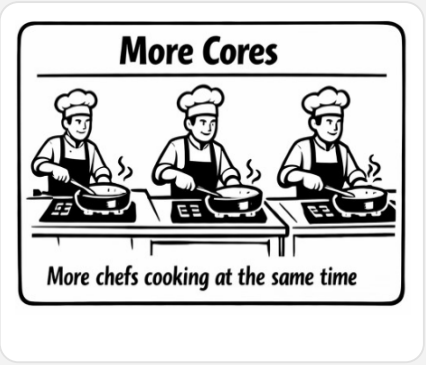
Chef works faster

CLOCK SPEED

TECHNICAL EXPLANATION

Clock speed is the rate of CPU clock cycles, measured in Hz or GHz. Raising it can allow the CPU to process more instructions in a given time.

Remember: Do not assume one whole instruction is completed on every clock tick.



More chefs together

NUMBER OF CORES

TECHNICAL EXPLANATION

Each CPU core can process instructions. More cores can carry out work in parallel and improve multitasking when tasks can use them.

Remember: One task cannot automatically use every core.



Ingredients kept close

CACHE SIZE

TECHNICAL EXPLANATION

Cache is small, fast memory holding frequently used data and instructions. A larger cache may reduce slower access to RAM.

Remember: More cache helps when the CPU can reuse the stored items.

01 THEORY

Using factors in combination

An exam might ask which CPU is better for a particular job. Explain why.

Compare two different processors

PROCESSOR A		PROCESSOR B	
Clock speed	**3.6 GHz**	Clock speed	**3.1 GHz**
Number of cores	**2 cores**	Number of cores	**6 cores**
Cache	**4 MB**	Cache	**12 MB**

Which would you choose?

Several independent programs running together

Processor B may be better: six cores offer more parallel capacity, and its larger cache may reduce memory delays. A has a higher clock speed, so it might perform better on some tasks that mainly use one core.

One task that cannot run in parallel

Processor A might be faster because of its higher clock speed, but GHz alone cannot prove that one entire processor is faster. The program and other factors matter.

Strong exam answer

Name the factor → explain how it changes CPU operation → link it to the task. For comparisons, explain a trade-off rather than claiming one CPU is always best.

02 KEY TERMINOLOGY

Words that earn marks

These terms connect your chef analogy to the correct OCR terminology.

CPU performance

How efficiently a processor handles work, including speed of execution.

Clock speed

The frequency of a CPU clock, measured in hertz (Hz).

Hertz (Hz)

One clock cycle per second; GHz means billions of cycles per second.

Cache

Small, fast memory storing frequently used data or instructions.

Cache size

The amount of data/instructions that can be held in cache.

CPU core

A processing unit within a processor that can execute instructions.

Parallel processing

Executing different parts of work at the same time.

Multitasking

Making progress on more than one task; multiple cores can help.

Tip: using the correct terms helps your explanation earn marks.

03 WORKED EXAMPLES

See how marks are earned

Follow the reasoning, then use the same approach for independent practice.

WORKED EXAMPLE 1 · 3 marks

Explain how increasing cache size can improve CPU performance.

1 **State what cache does**

Cache holds frequently used data and instructions.

2 **Explain the speed advantage**

The CPU accesses cache faster than it accesses RAM.

3 **Link to performance**

A larger cache can reduce time waiting for data, improving speed.

WORKED EXAMPLE 2 · 4 marks

A computer runs a web browser, a video call and a game together. Explain how four CPU cores may help compared with two.

1. More cores can process different instructions or tasks at the same time.
2. The browser, video call and game may make progress in parallel.
3. This can improve responsiveness and reduce waiting between tasks.
4. The benefit depends on whether the programs can use multiple cores.

Why these examples earn marks: each point explains a technical idea and connects it to the requested performance effect.

04 EXAM PRACTICE

Your turn · 1 of 4

Attempt these questions before checking the answers. Write enough for the marks shown.

Q1 · 1 mark · Define

What is meant by CPU performance?

Q2 · 1 mark · Define

Name one factor that affects CPU performance.

Q3 · 1 mark · Define

What is clock speed measured in?

04 EXAM PRACTICE

Your turn · 2 of 4

Attempt these questions before checking the answers. Write enough for the marks shown.

Q4 · 2 marks · Explain

State two ways increasing clock speed improves CPU performance.

Q5 · 2 marks · Explain

State two benefits of having multiple CPU cores.

Q6 · 3 marks · Explain

Explain how clock speed affects CPU performance.

04 EXAM PRACTICE

Your turn · 3 of 4

Attempt these questions before checking the answers. Write enough for the marks shown.

Q7 · 3 marks · Explain

Explain the role of cache memory in improving CPU performance.

Q8 · 3 marks · Explain

Explain how the number of CPU cores affects performance.

Q9 · 4 marks · Describe roles

Describe two factors that affect CPU performance.

04 EXAM PRACTICE

Your turn · 4 of 4

Attempt these questions before checking the answers. Write enough for the marks shown.

Q10 · 4 marks · Describe roles

Describe how cache size and number of cores improve CPU performance.

Q11 · 6 marks · Discuss / Explain

Explain how clock speed, cache size and number of cores affect CPU performance.

05 ANSWERS & MARK SCHEME

Check your work · 1 of 3

Mark the scientific idea, not only the exact wording. Use the model answer to improve your own.

Q1 · 1 mark

CPU performance is how quickly and efficiently a CPU handles instructions and work.

Marking points

Accurate definition mentioning the speed/efficiency of processing.

Q2 · 1 mark

Clock speed; cache size; or number of CPU cores.

Marking points

Any one valid listed characteristic.

Q3 · 1 mark

Hertz (Hz).

Marking points

Hz, kHz, MHz or GHz as a suitable frequency unit.

Q4 · 2 marks

Higher clock speed gives more clock cycles each second; this can allow instructions to be processed more quickly.

Marking points

One mark for more cycles per second; one for linked effect on speed/work completed.

05 ANSWERS & MARK SCHEME

Check your work · 2 of 3

Mark the scientific idea, not only the exact wording. Use the model answer to improve your own.

Q5 · 2 marks

Multiple cores can work on different tasks simultaneously; this can improve multitasking or responsiveness when software can run in parallel.

Marking points

One mark for parallel work; one for an explained performance/multitasking benefit.

Q6 · 3 marks

Clock speed is the rate of clock cycles per second, measured in Hz. A higher clock rate gives the processor more cycles in a given time, which can allow more instructions to be executed and reduce processing time.

Marking points

Clock frequency; change from increasing it; linked effect on performance.

Q7 · 3 marks

Cache holds frequently used data and instructions close to the CPU. It is faster to access than RAM, so it reduces delays fetching those items and may speed up processing.

Marking points

What cache holds; faster than RAM; fewer delays/more performance.

Q8 · 3 marks

Each CPU core can process instructions. With more cores, suitable tasks can run in parallel, helping the computer process multiple tasks at once and improve responsiveness.

Marking points

Meaning of cores; parallel operation; effect on suitable workload.

05 ANSWERS & MARK SCHEME

Check your work · 3 of 3

Mark the scientific idea, not only the exact wording. Use the model answer to improve your own.

Q9 · 4 marks

Clock speed: a higher rate provides more CPU cycles each second, so processing may complete sooner. Cache size: a larger cache can hold more frequently used data, reducing access to slower RAM.

Marking points

For each of two relevant factors, credit the characteristic and a valid explained effect.

Q10 · 4 marks

Larger cache means more frequently used information may be available quickly rather than retrieved from RAM, reducing waiting. Additional cores allow compatible tasks to be processed simultaneously, helping multitasking.

Marking points

Two linked explanatory points for cache and two for cores, as appropriate.

Q11 · 6 marks

Clock speed controls how many clock cycles occur per second; increasing it can allow instructions to be processed faster. A larger cache stores more frequently used data/instructions in fast memory and reduces slower RAM accesses. More CPU cores allow suitable work to proceed in parallel, improving performance when several tasks run. Actual improvement depends on how the program uses the CPU.

Marking points

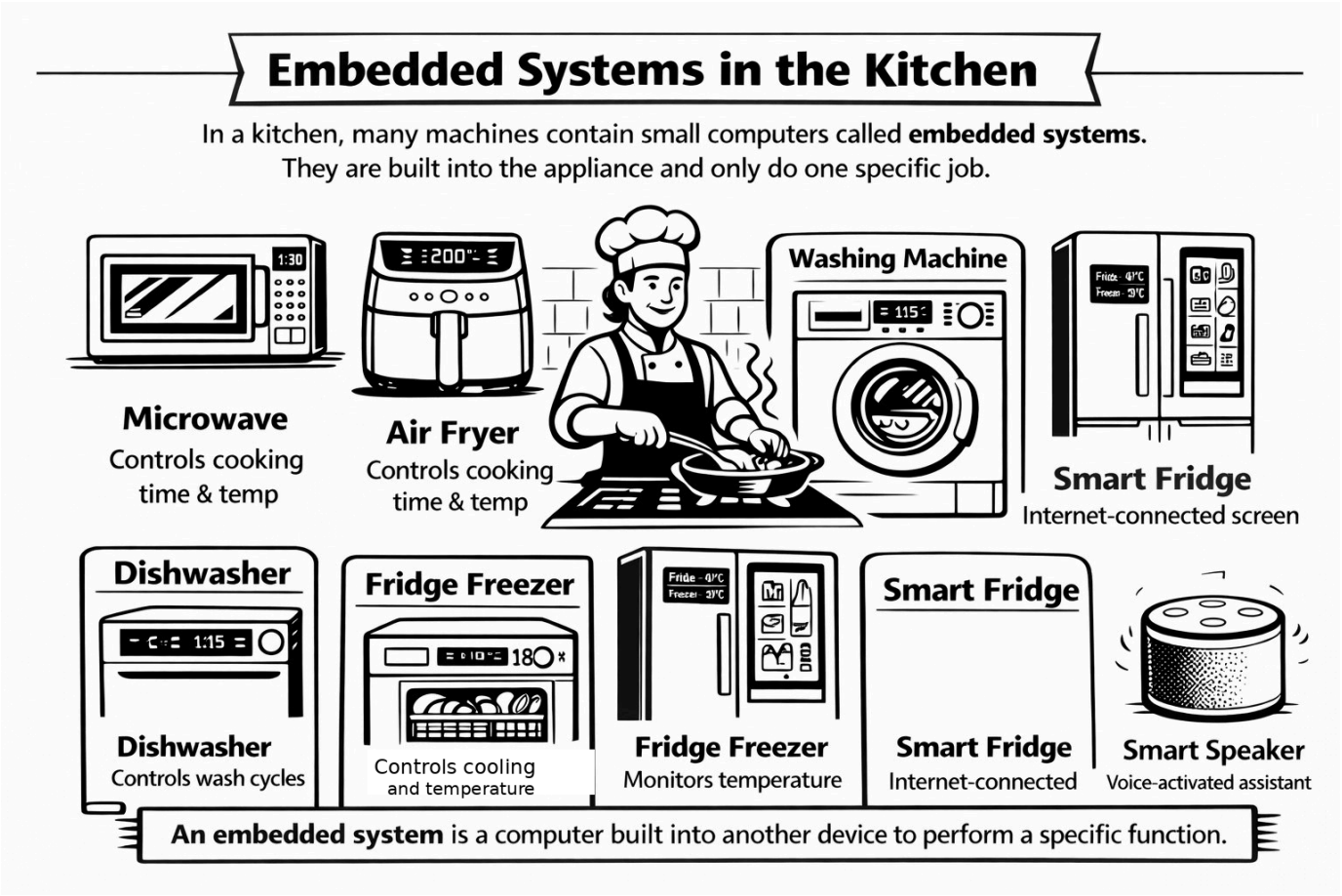
Indicative: explain clock speed and its effect; cache and its effect; cores and parallel work; link to overall performance and/or suitability of workload.

01 THEORY

Embedded systems in the kitchen

A familiar appliance can contain its own small, task-focused computer.

Embedded systems in the kitchen



Embedded system

A computer built into another device to control or monitor a dedicated function. It is not normally used like a general-purpose laptop.

Read the image carefully

Examples include a microwave managing heating time, a dishwasher controlling a wash cycle and a fridge controller monitoring temperature.

A fridge-freezer controller monitors the temperature and switches cooling on when needed.

01 THEORY

From appliance to exam answer

Three examples from the kitchen illustration.



MICROWAVE

INPUT

User sets cooking time

PROCESS

Controller counts down

OUTPUT

Turns heating off at the end



WASHING MACHINE

INPUT

User chooses wash cycle

PROCESS

Controller follows programme

OUTPUT

Operates pump, drum, heater



FRIDGE CONTROLLER

INPUT

Temperature sensor input

PROCESS

Controller compares to target

OUTPUT

Switches cooling as needed

A controller receives inputs, follows instructions and controls outputs.

01 THEORY

What makes them different?

Compare the function, interface and requirements of different devices.

Embedded

A washing-machine controller has a defined task. It reads inputs and controls a specific appliance.

Designed for a dedicated role

General-purpose

A laptop can run many different applications chosen by its user: documents, games, videos and coding.

Runs a wide range of programs

Typical characteristics — avoid overgeneralising

Designed for a specific role

Works within an appliance or a larger system.

Often a limited interface

Could have buttons, a small display or no screen.

Often reliable, low power

Requirements depend on the device and its environment.

Some are real-time

A response must meet a required time deadline.

Uses memory and firmware

Programs and temporary data need storage; memory sizes vary.

Exam habit: say “often” or “typically” unless the question specifies an absolute.

02 KEY TERMINOLOGY

Words behind the kitchen analogy

The exact lesson highlighter marks words used in exam explanations.

Embedded system

A computer integrated into a larger device for a dedicated task.

Dedicated function

The particular job or limited set of jobs a controller is designed to perform.

Controller

Processes inputs and issues outputs to manage an appliance.

Sensor

Detects a physical condition, such as temperature.

Input

A signal or information received by a system.

Output

An action or signal generated by the controller.

Real-time

Must respond within a specified deadline when timing matters.

User interface

The way a user operates a device or receives information.

Firmware / ROM

Program instructions kept in non-volatile storage (where appropriate).

RAM

Working memory for temporary values used while the system runs.

Exam precision: ROM and RAM capacity vary; not every embedded system is real-time.

03 WORKED EXAMPLES

Example 1 · Washing machine

Follow input → process → output, then turn the diagram into an answer.

Exam-style example (4 marks)

Explain how a washing machine can use an embedded system to control a wash cycle.

INPUT

User selects a wash programme; sensors may report water level.

▼

PROCESS

Embedded controller follows the selected sequence and sensor readings.

▼

OUTPUT

Motor, valves, pump or heater are switched on or off.

Model answer (4 linked points)

The washing machine contains a computer controller built into the appliance [1]. A user selects a wash programme [1]. The controller receives inputs such as sensor readings and follows the programmed sequence [1]. It controls outputs, for example the pump or motor, to carry out the cycle [1].

03 WORKED EXAMPLES

Example 2 · Microwave and laptop

Apply the characteristics without claiming that every device behaves identically.

Exam-style example (3 marks)

Explain why a microwave controller is an embedded system but a laptop is general-purpose.

1 Identify the system

The microwave contains a computer built inside another device.

2 Explain its purpose

It has a dedicated role: timing and controlling microwave operation.

3 Compare

A laptop allows users to run many unrelated programs.

Check the common misconceptions

Real-time does not mean “instant”: it means meeting a deadline when the application requires it.

ROM and RAM are not guaranteed to be “large” or “small”. Firmware may be held in non-volatile storage and RAM may hold temporary working data; the precise amounts depend on the device.

04 EXAM PRACTICE

Questions 1–5

Attempt all questions before looking at the answers.

Q1 1 mark

What is an embedded system?

Q2 1 mark

Name one characteristic of an embedded system.

Q3 1 mark

Where is the software for an embedded system usually stored?

Q4 2 marks

State two characteristics of an embedded system.

Q5 2 marks

State two reasons embedded systems use low power.

04 EXAM PRACTICE

Questions 6–8

Explain your reasons clearly; use the correct technical terms.

Q6 3 marks

Explain what is meant by a real-time embedded system.

Q7 3 marks

Explain why embedded systems have a limited user interface.

Q8 3 marks

Explain why embedded systems use a large ROM and a small amount of RAM.

04 EXAM PRACTICE

Questions 9–11

Use examples and connect characteristics to their purpose.

Q9 4 marks

Describe two characteristics of embedded systems.

Q10 4 marks

Describe how real-time operation and low power consumption are important in embedded systems.

Q11 6 marks

Explain the main characteristics of embedded systems.

05 ANSWERS & MARK SCHEME

Questions 1–5

Use these marking points to check and improve your answers.

Q1 · 1 marks

What is an embedded system?

- Built into another device [1].
- Performs a dedicated or specific function (helpful extra detail).

Q2 · 1 marks

Name one characteristic of an embedded system.

- Any one typical characteristic: dedicated purpose / built into larger device / restricted interface / power-efficient design / time-constrained operation where appropriate [1].

Q3 · 1 marks

Where is the software for an embedded system usually stored?

- ROM or another suitable non-volatile storage holding firmware [1].

Q4 · 2 marks

State two characteristics of an embedded system.

- Built into a larger system [1].
- Designed for a particular task or limited set of tasks [1].

Q5 · 2 marks

State two reasons embedded systems use low power.

- Lower consumption helps extend battery life in portable devices [1].
- Lower energy/heat demand benefits always-on or constrained appliances [1].

05 ANSWERS & MARK SCHEME

Questions 6–8

Use these marking points to check and improve your answers.

Q6 · 3 marks

Explain what is meant by a real-time embedded system.

- Timing of the response affects whether operation is correct [1].
- The controller responds to an event or input [1].
- It must complete the response within a specified deadline [1].

Q7 · 3 marks

Explain why embedded systems have a limited user interface.

- The system serves a limited range of user tasks [1].
- Many controls can therefore be pre-programmed or automatic [1].
- A few controls or indicators may be sufficient, depending on the appliance [1].

Q8 · 3 marks

Explain why embedded systems use a large ROM and a small amount of RAM.

- Firmware must be retained when power is removed, so it may be kept in ROM or other non-volatile storage [1].
- RAM stores temporary data and working values during operation [1].
- Memory is provisioned to suit the task; not every embedded system has “large ROM and small RAM” [1].

Self-mark and correct

Tick the correct points you made. Underline technical terms you missed, then improve one answer.

05 ANSWERS & MARK SCHEME

Questions 9–11

Use these marking points to check and improve your answers.

Q9 · 4 marks

Describe two characteristics of embedded systems.

- Embedded in a larger device [1] and carries out a dedicated task [1].
- Further characteristic with explanation (e.g. restricted interface because controls are task-specific) [1+1].

Q10 · 4 marks

Describe how real-time operation and low power consumption are important in embedded systems.

- A time-critical controller must complete a required response within a specified deadline [1], supporting safe or correct operation [1].
- Power-efficient design helps battery-powered or always-on devices run longer [1] and reduce energy or heat demands [1].

Q11 · 6 marks

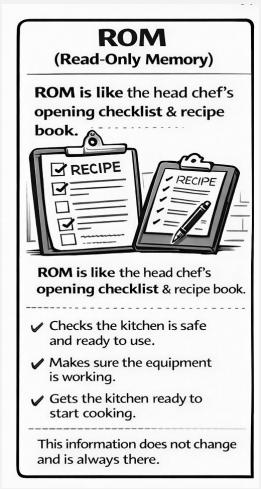
Explain the main characteristics of embedded systems.

- Embedded in a larger system, such as a washing machine [1].
- Designed for a dedicated control or monitoring role [1].
- Usually uses task-specific hardware or firmware [1].
- May use a limited interface because only relevant controls are needed [1].
- Some are real-time, meeting response deadlines [1].
- Often optimised for cost, size and power needs rather than general-purpose computing [1].

01 · THEORY

Primary memory

The chef’s kitchen helps us remember what RAM, ROM and virtual memory do.



ROM · opening instructions



RAM · current work

Three things this guide will make clear

01 **Start-up instructions**

ROM keeps the instructions needed to get the computer started.

02 **Working memory**

RAM holds programs and data that are being used right now.

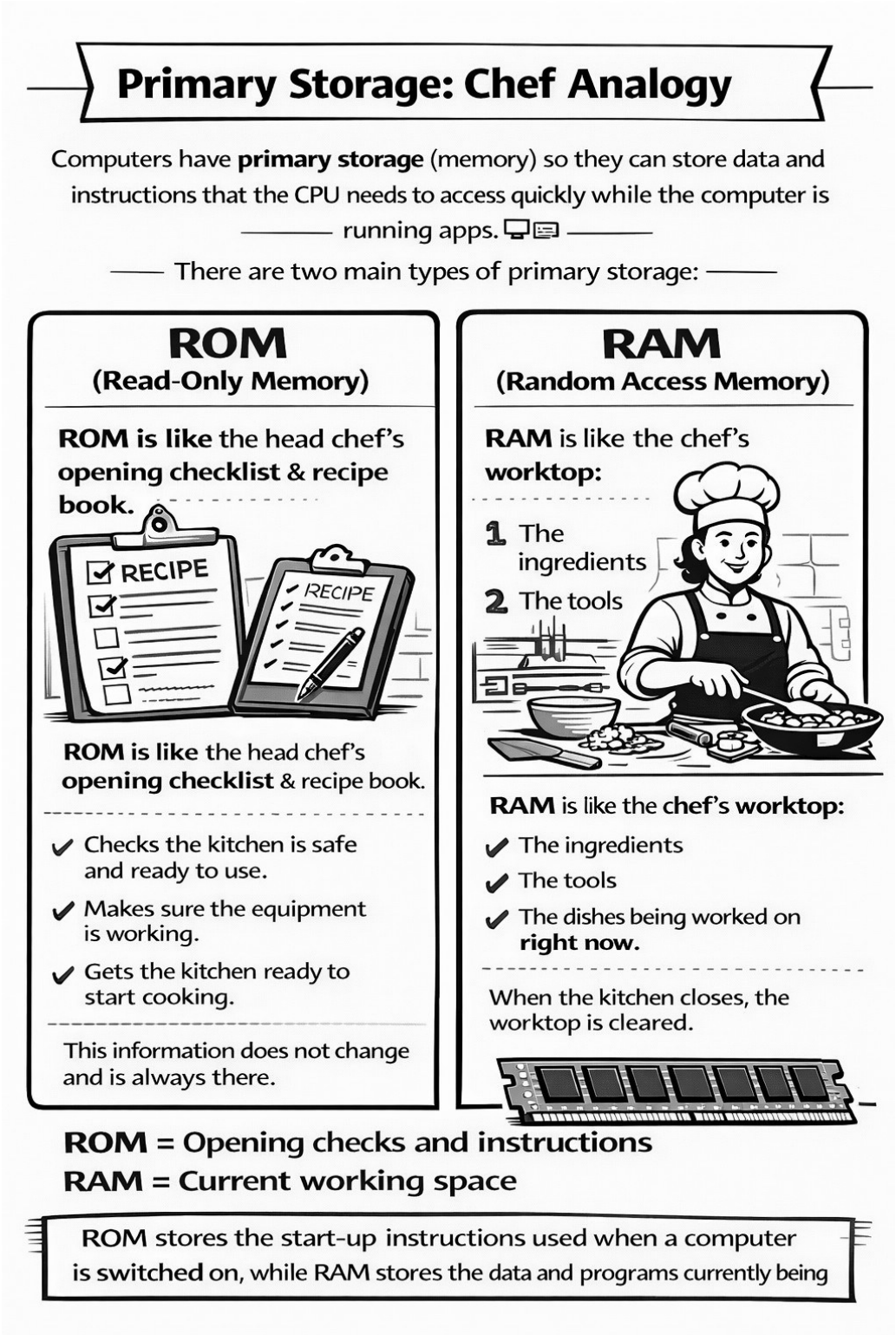
03 **When RAM is full**

Virtual memory temporarily uses slower secondary storage.

01 · THEORY

RAM and ROM in the kitchen

The illustration compares an opening checklist with a worktop.



01 · THEORY

From kitchen to examination

Use the visual analogy to understand the idea; use the precise technical words when answering.

ROM

Opening checklist in the analogy. In a computer, ROM is non-volatile memory that stores start-up instructions (firmware). The computer can read these instructions when it powers on.

Exam wording: ROM stores the start-up instructions and retains its contents when power is removed.

RAM

Chef’s worktop in the analogy. RAM stores the instructions and data for programs currently running. It can be read and written to, but its contents are lost when the computer powers off.

Exam wording: RAM is volatile, read/write memory for programs and data in current use.

Compare them — the details that earn marks

	RAM	ROM
Main use	Running programs and data	Start-up instructions
When power is off	Contents lost (volatile)	Contents retained (non-volatile)
Typical access	Read and write	Mainly read in normal use

01 · THEORY

Virtual memory: the second room

A slower place to hold some data when the worktop (RAM) is full.



01 · THEORY

How virtual memory works

The picture explains why it helps and why it makes some tasks slower.

1

RAM becomes full

A program needs space for more data or instructions.

↓

2

Move out less-needed data

The system transfers some contents from RAM to an area of secondary storage.

↓

3

Keep working

This frees space in RAM for the data currently required.

↓

4

Bring data back when needed

Retrieving it from secondary storage is slower than using RAM.

Do not confuse virtual memory with cache

Cache holds frequently needed data close to the CPU to improve access. Virtual memory uses slower secondary storage to free RAM space when needed.

02 · KEY TERMINOLOGY

Words that earn marks

These precise definitions connect the kitchen pictures to the OCR examination.

Primary storage

Memory that gives the processor access to data or instructions it needs.

RAM

Volatile read/write memory holding currently used programs and data.

ROM

Non-volatile memory used to store computer start-up instructions.

Volatile

Loses its stored contents when the power is switched off.

Non-volatile

Retains stored contents without electrical power.

Firmware

Instructions stored in non-volatile memory for a device to start or operate.

Virtual memory

An area of secondary storage used when available RAM space is insufficient.

Secondary storage

Non-volatile storage, such as an SSD or HDD, generally slower than RAM.

Tip · Use the term, then explain what it does in the question’s context.

03 · WORKED EXAMPLES

Worked Example 1 · RAM vs ROM

A model response with clearly explained reasoning, not a classroom “I Do”.

QUESTION · 4 MARKS

Describe two differences between RAM and ROM.

Step 1 · Choose two genuine contrasts

What they store

RAM stores currently used programs and data. ROM stores start-up instructions.

Whether contents persist

RAM is volatile, so its data is lost without power. ROM is non-volatile and retains its contents.

Step 2 · Write your answer in complete comparisons

Model response: RAM stores programs and data currently being used, while ROM stores the instructions needed to start the computer. RAM is volatile and loses its contents without power, whereas ROM is non-volatile and keeps them.

Why it earns marks: You identify two features of RAM and give the matching contrast for ROM (4 linked marking points).

03 · WORKED EXAMPLES

Worked Example 2 · RAM runs out

Use the second-room analogy, then write the computer-science explanation.

QUESTION · 6 MARKS

Explain the purpose of virtual memory and the effect it has on performance.

Picture	Ingredients not needed immediately move from the crowded worktop to the back room.
Technical process	When RAM is full, some data is moved to an area of secondary storage.
Return	When needed again, data is transferred back into RAM.
Consequence	Secondary storage is slower than RAM, so repeated transfers reduce performance.

Exam-quality six-mark explanation

Virtual memory allows a computer to continue running when available RAM is full. The system transfers some data from RAM to an area of secondary storage, freeing space for other data and programs. When the moved data is needed again, it is brought back into RAM. This is slower because secondary storage takes longer to access than RAM, so frequent swapping can reduce performance.

04 · EXAM PRACTICE

Your turn · Questions 1–5

Answer independently. The mark scheme is on separate pages.

Q1 What is RAM? **1 mark**

Q2 What does volatile mean? **1 mark**

Q3 What is ROM used for? **1 mark**

Q4 State two differences between RAM and ROM. **2 marks**

Q5 State two things stored in RAM. **2 marks**

04 · EXAM PRACTICE

Your turn · Questions 6–8

Answer independently. The mark scheme is on separate pages.

Q6 Explain why computers need RAM. **3 marks**

Q7 Explain why ROM is non-volatile and why that matters. **3 marks**

Q8 Explain what virtual memory is. **3 marks**

04 · EXAM PRACTICE

Your turn · Questions 9–11

Answer independently. The mark scheme is on separate pages.

Q9 Describe two differences between RAM and ROM. **4 marks**

Q10 Describe what happens when a computer uses virtual memory. **4 marks**

Q11 Explain the purpose of virtual memory and the effect it has on performance. **6 marks**

05 · ANSWERS & MARK SCHEME

Mark your answers · Q1–Q6

Mark for the computer-science idea, not just a memorised phrase.

Q1 · 1 mark

RAM is volatile main memory holding programs and data in current use.

Marking: 1: Holds running programs/data (main working memory).

Q2 · 1 mark

Volatile memory loses its stored data when power is switched off.

Marking: 1: Contents are lost without power.

Q3 · 1 mark

ROM stores start-up instructions or firmware used when the computer boots.

Marking: 1: Boot/start-up instructions.

Q4 · 2 marks

RAM is volatile whereas ROM is non-volatile. RAM stores current programs and data whereas ROM contains start-up instructions.

Marking: 1 for each valid contrasted difference (2 total).

Q5 · 2 marks

Running programs and data/files currently in use.

Marking: 1: Programs; 1: Data currently being processed.

Q6 · 3 marks

RAM temporarily holds programs and data while they run, allowing the CPU fast access to instructions and data required for processing.

Marking: 1: Stores active programs; 1: Stores active data/instructions; 1: CPU can access rapidly.

05 · ANSWERS & MARK SCHEME

Mark your answers · Q7–Q11

Use the marking ideas to identify anything you need to revisit.

Q7 · 3 marks

ROM is non-volatile, meaning its contents remain when the computer is turned off. This matters because start-up instructions are still available when power is switched on again.

Marking: Non-volatile (1); retains stored instructions without power (1); allows start-up on the next boot (1).

Q8 · 3 marks

Virtual memory uses an area of secondary storage when RAM does not have enough space. Data is transferred out of RAM into this area until it is needed again.

Marking: Insufficient RAM (1); uses secondary storage (1); temporarily transfers data (1).

Q9 · 4 marks

RAM is volatile and stores currently running programs and data. ROM is non-volatile and typically stores start-up instructions.

Marking: RAM active data/programs (1); ROM start-up instructions (1); RAM volatile (1); ROM non-volatile (1).

Q10 · 4 marks

When RAM is full, the operating system moves some less-needed data from RAM to a reserved area of secondary storage. This makes room in RAM for other data. If the moved data is needed again, it is brought back.

Marking: RAM reaches capacity (1); data moves to secondary storage (1); frees RAM space (1); returns when required (1).

Q11 · 6 marks

Virtual memory enables work to continue when available RAM is full. The computer moves some data from RAM to an area of secondary storage. This frees RAM space for active programs and data. When the moved data is needed, it is returned to RAM. Secondary storage is slower to access than RAM, so these transfers can slow the computer, particularly if frequent.

Marking: Why needed (1); uses secondary storage (1); moves data out of RAM (1); frees space (1); moves it back (1); explains reduced speed (1).

01 · THEORY

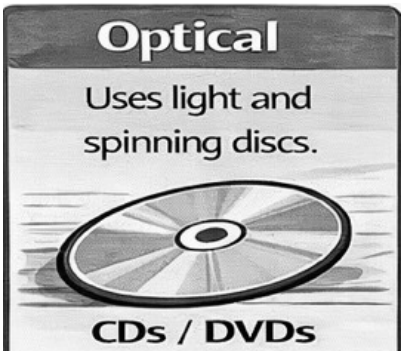
Why we need secondary storage

A device needs non-volatile storage to keep files when power is off. Use the diagram to compare optical, magnetic and solid-state devices.

Big idea · memory versus storage

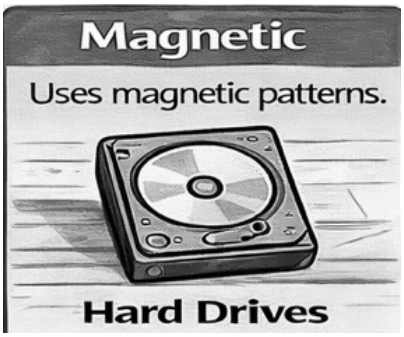
RAM holds the data currently being used; secondary storage keeps files and programs when power is off. It is non-volatile.

OPTICAL



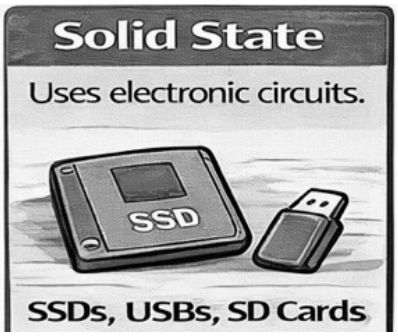
Discs: CD, DVD

MAGNETIC



Hard disk drives

SOLID STATE



SSDs, flash media

From picture to examination language

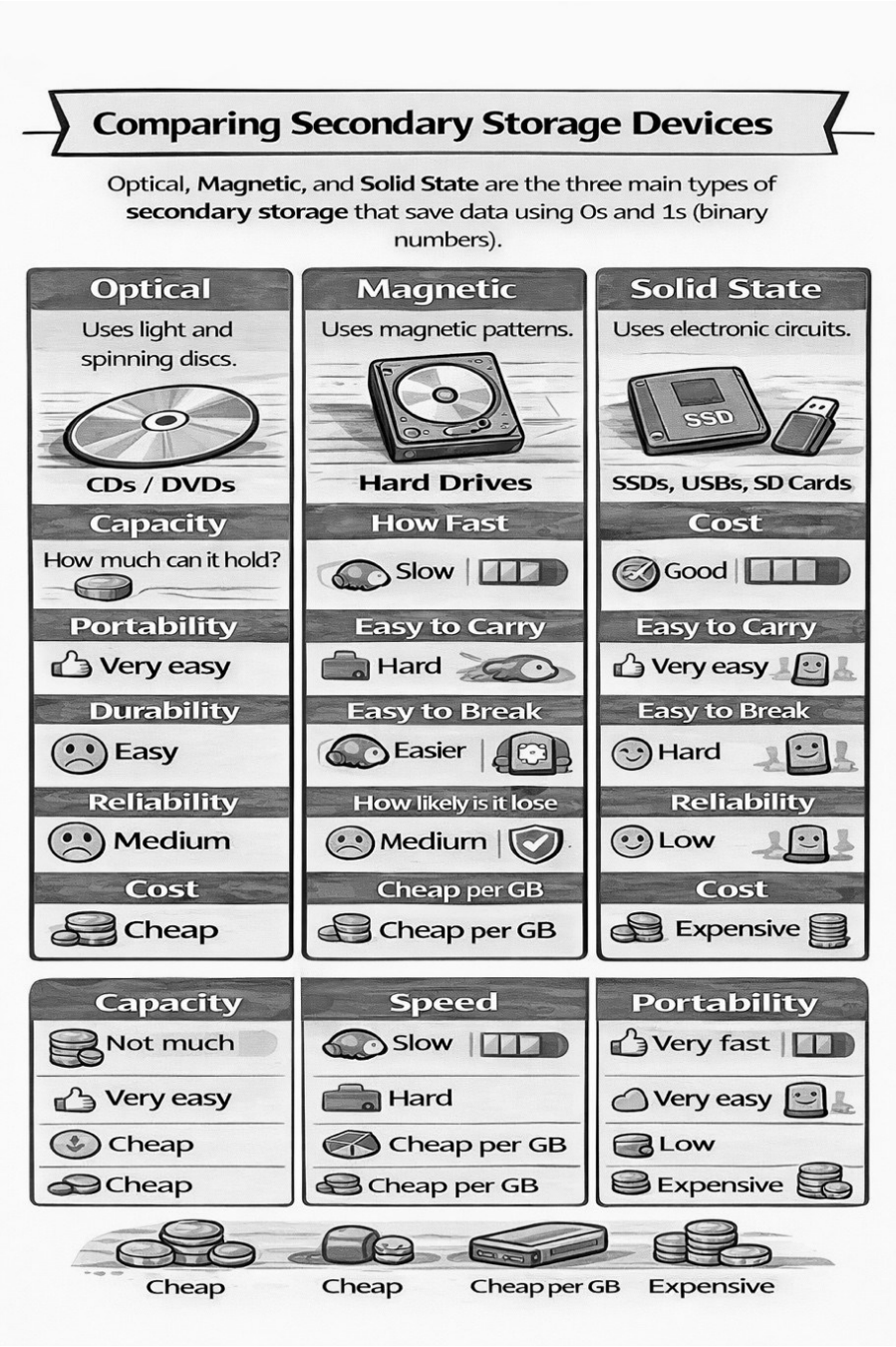
Recognise which type each device uses. When a question asks you to recommend one, justify it against the scenario: capacity, speed, portability, durability, reliability and cost.

Use the picture to recall each storage type. Check capacity, speed, durability, reliability and cost using the comparisons that follow.

01 · THEORY

Compare three types of secondary storage

Use the diagrams to compare optical, magnetic and solid-state storage. The devices shown are examples, not a fixed ranking chart.

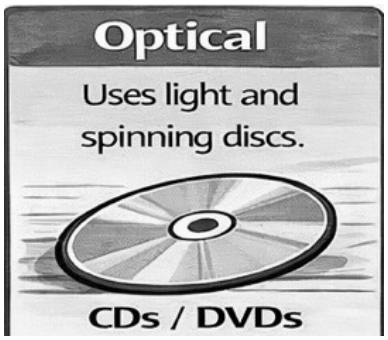


Devices vary in reliability and portability. For exam answers, compare capacity, speed, cost, durability, reliability and portability.

01 · THEORY

Three types, three different methods

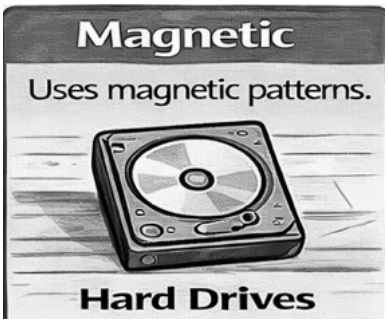
The image explains what the storage devices look like; the wording below is what earns marks.



Optical

Examples: CDs, DVDs, Blu-ray discs.

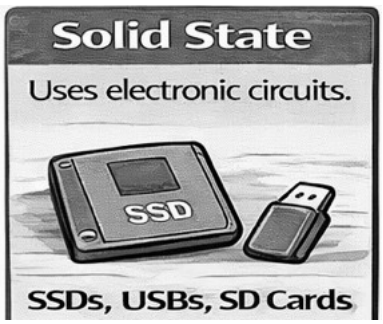
A laser reads the disc. Often convenient for distributing physical media; a disc can be scratched.



Magnetic

Example: hard disk drive (HDD).

A hard disk stores data as magnetic patterns. Typically offers large capacity at a relatively low cost per GB.



Solid state

Examples: SSD, memory card, USB flash drive.

Stores data electronically with no moving parts. Typically gives faster access and better shock resistance.

OCR does not require detailed knowledge of the component parts inside each type of storage device; concentrate on recognition, comparison and choice.

01 · THEORY

Which storage type should you choose?

There is no single best storage device. Compare the requirement with the characteristics that matter in the question.

Capacity	**Total amount stored.** Magnetic HDDs can offer very large capacities; check the particular device.
Speed	**How quickly data is accessed.** SSDs are commonly faster than HDDs; optical discs are often slower.
Portability	**Ease of moving or carrying it.** Flash drives and memory cards are small; optical discs can be carried but scratched.
Durability	**Resistance to physical damage.** SSDs have no moving parts; HDDs have moving mechanisms; discs can scratch.
Reliability	**Likelihood of retaining data and working.** Depends on age, use, environment and failure risk: no type is infallible.
Cost	**Price, often cost per GB.** HDDs often provide lower cost per GB; SSDs trade some price for speed.

Exam technique: Name a type, give a relevant characteristic, then link that characteristic to the user’s needs.

02 · KEY TERMINOLOGY

Words that earn marks

The exact translucent yellow strokes used in J277/H446 lessons sit behind each term.

Secondary storage	Non-volatile storage used for files and programs between sessions.
Non-volatile	Retains its stored data when the power is switched off.
Optical storage	Storage media read using light, such as CDs and DVDs.
Magnetic storage	Stores data using magnetic patterns, such as an HDD.
Solid-state storage	Electronic, non-volatile storage with no moving parts.
Capacity	The maximum quantity of data a device can store.
Access speed	How quickly data can be read or written.
Portability	How easily a device can be moved or carried.
Durability	How well a device withstands physical damage.
Reliability	How dependably data remains available over time.
Cost per GB	Cost relative to the amount of storage provided.

03 · WORKED EXAMPLES

Worked Example 1 · Backing up files

Use the actual storage characteristics, not blanket statements such as “solid state is always best”.

SCENARIO · 4 MARKS

A film club keeps a very large archive of videos at a fixed desk. It wants plenty of storage without spending too much. Recommend a type of secondary storage and justify it.

1 Identify the most important requirements

Very large capacity and a relatively low cost per GB.

2 Choose a suitable type

Magnetic storage, such as a large hard disk drive.

3 Link the type to the scenario

An HDD can provide high capacity economically, making it suitable for large video files.

4 State the sensible limitation

It has moving parts and may be slower than an SSD; portability matters less at a fixed desk.

03 · WORKED EXAMPLES

Worked Example 2 · Portable computing

Same method, different needs. A different choice can earn full marks when the explanation fits.

SCENARIO · 4 MARKS

A pupil takes a laptop to school each day. It must start quickly, survive frequent movement and use little power. Recommend a storage type and explain why.

Recommended type: solid-state storage

An SSD usually gives fast access for loading programs. With no moving parts, it is better suited to being carried than a typical spinning HDD.

Link reasons to the scenario

Fast access supports quick start-up; resistance to knocks supports a laptop that is frequently moved. You are explaining the needs, not just listing features.

Balanced judgement

For maximum capacity at minimum cost per GB, an HDD might still be sensible. The best answer depends on what the question values most.

Marking approach: award clear choice + justified points linked to the scenario. More than one type can sometimes be defended.

04 · EXAM PRACTICE

Your turn · Questions 1–5

Marks are shown for each question.

Q1 **What is secondary storage?** **1 mark**

Q2 **Give one example of secondary storage.** **1 mark**

Q3 **What does non-volatile mean?** **1 mark**

Q4 **State two reasons computers need secondary storage.** **2 marks**

Q5 **State two differences between primary and secondary storage.** **2 marks**

04 · EXAM PRACTICE

Your turn · Questions 6–9

Marks are shown for each question.

Q6 **Explain how optical storage stores data.** **3 marks**

Q7 **Explain how magnetic storage stores data.** **3 marks**

Q8 **Explain how solid-state storage stores data.** **3 marks**

Q9 **Describe two advantages of solid-state storage.** **4 marks**

04 · EXAM PRACTICE

Your turn · Questions 10–11

Marks are shown for each question.

Q10 **Describe two disadvantages of optical storage.** **4 marks**

Q11 **Compare optical, magnetic and solid-state storage.** **6 marks**

05 · ANSWERS & MARK SCHEME

Mark your answers · Q1–Q5

Check the explanation and whether each marking idea appears in your response. Correct weak vocabulary before retrying.

Q1 **What is secondary storage?** **1 mark**

Secondary storage is non-volatile storage used to store data and programs permanently.

Q2 **Give one example of secondary storage.** **1 mark**

Hard disk drive / SSD / USB flash drive / CD / DVD.

Q3 **What does non-volatile mean?** **1 mark**

Non-volatile means data is kept when the power is switched off.

Q4 **State two reasons computers need secondary storage.** **2 marks**

It stores files permanently and keeps data when the computer is turned off.

Q5 **State two differences between primary and secondary storage.** **2 marks**

Compared with RAM, secondary storage is non-volatile and typically slower to access. ROM is a non-volatile form of primary storage.

05 · ANSWERS & MARK SCHEME

Mark your answers · Q6–Q9

Check the explanation and whether each marking idea appears in your response. Correct weak vocabulary before retrying.

Q6 **Explain how optical storage stores data.** **3 marks**

Optical storage uses laser light to read data stored as pits and lands on a spinning disc.

Q7 **Explain how magnetic storage stores data.** **3 marks**

Magnetic storage stores data using magnetic patterns on a spinning disc that can be read and written by a magnetic head.

Q8 **Explain how solid-state storage stores data.** **3 marks**

Solid-state storage stores data using electronic circuits and has no moving parts.

Q9 **Describe two advantages of solid-state storage.** **4 marks**

Solid-state storage offers fast access, so files and programs load quickly. It has no moving parts, making it more resistant to damage from knocks when carried.

05 · ANSWERS & MARK SCHEME

Mark your answers · Q10–Q11

Check the explanation and whether each marking idea appears in your response. Correct weak vocabulary before retrying.

Q10 **Describe two disadvantages of optical storage.** **4 marks**

Optical discs usually hold less data than large HDDs/SSDs, so are unsuitable for very large backups. Access can be slower, so reading a large collection of files takes longer.

Q11 **Compare optical, magnetic and solid-state storage.** **6 marks**

Optical discs are portable and can be cheap per disc, but typically offer lower capacity and slower access. Magnetic HDDs can hold large amounts at low cost per GB, but moving parts can be damaged by knocks. Solid-state storage provides fast access and good shock resistance, but may cost more per GB. Choose according to the scenario.

Six-mark comparison checklist

Award credit for accurate, distinct comparisons between all three types. Include the six OCR factors when relevant: capacity, speed, portability, durability, reliability and cost. Avoid fixed claims that one type always wins.

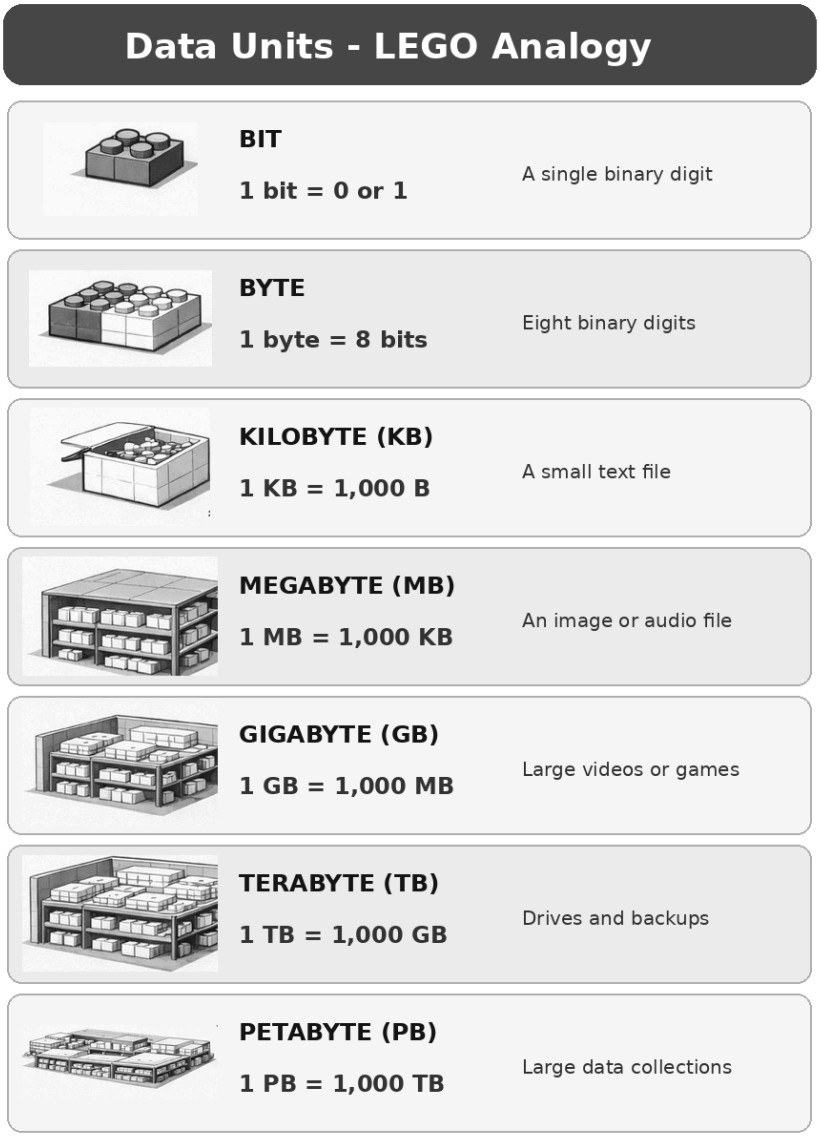
Self-mark strictly

For a 4-mark “describe two” answer, give two different advantages or disadvantages and develop each. For a 6-mark comparison, explain trade-offs: do not simply list three device names.

01 · THEORY

Data units are like LEGO

Start with a single binary digit, then group pieces to make larger units. The LEGO illustration makes the increase in storage units easier to remember.



From bytes onwards: 1,000 of each unit makes the next larger unit.
Bit to byte is different: eight bits make one byte.

From picture to examination language

A bit is one 0 or 1. 4 bits make a nibble; 8 bits make a byte. From bytes, OCR normally uses decimal steps of 1,000.

Use the LEGO blocks to picture the increasing size of each unit. The next page shows the exact conversions.

01 · THEORY

How big is each unit?

The LEGO stack grows; the table gives the exact units and conversions you should use in OCR answers.

Bit	**b**	A single 0 or 1
Nibble	**4 bits**	Half a byte
Byte	**8 bits**	Two nibbles
Kilobyte (KB)	**1,000 B**	1,000 bytes
Megabyte (MB)	**1,000 KB**	One thousand kilobytes
Gigabyte (GB)	**1,000 MB**	One thousand megabytes
Terabyte (TB)	**1,000 GB**	One thousand gigabytes
Petabyte (PB)	**1,000 TB**	One thousand terabytes

Move between units

Small → large: divide by 1,000 per step. Large → small: multiply by 1,000 per step. Bits ↔ bytes use 8, not 1,000.

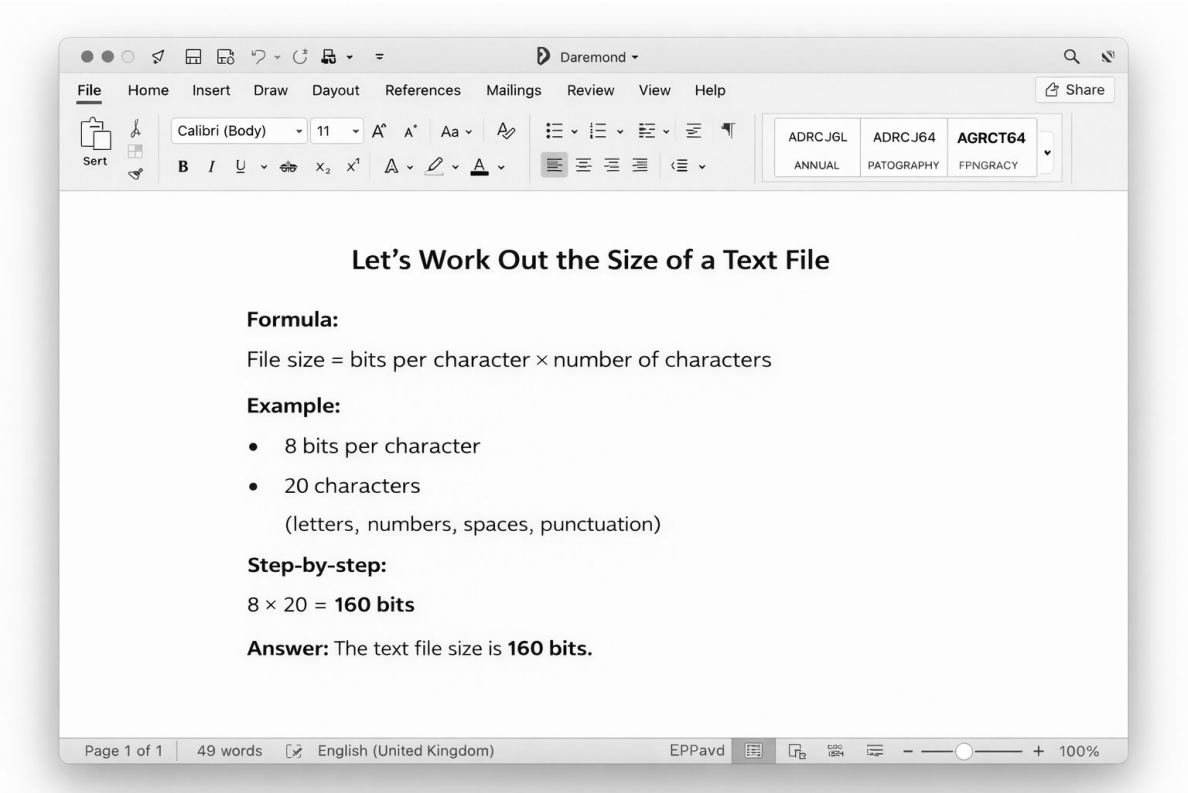
Capacity is not free space

Capacity is the fixed maximum for that device; free space changes as files are added or removed.

01 · THEORY

A text file is made of bits

The illustrated text-file calculation shows how a character count becomes a size in bits. It is an example, not a rule that every character is 8 bits.



Text file size formula

Number of characters × bits per character = file size in bits. Then divide by eight for bytes. A question may state an 8-bit character representation.

Worked example · character file size

20 characters × 8 bits = 160 bits. Divide by eight: 20 bytes. The number of bits per character must be provided or stated as an assumption.

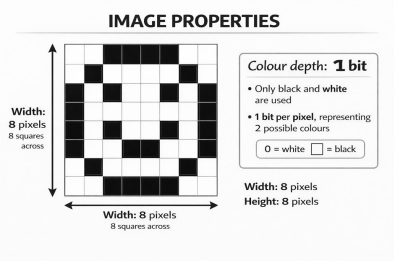
A character set assigns codes to letters and symbols. Unicode provides a much wider character repertoire than ASCII. Do not assume every Unicode character requires 16 or 32 bits.

01 · THEORY

Picture size: count pixels

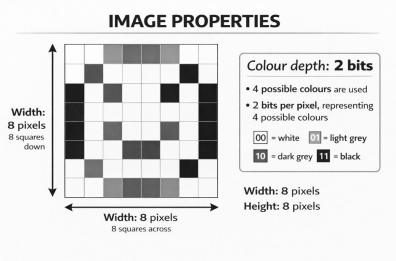
The pixel-grid images show that file size depends on image dimensions and the number of bits used for each pixel.

1-bit colour



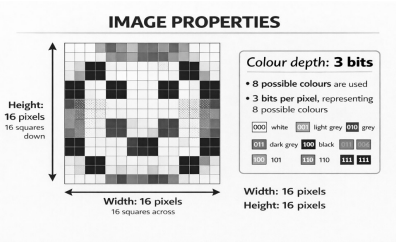
8 × 8 × 1 = 64 bits

2-bit colour



8 × 8 × 2 = 128 bits

3-bit colour



16 × 16 × 3 = 768 bits

Image file size (uncompressed)

Width (pixels) × height (pixels) × colour depth (bits per pixel) gives the number of bits. Divide by 8 for bytes.

How to work out a question

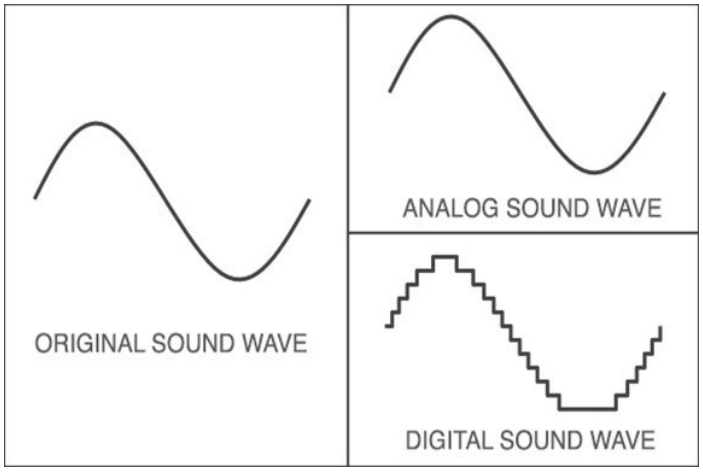
Write the formula; multiply dimensions to get pixels; multiply by colour depth; then convert bits → bytes → KB → MB if needed.

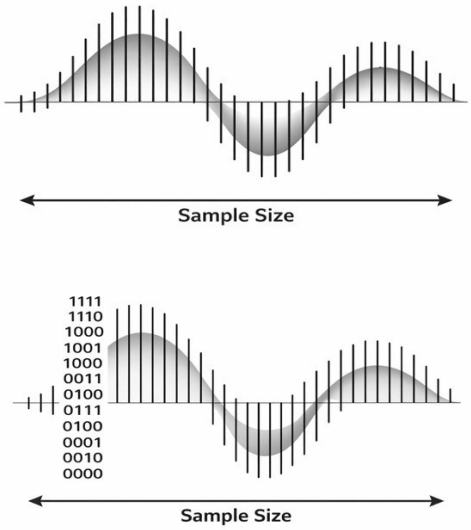
Exam reminder: this is the uncompressed pixel-data size and normally excludes file headers and metadata unless the question includes them.

01 · THEORY

Sound size: store each sample

The waveform diagrams show how an analogue sound is sampled and represented as binary data.





Sound file size (uncompressed, one channel)

Sample rate (samples/s) × duration (seconds) × bit depth (bits/sample) gives a file size in bits. Then divide by eight for bytes.

Sample rate	More measurements each second → larger file.
Duration	More seconds of recording → larger file.
Bit depth	More bits representing each sample → larger file.

Use the correct exam terms: sample rate is samples per second and bit depth is bits per sample.

02 · KEY TERMINOLOGY

Keywords · Part 1

Precise terms for OCR written answers, with the same three yellow highlighter pen strokes as the J277 lessons.

Bit	One binary digit, either 0 or 1.
Nibble	A group of four bits.
Byte	Eight bits.
Kilobyte	1,000 bytes using the OCR decimal convention.
Megabyte	1,000 kilobytes.
Gigabyte	1,000 megabytes.
Terabyte	1,000 gigabytes.
Petabyte	1,000 terabytes.

Check your units: bits vs bytes use 8; larger decimal units use 1,000.

02 · KEY TERMINOLOGY

Keywords · Part 2

Precise terms for OCR written answers, with the same three yellow highlighter pen strokes as the J277 lessons.

Capacity	Maximum quantity of data a device can store.
Free space	Unused portion of the current capacity.
Character set	Mapping between characters and their coded values.
Bits per character	Storage bits used for each character as given in a question.
Colour depth	Number of bits used to represent each pixel.
Sample rate	Number of audio samples taken each second.
Bit depth	Number of bits used to represent each audio sample.
Duration	How long the recording lasts, measured in seconds.

Check your units: bits vs bytes use 8; larger decimal units use 1,000.

03 · WORKED EXAMPLES

Worked Example 1 · LEGO to files

Follow each conversion rather than guessing whether to multiply or divide.

A · Convert 4,500 KB into MB

4,500 ÷ 1,000 = 4.5 MB. The unit is getting larger, so the number becomes smaller.

B · Convert 24 bits into bytes

24 ÷ 8 = 3 bytes. Use eight when moving from bits to bytes; do not divide by 1,000.

C · Calculate text storage

A 30-character file uses 8 bits per character. 30 × 8 = 240 bits. Then 240 ÷ 8 = 30 bytes. State the assumed character size.

The marking method

1. Write the relationship. 2. Substitute the numbers. 3. Calculate. 4. Give the right unit.

These are worked examples. Independent exam questions follow on separate pages, without model answers beside them.

03 · WORKED EXAMPLES

Worked Example 2 · Image data

The exam tells you the dimensions and colour depth; you must convert the result into the requested unit.

Question: An uncompressed image is 400 × 300 pixels with 4-bit colour depth. Calculate its size in KB (using 1,000 bytes per KB).

1. Count pixels

400 × 300 = 120,000 pixels

2. Find the number of bits

120,000 × 4 = 480,000 bits

3. Convert to bytes

480,000 ÷ 8 = 60,000 bytes

4. Convert to KB

60,000 ÷ 1,000 = 60 KB

Exam accuracy

The formula gives uncompressed pixel data. Compression or metadata can change the total if a question includes them.

03 · WORKED EXAMPLES

Worked Example 3 · Sound data

Use the sample rate, bit depth and duration in the single-channel uncompressed-audio formula.

Question: A mono sound recording uses 8,000 samples per second for 5 seconds, at 8 bits per sample. Calculate uncompressed size in KB.

1. Write the formula

Sample rate × duration × bit depth

2. Substitute the values

8,000 × 5 × 8 = 320,000 bits

3. Convert bits to bytes

320,000 ÷ 8 = 40,000 bytes

4. Convert bytes to KB

40,000 ÷ 1,000 = 40 KB

The bigger-file link

Higher sample rate, greater bit depth or longer duration increases the uncompressed data size if other factors stay fixed.

04 · EXAM PRACTICE

Your turn · Questions 1–5

Each question shows its mark allocation.

Q1 **What is a bit?** 1 mark

Q2 **How many bits are in a byte?** 1 mark

Q3 **What is meant by capacity?** 1 mark

Q4 **State two differences between capacity and free space.** 2 marks

Q5 **State two differences between ASCII and Unicode.** 2 marks

04 · EXAM PRACTICE

Your turn · Questions 6–9

Each question shows its mark allocation.

Q6 **Explain why computers store data in binary.** **3 marks**

Q7 **Explain what a character set is.** **3 marks**

Q8 **Explain how to calculate the size of a text file.** **3 marks**

Q9 **Calculate the size of a text file with 200 characters using ASCII.** **4 marks**

04 · EXAM PRACTICE

Your turn · Questions 10–11

Each question shows its mark allocation.

Q10 **Calculate the size of an image that is 1000×800 at 8-bit colour depth (in MB).** **4 marks**

Q11 **Explain how sample rate, bit depth and duration affect the size of a sound file.** **6 marks**

05 · ANSWERS & MARK SCHEME

Self-mark · Questions 1–5

Award marks for accurate terminology and working. Use these indicative points to self-mark your answers.

Q1 **What is a bit?** **1 mark**

A bit is one binary digit: 0 or 1.

Q2 **How many bits are in a byte?** **1 mark**

There are 8 bits in a byte.

Q3 **What is meant by capacity?** **1 mark**

Capacity is the maximum amount of data a storage device can hold.

Q4 **State two differences between capacity and free space.** **2 marks**

Capacity is the maximum amount a device can hold and is unchanged simply by deleting files; free space is the unused part of that capacity and changes as data is added or removed.

Q5 **State two differences between ASCII and Unicode.** **2 marks**

Unicode is able to represent a much wider range of characters and scripts than ASCII. The number of bits per character depends on the encoding; use the value specified in any file-size question.

05 · ANSWERS & MARK SCHEME

Self-mark · Questions 6–8

Award marks for accurate terminology and working. Use these indicative points to self-mark your answers.

Q6 **Explain why computers store data in binary.** **3 marks**

Electronic circuits can reliably distinguish two states, such as on/off or high/low voltage (1). Binary represents these using 0 and 1 (1), allowing data to be stored and processed as electronic states (1).

Q7 **Explain what a character set is.** **3 marks**

A character set is an agreed set of characters (1) where each character is associated with a code or numerical value (1), so letters and symbols can be represented in binary (1).

Q8 **Explain how to calculate the size of a text file.** **3 marks**

Multiply the number of characters by the number of bits per character (1). The product is a size in bits (1). Divide by 8 to get bytes, then convert to larger units as needed (1).

05 · ANSWERS & MARK SCHEME

Self-mark · Questions 9–10

Award marks for accurate terminology and working. Use these indicative points to self-mark your answers.

Q9 **Calculate the size of a text file with 200 characters using ASCII.** **4 marks**

Assume 8 bits per character, as in the Practice question (1). 200 x 8 = 1,600 bits (1). 1,600 / 8 = 200 bytes (1). Answer: 200 bytes (1).

Q10 **Calculate the size of an image that is 1000×800 at 8-bit colour depth (in MB).** **4 marks**

1,000 x 800 = 800,000 pixels (1). 800,000 x 8 = 6,400,000 bits (1). Divide by 8 = 800,000 bytes (1). Divide by 1,000 twice = 0.8 MB (1).

05 · ANSWERS & MARK SCHEME

Self-mark · Question 11

Award marks for accurate terminology and working. Use these indicative points to self-mark your answers.

Q11 **Explain how sample rate, bit depth and duration affect the size of a sound file.** **6 marks**

A higher sample rate means more samples each second, raising the total bits stored (2). Higher bit depth means more bits in each sample, raising the file size (2). Longer duration means more seconds of samples, raising the file size (2). The calculation is sample rate x duration (seconds) x bit depth, giving bits for one channel.

6-mark reasoning checklist

Sample rate: samples each second. Bit depth: bits per sample. Duration: seconds. Explain why increasing each raises uncompressed data size.

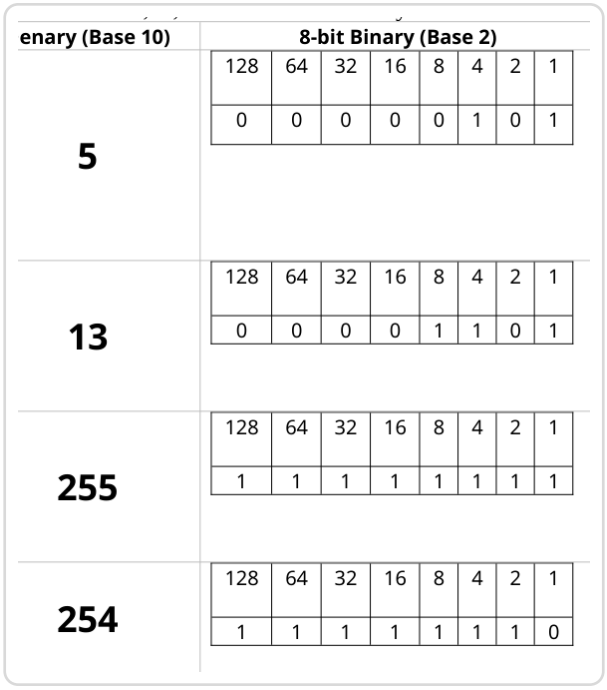
THEORY

Data Storage

Use the place-value diagrams to learn the conversion methods before attempting exam questions.

How to use this section

Study the picture first, follow the worked method, then use the exact terms. The last pages contain **all 15 Practice questions followed by separate mark-scheme answers.**



The eight-bit place-value number line

Three representations, one number

Binary uses base 2; denary uses base 10; hexadecimal uses base 16. The same number can be written in all three formats. For example, **254 = 11111110 (binary) = FE (hex)**.

The full topic also covers binary addition/overflow, shifts, character sets, image pixels/metadata and digital sound. These are all part of OCR J277 1.2.4.

THEORY

Binary and the number line

Use the 128–64–32–16–8–4–2–1 place values.

enary (Base 10)	8-bit Binary (Base 2)							
	128	64	32	16	8	4	2	1
5	0	0	0	0	0	1	0	1
	128	64	32	16	8	4	2	1
13	0	0	0	0	1	1	0	1
	128	64	32	16	8	4	2	1
255	1	1	1	1	1	1	1	1
	128	64	32	16	8	4	2	1
254	1	1	1	1	1	1	1	0

Use the place values: write 1 for a value used, otherwise write 0.

Example: 13 in 8 bits

128	64	32	16	8	4	2	1

8 + 4 + 1 = 13. The highest place (128) is the most significant bit; the smallest place (1) is the least significant bit.

Check the binary value

Check the place values: 1111110 = 126 (64+32+16+8+4+2), whereas 254 is **11111110 in eight bits. Always check the number of bits required.**

WORKED EXAMPLES

Convert to and from binary

Example 1 · 47 denary to binary

Work from left to right: 32 fits (15 left), then 8 (7 left), 4 (3 left), 2 (1 left) and 1. The unused weights become zero.

8-bit place-value number line

128	64	32	16	8	4	2	1

47 = 00101111. Each 1 means that place value is included in the total.

Example 2 · 10000001 binary to denary

8-bit place-value number line

128	64	32	16	8	4	2	1

Only the 128 and 1 columns contain a 1: **128 + 1 = 129**. You can use a shorter number line for fewer than eight digits, for example **1101 = 13**.

Exam wording

Say **place value**, **most significant bit (MSB)** and **least significant bit (LSB)**. Denary whole numbers tested here range from 0 to 255.

THEORY

Hexadecimal: two groups of four

Use the 8?4?2?1 nibble method to link binary, hexadecimal and colour codes.

128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	0
8	**4**	**2**	**1**	**8**	**4**	**2**	**1**
8+4+2+1 =15				8+4+2=14			
F				**E**			

128	64	32	16	8	4	2	1
1	0	1	0	0	0	1	1
8	**4**	**2**	**1**	**8**	**4**	**2**	**1**
8+2 =10				2+1=3			
A				**3**			

Split 11111110 into 1111 (F) and 1110 (E).

Hexadecimal uses digits **0–9** then **A=10, B=11, C=12, D=13, E=14, F=15**. Every four binary bits make one hexadecimal digit (a **nibble**).

Remember the colour-code analogy

A six-digit RGB colour such as **#00FF00** uses hexadecimal pairs for the red, green and blue channels: 00 red, FF green, 00 blue. Hex is a shorter notation for binary data, not a replacement for it.

WORKED EXAMPLES

Converting hexadecimal

Example 1 · 254 denary to hexadecimal

Convert 254 to eight bits

128	64	32	16	8	4	2	1

Split into two groups: **1111 | 1110**. The first nibble is 15 (F), the second is 14 (E). So **254 = FE**.

Example 2 · A4 hexadecimal to denary

A = 1010 and 4 = 0100

128	64	32	16	8	4	2	1

A=10 and 4=4, so the binary number is 10100100. Add weights with 1s: **128 + 32 + 4 = 164**. Alternatively **10 × 16 + 4 = 164**.

Check your method

Convert each nibble carefully, then check your answer using denary place values.

THEORY

Binary addition and overflow

Use column addition, carry bits and the eight-bit limit to check each result.

Example 1 · add 13 and 5 in eight bits

```
 00001101   (13)
+ 00000101   (5)
----------
 00010010   (18)
```

Add from the right. When 1+1 occurs, write **0** in that column and carry **1** to the next, just as carrying works in denary.

Example 2 · overflow

```
 11110000   (240)
+ 00110000   (48)
----------
1 00100000   (288 total)
```

An 8-bit register can only hold eight result bits, representing **0–255** for an unsigned number. The ninth carry bit cannot fit: this is an **overflow error**. Keeping only the low eight bits gives **00100000 = 32**, which is not the true sum.

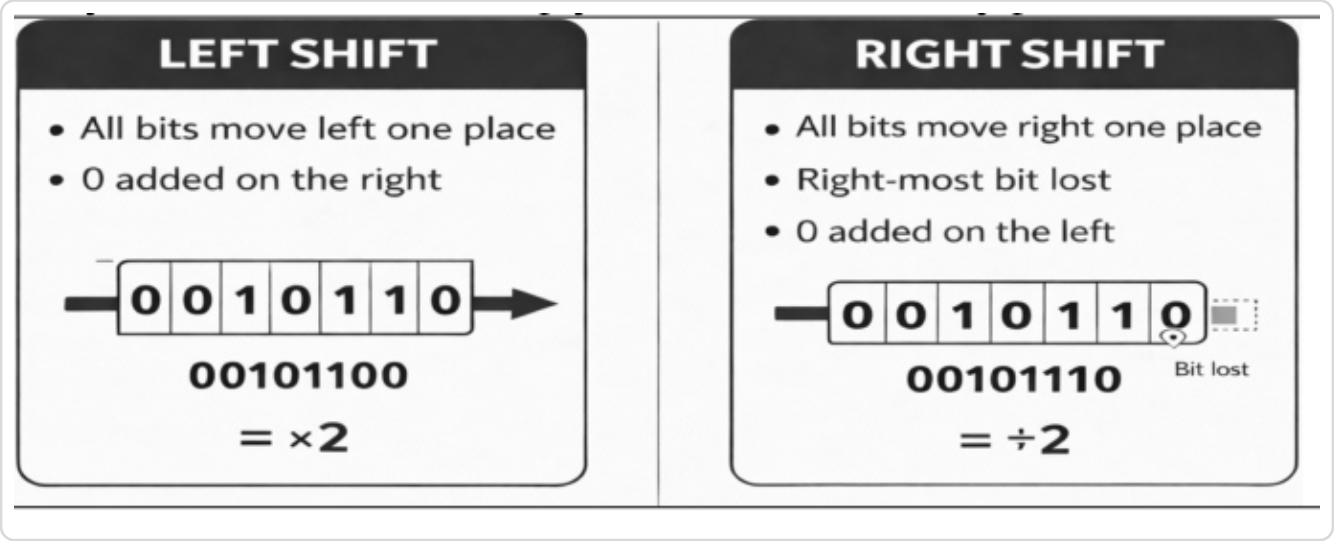
Worked model wording

An overflow error occurs when the result is too large for the available number of bits. The carry outside the eight-bit range is not stored in the eight-bit result.

THEORY

Binary shifts

Shifting binary digits changes their place values.



The empty positions created by a logical shift are filled with zero.

Before: 00010110 = 22

128	64	32	16	8	4	2	1

After a two-place left shift = 88

128	64	32	16	8	4	2	1

When no significant bits are lost, a two-place **left shift multiplies by 4**. A right shift of two places divides an unsigned value by four, discarding any bits shifted off the right edge.

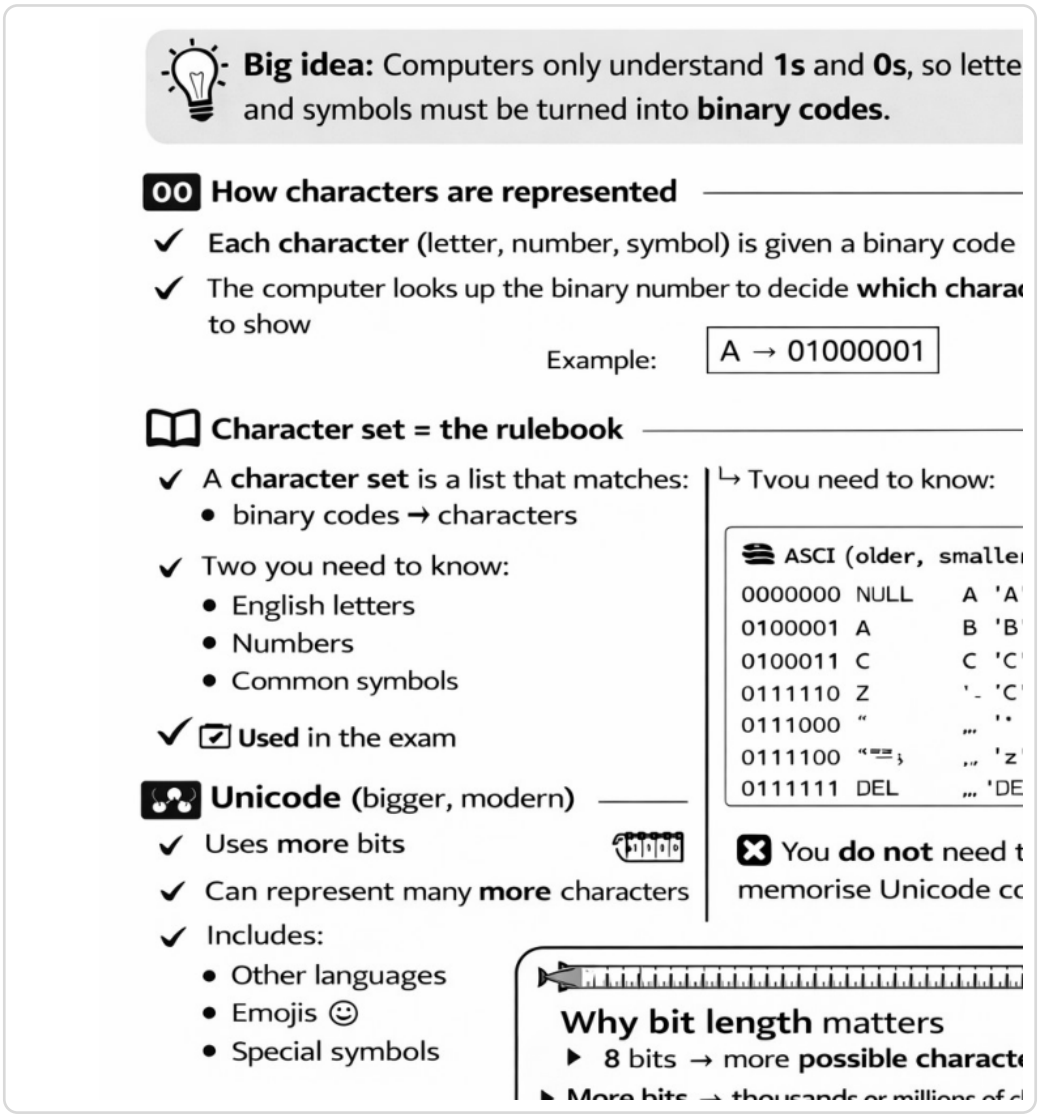
Important limit

Bits shifted beyond a fixed 8-bit boundary are **lost**. For example, 11000000 (192) shifted left one place becomes 10000000 (128) in eight bits, because the leftmost bit is discarded. Do not always claim shifting left doubles the value.

THEORY

Character sets: ASCII and Unicode

Use the annotated diagram to understand character encoding.



Use the character-set diagram to compare ASCII and Unicode.

A **character set** assigns numerical codes to characters. OCR gives eight-bit binary ASCII codes in questions; students do not need to memorise the character codes. The sequence is ordered: B has a code one greater than A.

Make the distinction precise

Unicode supports far more characters and languages than ASCII; a Unicode **encoding** can use different numbers of bytes. Avoid claiming every Unicode character always uses 16 or 32 bits.

WORKED EXAMPLES

Character codes follow a sequence

Worked example · the next three letters after C

Start with C = 01000011 in eight-bit ASCII. Find the codes for D, E and F without memorising them. Each next character increases the number by one.

C · 01000011

128	64	32	16	8	4	2	1

D · 01000100

128	64	32	16	8	4	2	1

E · 01000101

128	64	32	16	8	4	2	1

F · 01000110

128	64	32	16	8	4	2	1

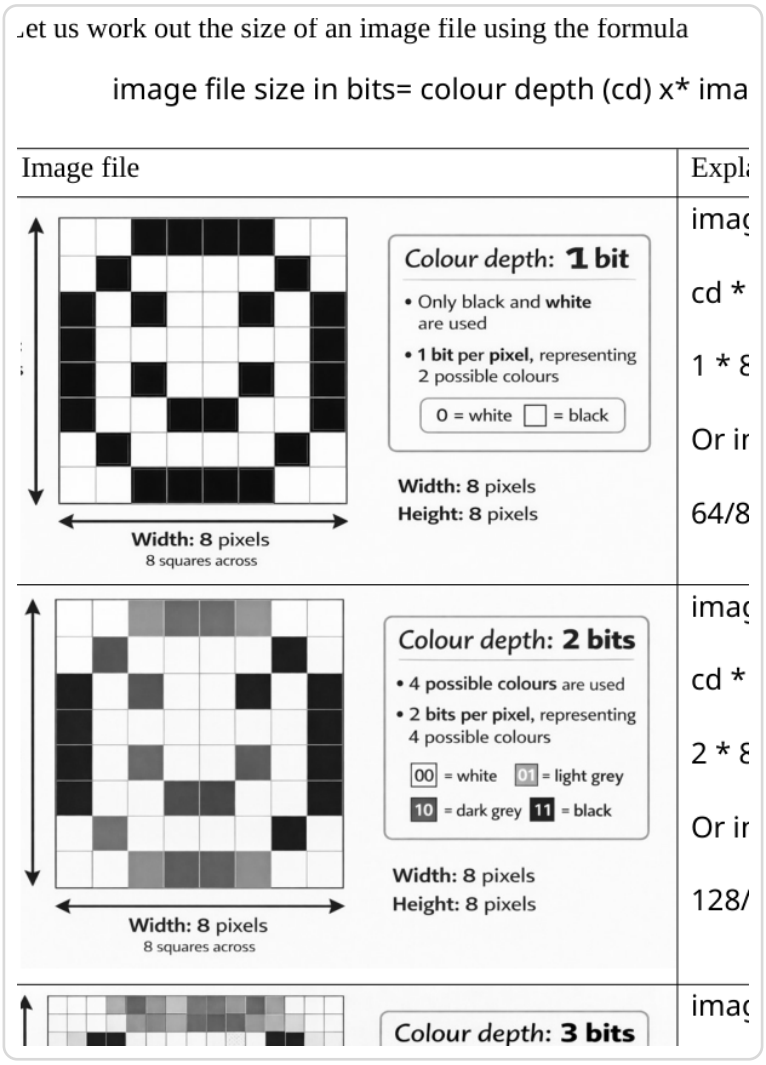
Exam wording

Explain that character codes are **ordered** and move forward by one for successive letters in this example. A character set determines how stored binary patterns are interpreted as characters.

THEORY

Images: pixels and colour depth

Use the pixel grids to understand resolution and colour depth.



These grids show pixels represented using different bit depths.

An image consists of **pixels**. Each pixel stores a binary colour code. **Resolution** tells you how many pixels the image contains; **colour depth** tells you how many bits are used to store each pixel's colour.

Increasing resolution can capture more spatial detail; increasing colour depth can represent more colours. Both generally increase uncompressed image data size.

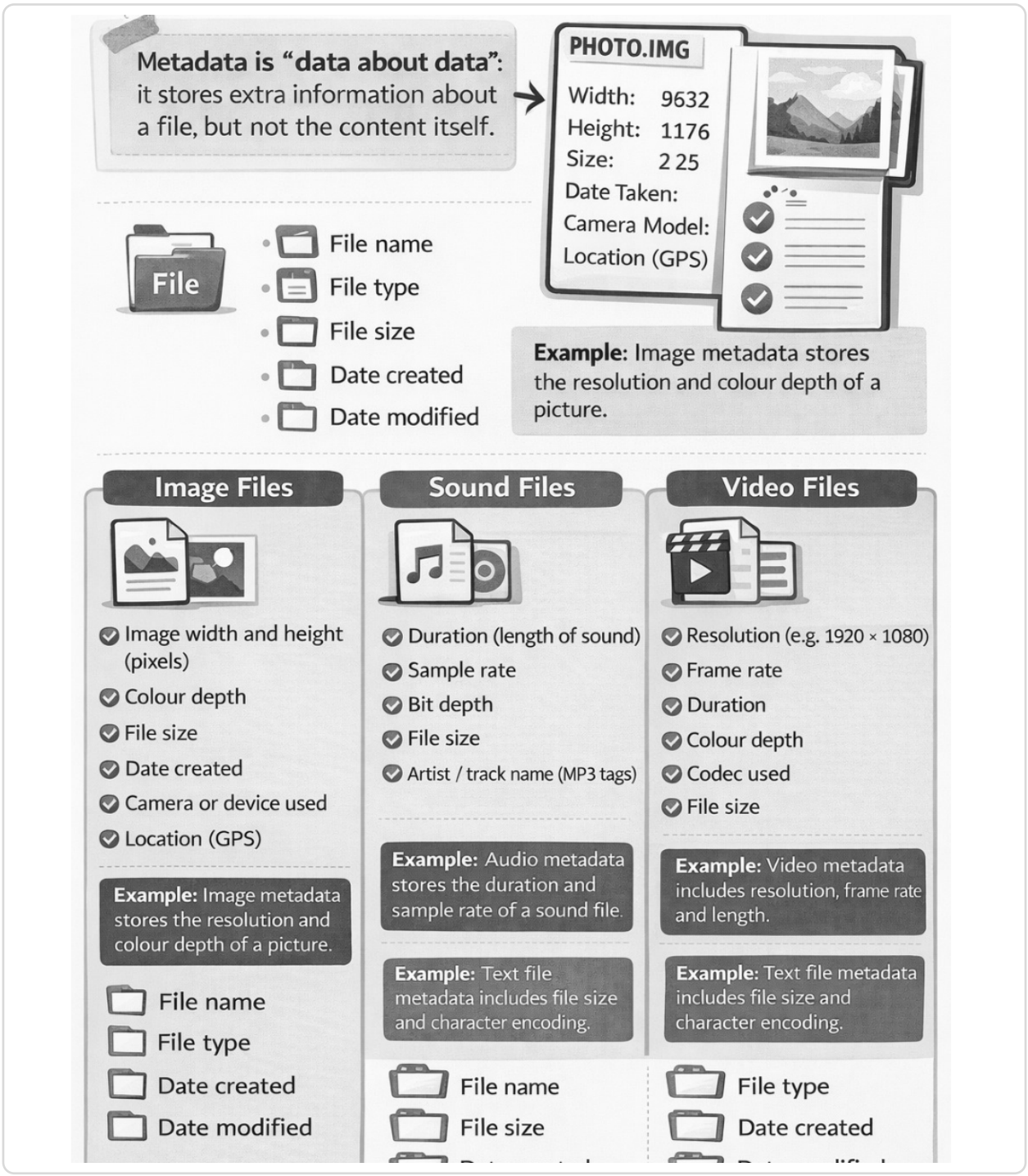
Synoptic link to 1.2.3

Uncompressed image data size in bits = width × height × colour depth. This calculation was introduced in the preceding units topic and is useful for explaining changes in file size.

THEORY

Metadata is data about the file

Use the examples to identify metadata stored with files.



Metadata can describe image, audio and video file properties.

Examples include image width and height, capture date, GPS location or file type. **Metadata describes the content or file; it is not the image pixels themselves.**

THEORY

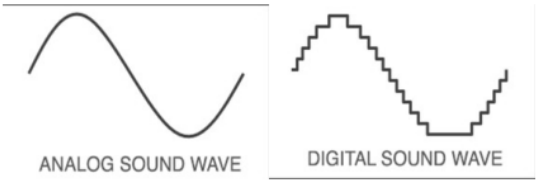
Sound becomes binary by sampling

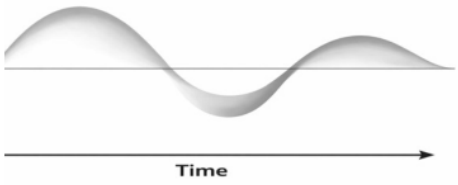
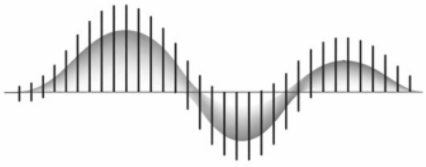
Compare the analogue and sampled sound waveforms.

	10,000 ÷ 1000 = **10 megabytes**

Analogue sound is smooth and continuous, with no steps, and is how sound exists in the real world.

Digital sound is how computers store sound by taking many snapshots (samples) and saving them as binary 0s and 1s) because computers cannot store sound/analogue waves.



Term	Definition	Image
Duration	How long the sound lasts (time along the bottom) Measured in **seconds**	
Sample rate	How often the sound is measured each second (how close the vertical sample lines are) Measured in **Hertz**	

These waveform diagrams help distinguish samples and bit depth.

Sample rate is measurements per second, measured in hertz (Hz). **Bit depth** is the number of bits stored for each amplitude sample. **Duration** is how long the recording lasts.

More samples per second or more possible amplitude levels can improve the digital representation, but they also increase raw file size. Longer recordings store more samples.

Exam-accurate units

Bit depth is measured in **bits per sample**, not “the number of 1s and 0s used to represent height.” Sample rate is measured in **Hz**, not bits.

KEY TERMINOLOGY

Numbers and operations

Binary	Base-2 number system using 0 and 1.
Denary	Base-10 number system using digits 0 to 9.
Hexadecimal	Base-16 number system using 0–9 and A–F.
Bit	One binary digit, either 0 or 1.
Nibble	A group of four bits; equivalent to one hex digit.
MSB	Most significant bit: the highest place value.
LSB	Least significant bit: the lowest place value.
Binary shift	Moves bits left or right within a fixed width.
Overflow	A result outside the number range available in the given bits.
Character set	An agreed mapping between numbers and characters.

KEY TERMINOLOGY

Characters, images and sound

ASCII	Character-code standard; OCR represents ASCII as eight bits in exam questions.
Unicode	Standard supporting characters from many writing systems.
Pixel	A picture element storing a colour code.
Resolution	Width and height of an image in pixels.
Colour depth	Number of bits used for each pixel’s colour.
Metadata	Information about the file, such as dimensions or capture date.
Sampling	Measuring an analogue sound signal at regular time intervals.
Sample rate	Number of amplitude samples taken each second, in Hz.
Bit depth	Number of bits available to store each audio sample.
Duration	Length of a recording, in seconds.

WORKED EXAMPLES

Put the methods together

Example 1 · mixed binary and hex question

Convert **01101101** into denary and hexadecimal. Add the weights where the bits are 1: 64+32+8+4+1 = **109**. Split 0110 | 1101 to get 6 and D: **6D**.

8-bit place-value number line

128	64	32	16	8	4	2	1

Example 2 · shifting a binary integer

Shift **00010110** two places left, keeping eight bits. Move every bit two places left and fill the two empty spaces on the right with 0s: **01011000**.

8-bit place-value number line

128	64	32	16	8	4	2	1

Its value changes from 22 to **88**, which is four times as large because no significant bits were lost.

WORKED EXAMPLES

Explain representation changes

Images · changing colour depth

A small bitmap is **8 × 8 pixels**. At 1 bit per pixel, its raw pixel data is 64 bits. At 2 bits per pixel it needs 128 bits. Double the colour depth → double the raw pixel-data size if the pixel dimensions stay the same.

Sound · changing sample rate

A mono clip lasting five seconds sampled at 10 Hz and 8-bit depth stores **10 × 5 × 8 = 400 bits** of raw samples. Doubling only the sample rate to 20 Hz doubles the raw size to **800 bits**.

Character sets · counting available codes

With 8 bits, there are **2^8 = 256** unique binary patterns. An encoding using more available patterns can represent a larger collection of distinct symbols. The actual byte count per encoded character depends on the encoding.

Practice first, answers later

Attempt all 15 questions. Keep the answers separate until you have finished.

EXAM PRACTICE

Practice questions

Try all 15 questions (43 marks). Answers follow the practice section.

Q1 **1 mark**

What is binary?

Q2 **1 mark**

What does hexadecimal use instead of only numbers?

Q3 **1 mark**

What is a binary shift?

Q4 **2 marks**

State two reasons hexadecimal is used.

Q5 **2 marks**

State two rules of binary shifts.

Q6 **3 marks**

Explain how to convert a denary number into binary.

EXAM PRACTICE

Questions — continued

Answers are in a separate section. Complete each question before turning to the mark scheme.

Q7 **3 marks**

Explain how to convert binary to denary.

Q8 **3 marks**

Explain the effect of a left binary shift by 2 places.

Q9 **3 marks**

Explain what happens to bits that move off the end in a binary shift.

Q10 **3 marks**

Explain what metadata is and give one example of metadata stored in a file.

Q11 **3 marks**

Explain why hexadecimal does not replace binary.

EXAM PRACTICE

Questions — continued

Answers are in a separate section. Complete each question before turning to the mark scheme.

Q12 **4 marks**

Convert 13 into 8-bit binary.

Q13 **4 marks**

Convert the hexadecimal number A4 into denary.

Q14 **4 marks**

The ASCII code for C is 01000011. Work out the ASCII codes for D, E and F.

Q15 **6 marks**

Explain how denary numbers are converted into hexadecimal.

EXAM PRACTICE

Additional OCR coverage

These five additional questions practise specification content not covered by the preceding 15 questions.

QE1 **3 marks**

Add 00001101 and 00000101 using eight-bit binary addition. Show the carry when it is needed.

QE2 **3 marks**

An eight-bit register adds 11110000 and 00110000. Explain why an overflow error occurs, and state the eight-bit result.

QE3 **3 marks**

Explain what the terms MSB and LSB mean when interpreting 10100001.

QE4 **4 marks**

An image doubles in both width and height but keeps its colour depth. Explain the effects on the number of pixels, potential detail, and uncompressed size.

QE5 **3 marks**

Explain the difference between sample rate and bit depth in digital sound.

ANSWERS & MARK SCHEME

Practice questions: model answers

Indicative marking points for the 15 practice questions.

Q1 · 1 mark

Binary is a **base-2** number system using only **0** and **1** (1).

Q2 · 1 mark

It uses the letters **A–F** to represent values **10–15** (1).

Q3 · 1 mark

A binary shift moves the bits in a binary number left or right by a specified number of places (1).

Q4 · 2 marks

Two points: hexadecimal represents a long binary pattern with fewer symbols (1), and is easier for people to read, remember or communicate (1).

Q5 · 2 marks

An unsigned left shift by one **doubles** a value if no significant bits are lost (1). A right shift by one **halves** the integer value, discarding any remainder (1).

Q6 · 3 marks

Use the **128–64–32–16–8–4–2–1 place values** (1). Choose which weights sum to the denary number (1). Write 1 under selected weights, otherwise 0, keeping eight places where required (1).

Q7 · 3 marks

Identify the binary place values (1). Select only those positions whose bit is 1 (1). Add their weights to obtain the denary value (1).

Q8 · 3 marks

Every bit moves two places to the left (1), zero bits fill the empty right-hand positions (1), and the unsigned value is multiplied by 4 **only if no significant bits are lost** (1).

ANSWERS & MARK SCHEME

Answers — continued

Use these indicative marking points to self-mark after attempting the questions.

Q9 · 3 marks

A bit shifted past the fixed-width boundary is **discarded** (1), it is absent from the stored result (1), and the information cannot be recovered from that result (1).

Q10 · 3 marks

Metadata is information **about** the image/file rather than its pixel/audio contents (1). Give a plausible item, e.g. image width or height (1); explain that it helps software interpret or display the file (1).

Q11 · 3 marks

Computers process and store patterns of binary bits (1). Hexadecimal is a shorter way for people to write those bit patterns (1). Each hex digit maps exactly onto four bits (1).

Q12 · 4 marks

Write the eight place values (1). Use 8, 4 and 1 (1). Fill the other places with zeros (1). Final eight-bit answer **00001101** (1).

Q13 · 4 marks

A = 10 (1). Convert A and 4 to nibbles **1010 0100** (1). Use place values or 10×16 +4 (1). Final answer: **164** denary (1).

Q14 · 4 marks

The letter codes follow an ordered sequence (1). **D = 01000100** (1), **E = 01000101** (1), and **F = 01000110** (1).

Q15 · 6 marks

Identify the denary input and use an eight-bit binary representation (1). Select 128–1 place values correctly (1). Split the eight bits into two four-bit groups (1). Work out each nibble’s value using 8–4–2–1 (1). Convert 10–15 into hex letters A–F where needed (1). Present the resulting two-digit hexadecimal number in the correct order (1).

ANSWERS & MARK SCHEME

Additional OCR coverage

These answers apply only to the five additional questions.

QE1 · 3 marks

Show correct alignment (1), binary carrying within the rightmost columns (1), and the result **00010010 (18)** (1).

QE2 · 3 marks

240+48 = 288 (1). This exceeds 255, the unsigned eight-bit maximum, producing a ninth carry bit (1). Only eight result bits fit: **00100000** (1).

QE3 · 3 marks

MSB is the leftmost bit/highest place value (1); LSB is the rightmost/lowest place value (1); in 10100001 their weights are 128 and 1 respectively (1).

QE4 · 4 marks

Doubling width doubles each row of pixels (1), doubling height doubles the number of rows (1), the total pixel count therefore quadruples (1), and with unchanged colour depth, the uncompressed pixel data also quadruples (1).

QE5 · 3 marks

Sample rate is the number of amplitude measurements taken per second and is measured in Hz (1). Bit depth is the number of bits used for each amplitude measurement (1). Raising either can increase raw file size and potentially representation quality, but they affect different parts of the recording (1).

Exam reminder: Use accurate technical definitions alongside visual analogies.
Check units and worked calculations against the correct method on these pages.
Show your working wherever the question requires it.

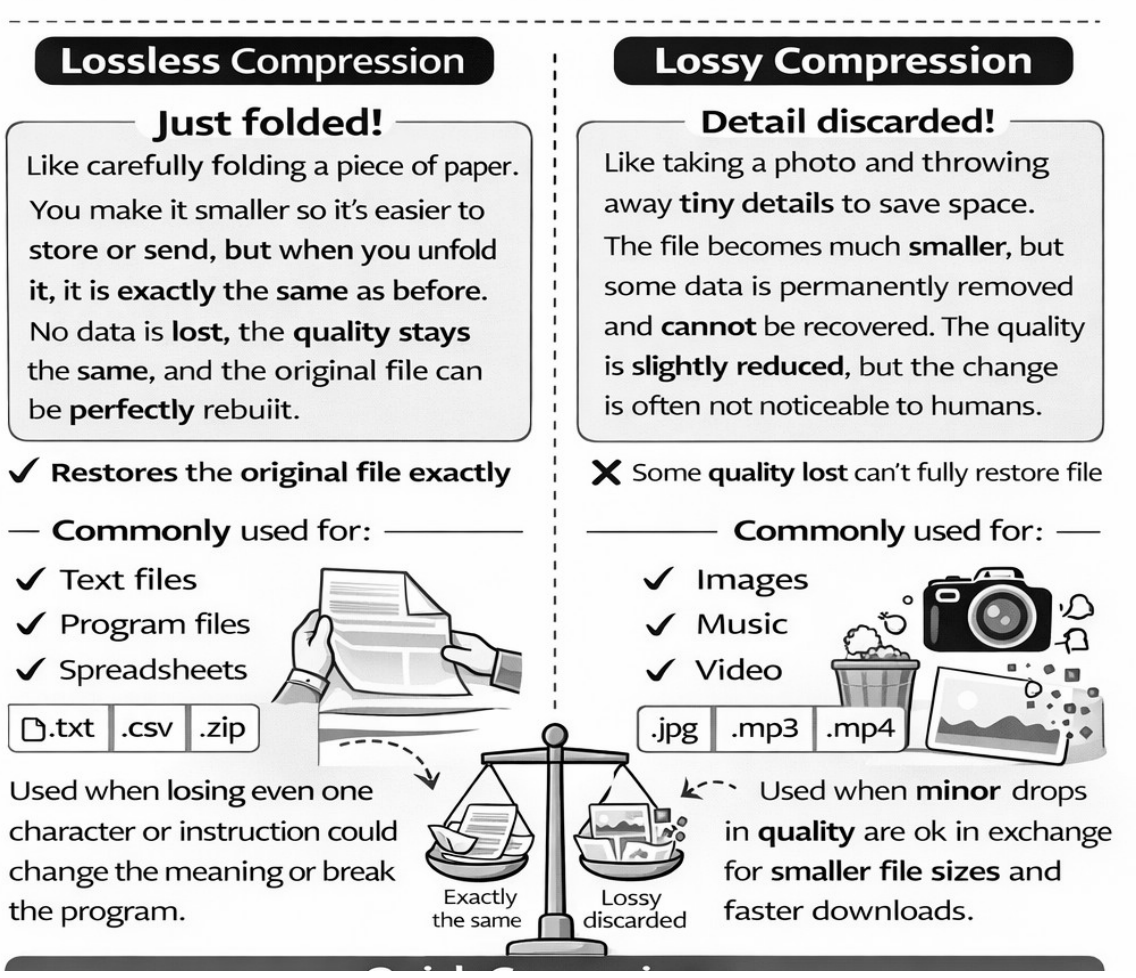
THEORY

Compression: a smaller file

Begin with the visual analogy, then use the technical explanation in an OCR exam answer.

THE BIG IDEA

Compression reduces the amount of data needed to represent a file. This can save storage space and reduce how much data must be transmitted.



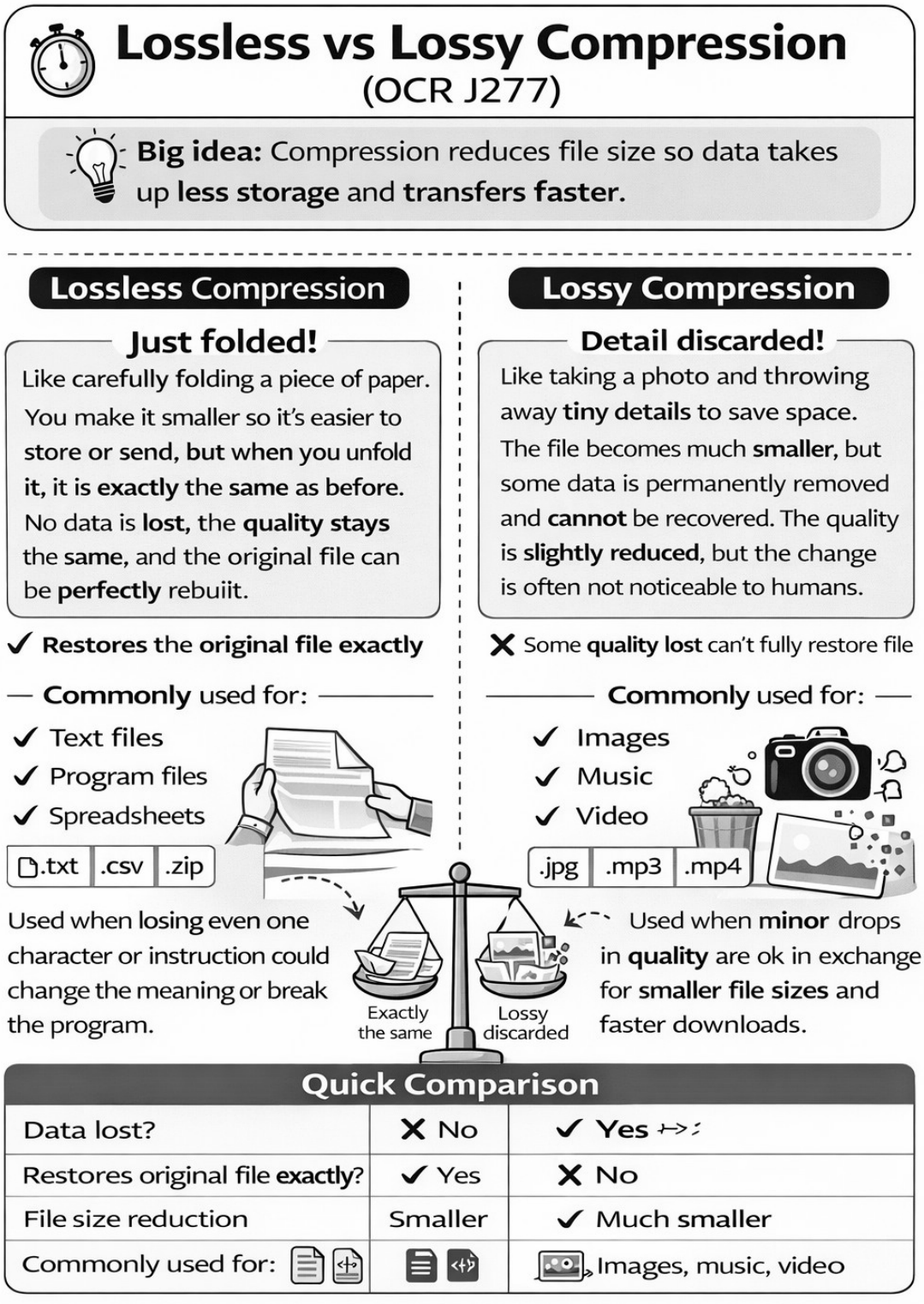
Lossless: like folding paper; recover the original exactly.
Lossy: like throwing away some detail; smaller file, but exact recovery is impossible.

Study the picture, then compare the two compression methods.

THEORY

Compression explained with an analogy

The folded-paper and photograph examples show lossless and lossy compression.



THEORY

What the analogy tells us

Start with the picture, then use the correct technical words in your explanation.

1 LOSSLESS / THE FOLDED DOCUMENT

The original information is encoded more efficiently. When decompressed, every bit of the original data can be reconstructed. That makes it appropriate for source code, text documents and other files where missing information would matter.

Exam wording: lossless compression reduces file size while allowing exact restoration, with no original data lost.

2 LOSSY / THE PHOTO WITH DETAIL REMOVED

Some information is discarded permanently, often details which are less noticeable in photographs, sound or video. This can produce smaller files, but the original cannot be restored exactly. Visible or audible quality can be reduced.

Exam wording: lossy compression permanently removes some data, trading the ability to restore the original exactly for reduced file size.

CHECK THE DIFFERENCE BETWEEN LOSSY AND LOSSLESS

Lossy can achieve greater size reductions, but the amount saved varies with the content, format, method and settings. Do not memorise “lossless = small” and “lossy = always much smaller” as universal rules.

THEORY

How to choose a compression type

The best method depends on whether the content must be recovered exactly and how much quality can be sacrificed.

Question to ask	Lossless	Lossy
Can all original data be recovered?	Yes	No
What happens to data?	No permanent loss	Some data discarded
What happens to quality?	Original preserved	May be reduced
When is it useful?	Code, documents, archives	Photos, music, video
Is file size reduction fixed?	No; depends on file	No; depends on file

WHY COMPRESS?

Smaller files require less storage and usually less data to send across a network. Transfer time can therefore fall when connection conditions are otherwise similar. Compression and decompression can also require processing.

KEY TERMINOLOGY

Words to use in exam answers

The highlighted words use the same three translucent SVG pen strokes as the J277 and H446 lessons.

Compression	Reducing the size of a file or the amount of data needed to represent it.
Lossless	Compression that allows the original information to be reconstructed exactly.
Lossy	Compression that permanently discards some information.
Decompression	Reconstructing data from its compressed representation.
File size	The amount of storage space used by a file.
Storage capacity	The total amount of data a storage device can hold.
Transfer time	Time taken to send or receive a file.
Quality	How faithfully media represents the original image or sound.
Exact restoration	Reconstructing all of the original data without loss.

WORKED EXAMPLES

Worked example 1 · School files

Scenario: a school is compressing examination answers and its programming coursework.

DECIDE

Should the school choose lossless or lossy compression? Explain two reasons for your decision.

STEP 1 IDENTIFY THE REQUIREMENT

Text and source code must be reproduced exactly. Even one changed character could alter a sentence or a program.

STEP 2 CHOOSE THE COMPRESSION TYPE

Select lossless compression, because no original information is permanently discarded.

STEP 3 LINK TO THE BENEFIT

The files take up less storage and can be transferred using less data, while still being restored exactly.

Model answer: Use lossless compression because the original examination and program files must be recovered exactly. It reduces file size without removing data that could change the answers or break code.

WORKED EXAMPLES

Worked example 2 · Website photos

Scenario: a school website contains many photographs and must load efficiently on mobile connections.

DECIDE

Explain why lossy compression might be chosen for the published photographs, and give one disadvantage.

STEP 1 IDENTIFY THE NEED

Many large images use storage and must be transferred to visitors. Smaller image files can help the pages load faster.

STEP 2 CHOOSE THE TRADE-OFF

Lossy compression may remove image detail that viewers are unlikely to notice, reducing the transmitted file size.

STEP 3 STATE THE DISADVANTAGE

Some original image information is permanently removed. High compression settings may cause visible artefacts; the original cannot be reconstructed exactly.

Model answer: Lossy compression makes web images smaller so less data needs transferring. However, some visual information is permanently discarded and image quality may be reduced. The school should retain the original separately if needed.

EXAM PRACTICE Practice questions · Q1?Q6

Answer these independently. The solutions and indicative marking points are on later pages.

Q1 What is compression? **1 mark**

Q2 What is lossless compression? **1 mark**

Q3 What is lossy compression? **1 mark**

Q4 State two benefits of compression. **2 marks**

Q5 State two differences between lossless and lossy compression. **2 marks**

Q6 Explain why lossless compression is used for text files. **3 marks**

EXAM PRACTICE Practice questions · Q7?Q10

Explain each decision using the compression type, what happens to the data and the practical benefit.

Q7 Explain why lossy compression is suitable for images, sound, or video. **3 marks**

Q8 Explain how compression improves data transfer. **3 marks**

Q9 Describe the difference between lossless and lossy compression. **4 marks**

Q10 Describe two situations where lossy compression would be used. **4 marks**

EXAM PRACTICE Practice question · Q11

This is a 6-mark explain/discuss response. Plan advantages, disadvantages and a concluding judgement before writing.

Q11 Explain the advantages and disadvantages of lossy compression. **6 marks**

SUGGESTED ANSWER STRUCTURE (NOT A MODEL ANSWER)

1. Explain how lossy compression reduces data and file size.
2. Link smaller files to a practical storage or transfer benefit.
3. Describe permanent data loss or possible quality reduction.
4. Say when that trade-off is acceptable.

ANSWERS & MARK SCHEME

Model answers · Q1–Q6

Use these after attempting the questions. Marking points are indicative; OCR mark schemes may accept equivalent wording.

Q1 · 1 MARK

What is compression?

Model answer: Compression reduces the size of a file so it takes up less storage space.

Q2 · 1 MARK

What is lossless compression?

Model answer: Lossless compression reduces file size without losing any data.

Q3 · 1 MARK

What is lossy compression?

Model answer: Lossy compression reduces file size by permanently removing some data.

Q4 · 2 MARKS

State two benefits of compression.

Model answer: Compression reduces storage space needed and allows files to be transferred faster.

Q5 · 2 MARKS

State two differences between lossless and lossy compression.

Model answer: Lossless compression does not lose data and can be restored exactly, while lossy compression loses data and cannot be restored exactly.

Q6 · 3 MARKS

Explain why lossless compression is used for text files.

Model answer: Text files must be restored exactly, as losing data could change words or instructions.

ANSWERS & MARK SCHEME

Model answers · Q7–Q11

Give yourself credit for correct equivalent points, not only these exact sentences.

Q7 · 3 MARKS

Explain why lossy compression is suitable for images, sound, or video.

Model answer: Some data can be removed without noticeable loss of quality, greatly reducing file size.

Q8 · 3 MARKS

Explain how compression improves data transfer.

Model answer: Smaller files require less data to be sent, so they transfer more quickly over a network.

Q9 · 4 MARKS

Describe the difference between lossless and lossy compression.

Model answer: Lossless compression reduces file size without losing data and allows exact restoration, while lossy compression permanently removes some data and cannot be restored exactly.

Q10 · 4 MARKS

Describe two situations where lossy compression would be used.

Model answer: Lossy compression is used for images and video where a small reduction in quality is acceptable.

Q11 · 6 MARKS

Explain the advantages and disadvantages of lossy compression.

Model answer: Lossy compression greatly reduces file size, improving storage and transfer speeds. However, data is permanently lost and quality may be reduced, meaning the original file cannot be restored exactly.

THEORY

LAN and WAN · the road analogy

Use the road comparison to distinguish networks and their connections.

WHAT IS A NETWORK?

Two or more computers or devices connected to share data, resources or services. A network may use cables, wireless links, or both.

THE ROAD ANALOGY

LAN · PRIVATE ROAD OR SCHOOL CAR PARK

Picture: A small, local route between nearby places.

Technical: A Local Area Network covers a limited geographical area, such as one school site.

WAN · MOTORWAY BETWEEN CITIES

Picture: Roads link distant places and many local routes.

Technical: A Wide Area Network spans a large area and can connect several LANs.

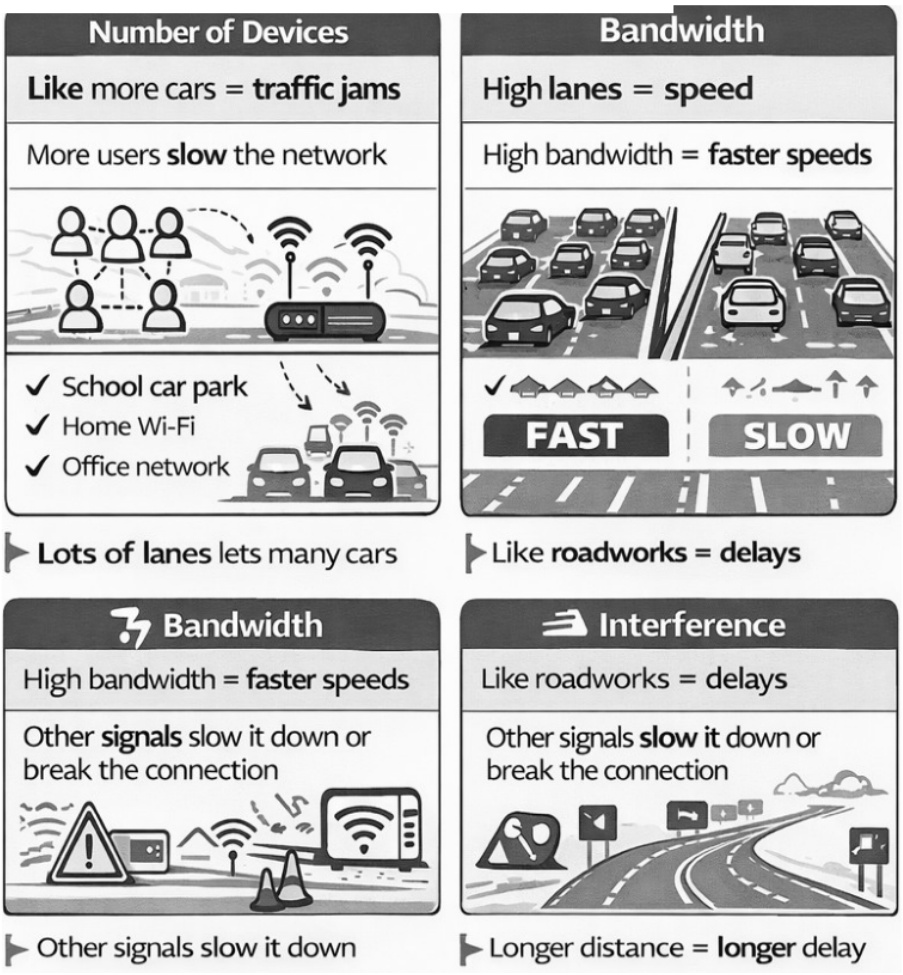
EXAM ACCURACY

LAN does not always mean “fast”, and WAN does not always mean “slow”. Actual speed depends on the connections, load, equipment and bandwidth.

THEORY

Network performance · road traffic

Each cartoon represents a different network idea. Identify what each illustrates.



BANDWIDTH = ROAD CAPACITY

Bandwidth is the maximum amount of data that can be transferred per unit of time. More bandwidth can help when multiple devices share a connection.

NUMBER OF DEVICES AND INTERFERENCE

More simultaneous traffic can cause congestion. Interference or unreliable wireless conditions may also lower actual transfer speed. Latency describes delay; it is not the same as bandwidth.

THEORY

Client–server and peer-to-peer

A school file server is a useful comparison with computers sharing files directly.

CLIENT–SERVER

Clients request services from a server. The server can provide files, accounts or web pages.

ADVANTAGE

Central accounts and backups; consistent access and management.

LIMITATION

Dedicated services cost money; some server failures can interrupt access.

PEER-TO-PEER

Each computer can directly provide resources to others and also use shared resources.

ADVANTAGE

Straightforward for a very small network; no dedicated server required.

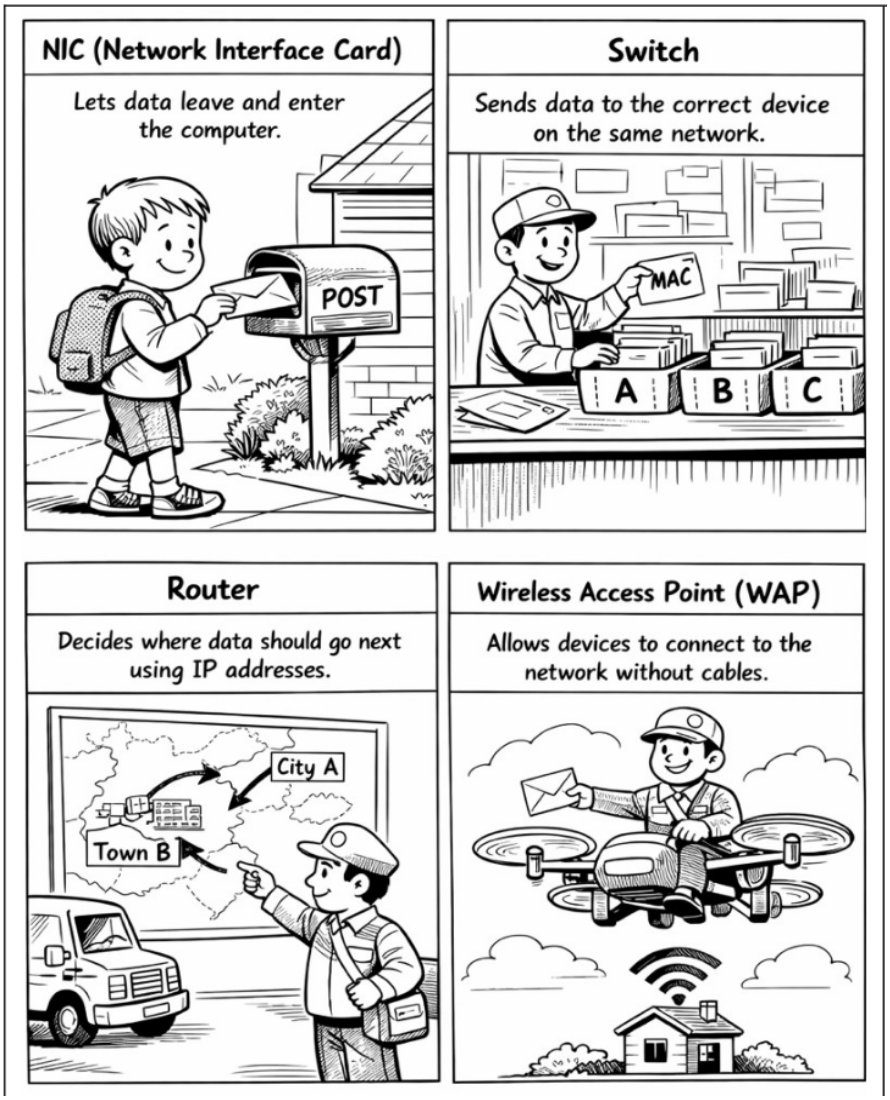
LIMITATION

No automatic central management or backup; peers must be available to share.

THEORY

Network hardware · illustrated

The postal, switch, routing and wireless delivery illustrations are kept intact.



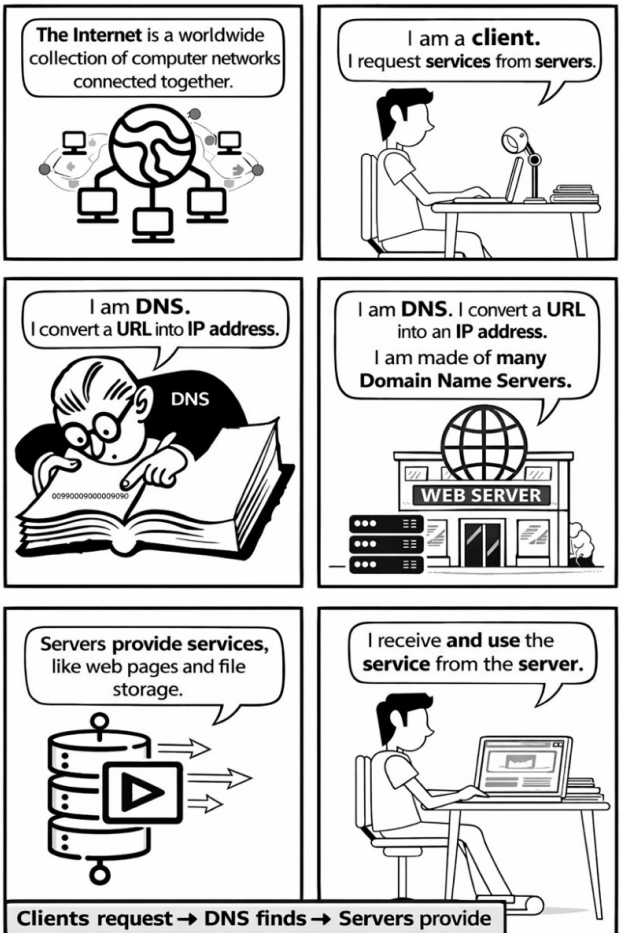
TECHNICAL ROLES · THE EXAM WORDS

NIC: provides network connectivity. Switch: connects devices on a LAN and forwards frames using MAC addresses. Router: connects networks and forwards IP packets between them. WAP: allows wireless devices to join a network. A single home router may combine several roles.

THEORY

The Internet, DNS and servers

Use the diagrams to identify connected networks, clients, servers and DNS.



HOW DNS HELPS YOU VISIT A WEBSITE

A browser uses the domain name in a web address to look up an IP address (unless already cached). DNS uses a distributed system of name servers; the browser can then contact the appropriate server.

CLIENTS AND SERVERS

A client requests a service, and a server responds. Web servers provide web content; file servers store and make files available. The Internet connects many networks; it is not one single server.

THEORY

Cloud services and network connections

Compare cloud storage, cloud software and cloud processing.

CLOUD STORAGE

Files reside on remote servers and can be accessed over a network, e.g. a shared school drive.

CLOUD SOFTWARE

Applications or functions are supplied remotely, e.g. a browser-based office service.

CLOUD PROCESSING

Remote servers carry out computational work, e.g. some online AI tools or rendering systems.

EXAM BALANCE · ADVANTAGES AND DISADVANTAGES

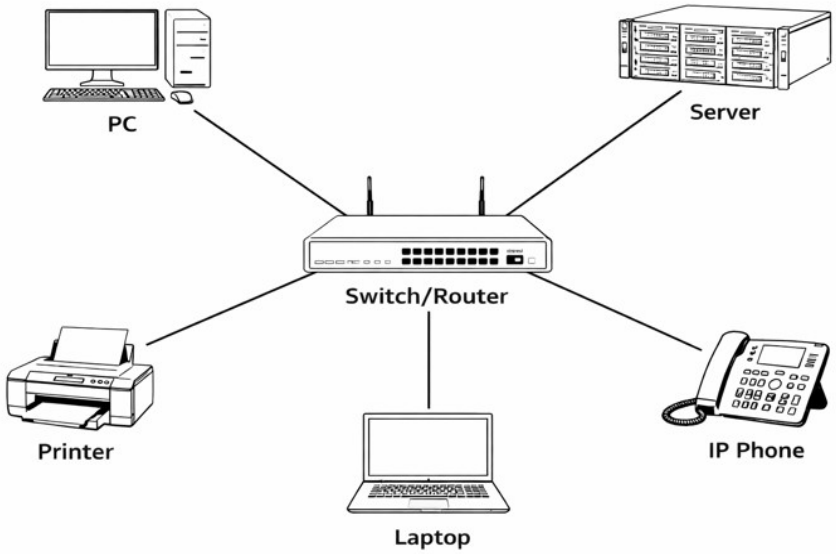
Potential advantages: access from different locations; shared work; less local storage; flexible capacity. Potential disadvantages: reliance on a connection and provider; ongoing cost; privacy/security considerations; reduced access during outages. Not every app or game sends all its work to a cloud server.

THEORY

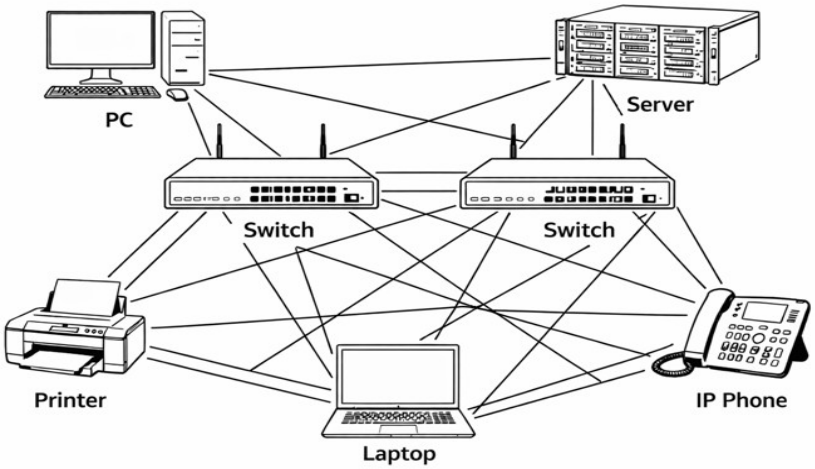
Star and mesh network topologies

Study the star and mesh diagrams, then describe their features.

STAR NETWORK / SCHOOL EXAMPLE



MESH NETWORK / MULTIPLE ROUTES



- Devices (or switches/routers) connected to multiple others
- Data can travel directly between nodes or wo alternative routes
- This makes them fault toralant

Star: devices link to a central network device; simple to manage but failure of that device can disconnect its attached devices. Mesh: nodes have multiple links, giving alternative paths, but requiring more connections and greater complexity. A mesh may be partial, not fully connected.

KEY TERMINOLOGY

Keywords for 1.3.1 · 1 of 2

The three yellow translucent pen strokes are taken from the shared J277/H446 lesson CSS.

Network	Connected devices sharing information, services or resources.
LAN	Local Area Network: a network covering a relatively small area.
WAN	Wide Area Network: links devices or networks across large distances.
Bandwidth	Maximum rate of data transfer, commonly measured in bits per second.
Latency	Time delay before data begins to reach its destination.
Client	Device or software that requests a service.
Server	Device or software providing a service to clients.
Peer-to-peer	Model where computers share resources directly with one another.

KEY TERMINOLOGY

Keywords for 1.3.1 · 2 of 2

The three yellow translucent pen strokes are taken from the shared J277/H446 lesson CSS.

NIC	Network interface component that connects a device to a network.
Switch	Connects LAN devices and forwards frames using MAC information.
Router	Forwards packets between networks using IP addresses.
WAP	Wireless access point, allowing wireless connections to a network.
DNS	Domain Name System: resolves domain names to IP addresses using name servers.
Cloud	Remote network-based storage, software or processing services.
Star topology	Network layout in which nodes connect to a central device.
Mesh topology	Layout where nodes have multiple connections and alternative paths.

WORKED EXAMPLES

Worked example 1 · The school network

Explain which network type and hardware a new school building might use.

SCENARIO

A school opens a second block on the same site. It wants classroom computers, a shared printer, Wi-Fi for tablets and Internet access.

STEP 1 · NAME THE TYPE

A LAN is appropriate because the school devices are on one local site.

STEP 2 · CONNECT THE COMPUTERS

NICs provide device connectivity. A switch joins wired devices within the LAN. A WAP allows tablets to connect wirelessly.

STEP 3 · CONNECT OTHER NETWORKS

A router connects the school LAN to an external network such as the Internet.

STEP 4 · LINK TO A REASON

Central devices help management, but the selected switch/WAP capacity and number of connected devices affect performance.

WORKED EXAMPLES

Worked example 2 · DNS lookup

Trace what happens when a student enters a web address into a browser.

STUDENT ENTERS A WEBSITE ADDRESS

The browser needs an IP address for the website domain name. The domain name is part of the URL.

1 · ASK

If a suitable answer is not already cached, a DNS lookup is made.

2 · LOOK UP

Distributed DNS name servers help resolve the domain name to an IP address.

3 · RETURN

The DNS lookup returns the address information to the client.

4 · CONNECT

The client can then use the returned address to contact the relevant web server.

WORKED EXAMPLES

Worked example 3 · Star versus mesh

Choose a topology for a school classroom and for a resilient network between sites.

SCHOOL CLASSROOM · STAR

A central switch connecting computers is usually practical: cabling and management are relatively straightforward. If that central switch fails, its connected computers lose their connection.

SITES NEEDING ALTERNATIVE ROUTES · MESH

Some networks benefit from multiple links between routers. Traffic can use another available route when one link fails. That improves resilience but adds hardware and complexity.

EXAM JUDGEMENT

There is no one best topology for every situation. Choose by cost, complexity, central management, reliability and need for alternative routes. A partial mesh can provide redundancy without connecting every device to every other device.

EXAM PRACTICE

Practice questions · set 1 of 5

Answer independently. Use the marks shown beside each question.

Q1 What is a computer network? **1 mark**

Q2 What does LAN stand for? **1 mark**

Q3 What does WAN stand for? **1 mark**

Q4 State two differences between a LAN and a WAN. **2 marks**

Q5 State two uses of a computer network. **2 marks**

EXAM PRACTICE

Practice questions · set 2 of 5

Answer independently. Use the marks shown beside each question.

Q6 Explain the difference between a client–server network and a peer-to-peer network. **3 marks**

Q7 Explain the role of a Network Interface Card (NIC). **3 marks**

Q8 Explain the role of a switch in a LAN. **3 marks**

Q9 Explain the difference between copper cable and fibre- optic cable. **3 marks**

Q10 Explain one disadvantage of wireless transmission. **3 marks**

EXAM PRACTICE

Practice questions · set 3 of 5

Answer independently. Use the marks shown beside each question.

Q11 Explain what DNS is used for. **3 marks**

Q12 Explain what is meant by the Cloud. **3 marks**

Q13 Explain one advantage of a star topology. **3 marks**

Q14 Explain one advantage of a mesh topology. **3 marks**

Q15 Describe the role of a router in a network. **4 marks**

EXAM PRACTICE

Practice questions · set 4 of 5

Answer independently. Use the marks shown beside each question.

Q16 Describe two advantages of a client– server network. **4 marks**

Q17 Describe two disadvantages of a peer-to- peer network. **4 marks**

Q18 Describe how DNS works. **4 marks**

Q19 Describe two cloud services. **4 marks**

EXAM PRACTICE

Practice questions · set 5 of 5

Answer independently. Use the marks shown beside each question.

Q20 Explain how the Internet works as a client–server network. **6 marks**

Q21 Compare star and mesh network topologies. **6 marks**

ANSWERS & MARK SCHEME

Indicative answers · set 1 of 5

Compare with your own answer. Equivalent accurate wording may also earn credit.

Q1 · 1 MARK

What is a computer network?

Model answer: A network is two or more computers connected together to share data and resources.

Q2 · 1 MARK

What does LAN stand for?

Model answer: Local Area Network.

Q3 · 1 MARK

What does WAN stand for?

Model answer: Wide Area Network.

Q4 · 2 MARKS

State two differences between a LAN and a WAN.

Model answer: A LAN usually covers a smaller geographic area (e.g. one school site), whereas a WAN spans a much larger area and may link several LANs. Speed is not determined by network type alone.

Q5 · 2 MARKS

State two uses of a computer network.

Model answer: Networks allow file sharing and sharing hardware such as printers.

ANSWERS & MARK SCHEME

Indicative answers · set 2 of 5

Compare with your own answer. Equivalent accurate wording may also earn credit.

Q6 · 3 MARKS

Explain the difference between a client–server network and a peer-to-peer network.

Model answer: In a client–server network, servers provide services and clients request them. In a peer-to-peer network, all computers have the same role.

Q7 · 3 MARKS

Explain the role of a Network Interface Card (NIC).

Model answer: A NIC allows a device to connect to a network and send and receive data, and it has a unique MAC address.

Q8 · 3 MARKS

Explain the role of a switch in a LAN.

Model answer: A switch connects devices in a LAN and sends data only to the intended device using MAC addresses.

Q9 · 3 MARKS

Explain the difference between copper cable and fibre- optic cable.

Model answer: Copper uses electrical signals and is slower, while fibre uses light signals and is much faster over long distances.

ANSWERS & MARK SCHEME

Indicative answers · set 3 of 5

Compare with your own answer. Equivalent accurate wording may also earn credit.

Q10 · 3 MARKS

Explain one disadvantage of wireless transmission.

Model answer: Wireless transmission can suffer from interference and security issues.

Q11 · 3 MARKS

Explain what DNS is used for.

Model answer: DNS converts human-readable website names into IP addresses so computers can locate servers.

Q12 · 3 MARKS

Explain what is meant by the Cloud.

Model answer: The Cloud is the use of remote servers on the Internet to store data, run software, or process data.

Q13 · 3 MARKS

Explain one advantage of a star topology.

Model answer: Star networks are easy to manage because all devices connect to a central switch.

Q14 · 3 MARKS

Explain one advantage of a mesh topology.

Model answer: Mesh networks are fault tolerant because data can travel along multiple routes.

ANSWERS & MARK SCHEME

Indicative answers · set 4 of 5

Compare with your own answer. Equivalent accurate wording may also earn credit.

Q15 · 4 MARKS

Describe the role of a router in a network.

Model answer: A router forwards data between different networks using IP addresses and connects a LAN to the Internet.

Q16 · 4 MARKS

Describe two advantages of a client– server network.

Model answer: Client–server networks allow central management and stronger security, with data backed up in one place.

Q17 · 4 MARKS

Describe two disadvantages of a peer-to- peer network.

Model answer: Peer-to-peer networks have weaker security and no central backup, and files may be unavailable if a computer is off.

Q18 · 4 MARKS

Describe how DNS works.

Model answer: A client sends a URL to a DNS server, which looks up the IP address and returns it so the client can contact the web server.

ANSWERS & MARK SCHEME

Indicative answers · set 5 of 5

Compare with your own answer. Equivalent accurate wording may also earn credit.

Q19 · 4 MARKS

Describe two cloud services.

Model answer: Cloud storage stores files online, while cloud software allows applications to run in a web browser without installation.

Q20 · 6 MARKS

Explain how the Internet works as a client–server network.

Model answer: The Internet is a network of networks where clients request services from servers. DNS converts URLs into IP addresses, and servers provide services such as web pages and file storage.

Q21 · 6 MARKS

Compare star and mesh network topologies.

Model answer: Star networks are simple and easy to manage but rely on a central device. Mesh networks allow multiple routes, making them fault tolerant and reliable but more complex and expensive.

THEORY

Wired and wireless connections

Use the comparison diagram to connect each visual example to GCSE terminology.

WHAT STUDENTS NEED TO DECIDE

A wired connection uses a cable, while a wireless connection sends signals through the air. Choose by scenario: speed, interference, mobility, installation, reliability and security.

WIRED · ETHERNET

The cable is the fixed delivery route.

Often consistent for fixed equipment; cables limit movement and take time to install.

WIRELESS · WI-FI / BLUETOOTH

The connection works without a network cable to the device.

Useful for moving devices, but range, congestion and interference can affect performance.

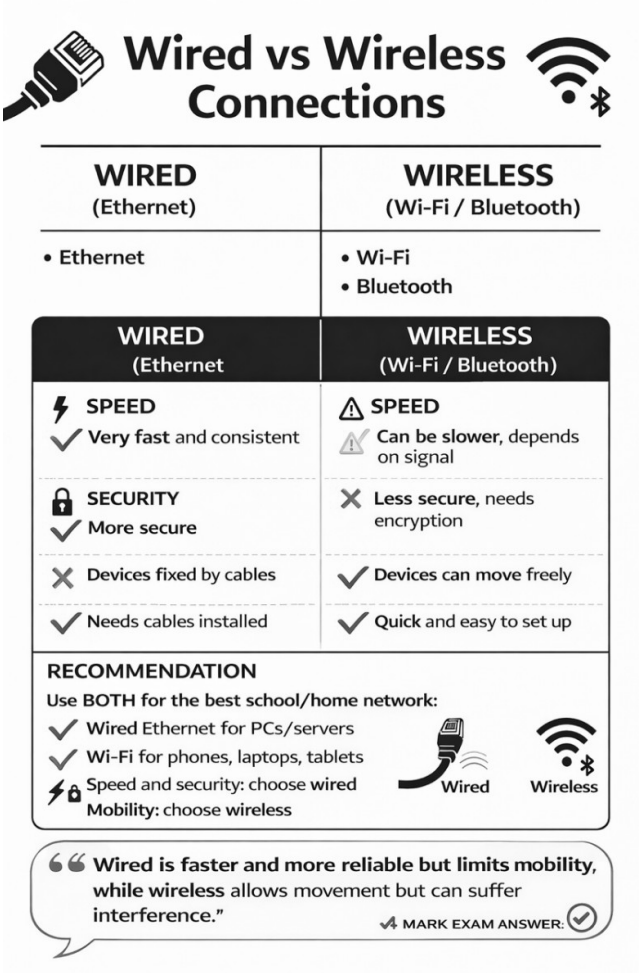
THE ANALOGY HAS LIMITS

A wired connection may be faster or more secure in some circumstances, but this is not guaranteed. Compare configuration, interference and encryption.

THEORY

Comparing wired and wireless networks

Use the categories to compare trade-offs rather than treating them as universal rankings.
universal rankings.



EXAM-ACCURATE COMPARISON

Wired: requires cables, usually offers stable high bandwidth to stationary devices. Wireless: allows movement and easier installation for mobile devices, but is affected by signal strength, interference and shared airtime.

RECOMMENDATION

For a school: Ethernet for desktop PCs and servers; Wi-Fi for tablets and laptops. Security depends on encryption, configuration and access control, not just whether a cable is present.

THEORY

Connections, signals and decisions

Select the connection for the task, then give a linked justification.

ETHERNET / COPPER

Electrical signals travel along copper cabling. Common for wired school LAN connections.

FIBRE OPTIC

Pulses of light travel through fibre. Useful for high-capacity links and long distances, but installation costs vary.

WI-FI

Uses radio for devices to connect to a wireless LAN through an access point.

BLUETOOTH

Short-range radio connection commonly used between nearby devices or peripherals.

THEORY

IP addresses and MAC addresses

Three distinct ideas: the IP version, what an address identifies, and who uses it.

IPv4

32 bits. Typically shown as four decimal values from 0 to 255 separated by dots. Example: 192.168.1.25. It is a logical network address used when delivering IP packets.

IPv6

128 bits. Typically expressed as eight groups of hexadecimal digits separated by colons; groups may be shortened. Example: 2001:db8::1. It addresses IPv4 address-space limitations.

MAC ADDRESS

A link-layer device/interface identifier used for local frame delivery, often represented using six groups of hexadecimal digits (48 bits), e.g. 00:1A:2B:3C:4D:5E. Many devices permit address randomisation or change.

NOT THE SAME AS DNS

An IP address identifies an interface/network destination for routing. A MAC address supports local link communication. DNS resolves domain names to address records; it does not turn an entire URL into one IP address.

THEORY

Encryption, standards and protocols

Data is vulnerable to interception in transit. Protocols and standards let systems communicate in understood

ENCRYPTION

Encryption transforms readable information into ciphertext using an algorithm and key. Without the correct key, an interceptor should not be able to understand the protected content. Encryption protects confidentiality; it does not alone prevent every attack.

STANDARDS

Agreed specifications let products from different manufacturers interoperate. For example, Wi-Fi and Ethernet standards allow compatible network equipment to exchange information.

COMMUNICATION PROTOCOLS

A protocol is a set of rules for communication, defining how messages are formatted, sent, received or handled. Different protocols perform different jobs: HTTP, HTTPS, TCP, IP, SMTP and others.

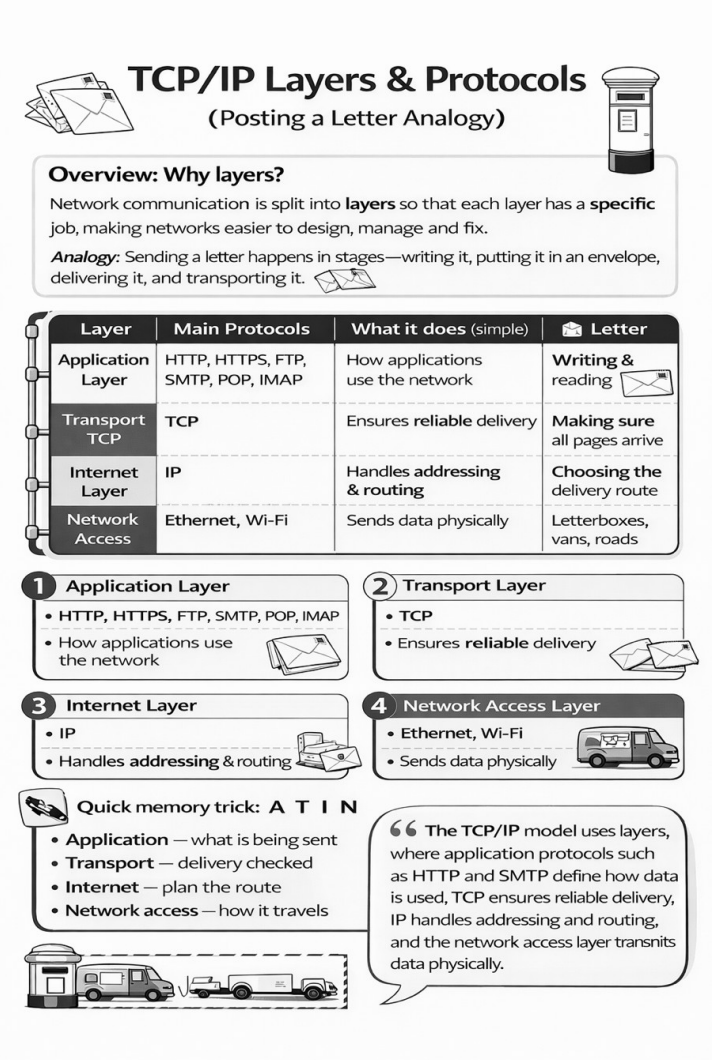
HTTPS IS MORE THAN JUST "HTTP + ENCRYPTION"

HTTPS protects HTTP traffic using TLS, also providing server authentication and integrity safeguards when correctly configured. A padlock does not prove the website itself is trustworthy.

THEORY

Protocols · postal delivery analogy

Use the postal analogy to remember each purpose, then explain it in technical terms.



WHAT THE ANALOGY EXPLAINS

Application: writing/reading the letter (the service requested). Transport: checking the pages arrive in order. Internet: selecting destination and route. Network access: actual movement across the link. OCR note: the names and functions of individual TCP/IP layers are not required for memorisation in J277, but the benefit of using layers is required.

THEORY

Which protocol does what?

Use the postal analogy,

HTTP	Request and transfer web resources; not encrypted by default.
HTTPS	HTTP over TLS; protects web communications in transit.
FTP	Transfer files between networked systems.
SMTP	Send and relay email messages.
POP3	Retrieve/download messages from a mail server; copies can remain depending on settings.
IMAP	Access and synchronise email kept on the server across devices.
TCP	Reliable ordered byte-stream delivery with error detection and retransmission.
IP	Logical packet addressing and forwarding between networks.

KEY TERMINOLOGY

Key words for 1.3.2 · 1 of 2

Highlights use the three exact translucent yellow SVG strokes from the shared J277/H446 lesson CSS.

Ethernet	Wired local-network technology using defined standards.
Wi-Fi	Wireless local networking using radio signals.
Bluetooth	Short-range wireless communication commonly between nearby devices.
Fibre optic	Cable transmitting light pulses to carry digital data.
Interference	Unwanted signals that disrupt or reduce transmission quality.
Encryption	Transforms plaintext into ciphertext using cryptographic methods/keys.
Ciphertext	Encrypted information that should be unreadable without the required key.
Standard	Agreed specification allowing compatible systems to work together.
Protocol	Rules that systems follow when communicating.

KEY TERMINOLOGY

Key words for 1.3.2 · 2 of 2

Highlights use the three exact translucent yellow SVG strokes from the shared J277/H446 lesson CSS.

IPv4	32-bit Internet Protocol address format, commonly four decimal octets.
IPv6	128-bit Internet Protocol address format, commonly hexadecimal groups.
MAC address	Link-layer identifier used in the delivery of local network frames.
HTTP / HTTPS	Web transfer protocol / HTTP protected using TLS.
SMTP	Protocol for sending or relaying email.
POP3	Protocol commonly used for downloading email from a server.
IMAP	Protocol for synchronising email held on a server.
TCP	Protocol offering reliable ordered delivery for applications.
IP	Protocol providing logical packet addressing and routing between networks.

WORKED EXAMPLES

Example 1 · Choosing connections

A school needs reliable fixed PCs and roaming tablet devices.

SCENARIO

A computer room has 28 desktop PCs. Pupils also use tablets across classrooms. Recommend how each device type should connect.

STEP 1 · FIXED COMPUTERS

Choose wired Ethernet: no need to move the PCs and a cable can give stable, high-capacity links.

STEP 2 · MOBILE TABLETS

Choose Wi-Fi: pupils can move between suitable access points without network cables.

STEP 3 · THINK ABOUT REAL LIMITATIONS

Wi-Fi performance depends on signal strength, interference and user numbers. Ethernet needs installation and switch ports.

STEP 4 · EXPLAIN SECURITY

Apply suitable network controls and encryption. Never assume wired is automatically secure, or Wi-Fi automatically unsafe.

WORKED EXAMPLES

Example 2 · Addressing and delivery

Do not confuse a MAC address, an IP address and DNS name resolution.

STEP 1 · IDENTIFY IP VERSION

192.168.5.19 is an IPv4 address: four decimal octets, each between 0 and 255.

STEP 2 · RECOGNISE AN IPv6 FORMAT

2001:db8::5 is a valid shortened IPv6 notation using hexadecimal and colons.

STEP 3 · CONNECT THE MAC ADDRESS

A switch uses the local link-layer destination MAC address to forward Ethernet frames on a LAN.

STEP 4 · CONNECT IP ROUTING

A router examines logical IP destinations to forward packets between networks.

STEP 5 · CONNECT DNS

DNS can resolve a domain name to an IP address; it does not translate an entire webpage URL.

WORKED EXAMPLES

Example 3 · Posting a secure letter

Use the analogy to explain separate jobs performed by network protocols.

SCENARIO

A student sends an email containing a homework file, then opens a website to check the deadline.

SMTP · PUTTING THE LETTER IN THE POST

SMTP sends and relays messages. It is not the protocol normally used to read a synced inbox.

IMAP · CHECKING THE POSTBOX

IMAP lets the student access and keep mail synchronised on several devices.

TCP/IP · GETTING THE LETTER THERE

TCP can support reliable ordered transport; IP carries addressing and routing information between networks.

HTTPS · THE SECURE ENVELOPE

HTTPS uses TLS to protect information exchanged with the website. Never claim it guarantees the content or site is harmless.

EXAM PRACTICE

Independent questions · 1 of 4

Answer the question before opening the mark scheme. Added OCR-coverage items are labelled E1–E4.

Q1 **1 mark**

What is encryption?

Q2 **1 mark**

Name one communication protocol.

Q3 **1 mark**

What does HTTPS provide that HTTP does not?

Q4 **2 marks**

State two reasons encryption is needed on Wi-Fi networks.

Q5 **2 marks**

State two benefits of network standards.

EXAM PRACTICE

Independent questions · 2 of 4

Answer the question before opening the mark scheme. Added OCR-coverage items are labelled E1–E4.

Q6 **3 marks**

Explain why data sent over the Internet should be encrypted.

Q7 **3 marks**

Explain the purpose of communication protocols.

Q8 **3 marks**

Explain the difference between HTTP and HTTPS.

Q9 **3 marks**

Explain the difference between SMTP and POP.

Q10 **3 marks**

Explain the difference between POP and IMAP.

EXAM PRACTICE

Independent questions · 3 of 4

Answer the question before opening the mark scheme. Added OCR-coverage items are labelled E1–E4.

Q11 **4 marks**

Describe the role of TCP in data transmission.

Q12 **4 marks**

Describe the role of IP in network communication.

Q13 **4 marks**

Describe two differences between wired and wireless networks.

Q14 **6 marks**

Explain how protocols and standards allow data to be transmitted successfully across networks.

EXAM PRACTICE

Independent questions · 4 of 4

Answer the question before opening the mark scheme. Added OCR-coverage items are labelled E1–E4.

E1 · EXTRA OCR COVERAGE **2 marks**

State the number of bits in an IPv4 address and describe the usual written format.

E2 · EXTRA OCR COVERAGE **2 marks**

State the number of bits in an IPv6 address and describe how its value is written.

E3 · EXTRA OCR COVERAGE **3 marks**

Explain why a switch uses a MAC address but a router uses an IP address when forwarding traffic.

E4 · EXTRA OCR COVERAGE **3 marks**

A school needs fixed desktop computers in a lab and mobile tablets in classrooms. Recommend suitable connections and justify each.

ANSWERS & MARK SCHEME

Indicative answers · 1 of 4

These are model points for self-marking, not a replacement for the official OCR mark scheme.

Q1 **1 mark**

Encryption is the process of scrambling data so that only authorised users can read it.

Q2 **1 mark**

HTTP / HTTPS / FTP / SMTP / TCP / IP.

Q3 **1 mark**

Encryption / security.

Q4 **2 marks**

Wi-Fi data travels through the air and can be intercepted, so encryption prevents unauthorised access.

Q5 **2 marks**

Standards allow devices from different manufacturers to work together and ensure reliable communication.

ANSWERS & MARK SCHEME

Indicative answers · 2 of 4

These are model points for self-marking, not a replacement for the official OCR mark scheme.

Q6 **3 marks**

Data passes through many routers and could be intercepted, so encryption prevents unauthorised users from reading it.

Q7 **3 marks**

Protocols are sets of rules that control how data is transmitted so devices can communicate correctly.

Q8 **3 marks**

HTTP is used to view web pages, while HTTPS encrypts the data to provide secure communication.

Q9 **3 marks**

POP3 downloads messages from a server. Whether they remain on the server depends on settings.

Q10 **3 marks**

POP downloads emails to one device, while IMAP synchronises emails across multiple devices.
Exam accuracy: State that IMAP normally synchronises mail stored on the server, while POP3 commonly retrieves/downloads it.

ANSWERS & MARK SCHEME

Indicative answers · 3 of 4

These are model points for self-marking, not a replacement for the official OCR mark scheme.

Q11 4 marks

TCP ensures data is delivered reliably and in the correct order by checking that packets arrive successfully.

Q12 4 marks

IP is responsible for addressing and routing data so it reaches the correct destination.

Q13 4 marks

Wired networks use cables and are more reliable, while wireless networks use radio waves and are more flexible but can suffer interference.
Exam accuracy: Avoid universal performance rankings; wired is generally stable and requires cables, while wireless permits mobility but can face interference.

Q14 6 marks

Standards ensure devices from different manufacturers can communicate. Protocols such as HTTP, TCP, and IP define how data is requested, delivered reliably, and routed to the correct destination, making communication efficient and reliable.

E1 · EXTRA 2 marks

IPv4 uses 32 bits (1); the usual format is four decimal numbers separated by dots, with each value from 0 to 255 (1).

ANSWERS & MARK SCHEME

Indicative answers · 4 of 4

These are model points for self-marking, not a replacement for the official OCR mark scheme.

E2 · EXTRA — **2 marks**

IPv6 uses 128 bits (1); it is generally written as eight groups of hexadecimal digits separated by colons, with shortening permitted (1).

E3 · EXTRA — **3 marks**

A switch uses link-layer MAC addresses to deliver frames locally (1). A router uses logical IP addresses to forward packets between networks (1). Their jobs differ by network scope/layer (1).

E4 · EXTRA — **3 marks**

Use wired Ethernet for fixed desktops for a stable, high-capacity connection (1); use Wi-Fi for mobile tablets to allow movement (1). Explain that encryption, access point capacity and signal strength must also be considered (1).

THEORY

Six ways a system can be attacked

Picture first, then the memorable analogy, then the precise exam explanation.

THE CENTRAL QUESTION

For each threat, explain how the attack happens and what the attacker wants or what harm is caused. That is the explicit OCR 1.4.1 requirement.

MALWARE	Malicious software that may steal, change or damage data.
SOCIAL ENGINEERING	Deceiving people into sharing information or taking unsafe actions.
BRUTE FORCE	Trying many credentials to guess a valid one.
DENIAL OF SERVICE	Making a service unavailable by overwhelming resources.
INTERCEPTION & THEFT	Capturing information during transmission.
SQL INJECTION	Abusing unsafe database input handling to change a query.

THEORY

Six cyber threats · illustrated

Study the six attacks using the illustrated examples. Then practise the precise explanation needed in an exam.



FROM IMAGE TO EXAM ANSWER

The illustration shows six different threats. For each one, explain how it works and what effect it may have.

THEORY

Three threats · people and passwords

Keep the existing illustration and analogies; always finish with the exam explanation.

MALWARE

Analogy: A computer catching a virus.

How: Malicious code is introduced onto a device, e.g. through an unsafe file or download.

Purpose / effect: It may steal information, damage files, spy on activity or disrupt use.

SOCIAL ENGINEERING / PHISHING

Analogy: Someone pretending to be trusted.

How: A message or page impersonates a trusted source to persuade someone to take an action.

Purpose / effect: A person may disclose credentials, send data or grant unauthorised access.

BRUTE-FORCE ATTACK

Analogy: Trying every key on a keyring.

How: Software repeatedly submits potential passwords or other credentials.

Purpose / effect: The attacker hopes to guess a valid login and access an account.

THEORY

Three threats · networks and databases

Explain what the attack changes and what its victim experiences.

DENIAL OF SERVICE

Analogy: Blocking a doorway so others cannot enter.

How: An attacker overwhelms a server or network service with traffic or requests.

Purpose / effect: Legitimate users cannot access the service or experience severe delays.

DATA INTERCEPTION & THEFT

Analogy: Listening in on a private conversation.

How: An unauthorised person captures data while it travels between devices.

Purpose / effect: Sensitive information such as credentials may be exposed or stolen.

SQL INJECTION

Analogy: A cheat code changing the rules - only a memory aid.

How: If application input is incorrectly combined into an SQL query, the intended query can be altered.

Purpose / effect: An attacker may obtain, change or delete database information.

KEY TERMINOLOGY

Key words for 1.4.1 · 1 of 2

Use the correct technical terms when explaining how each attack works and what it aims to achieve.

Malware

Software designed to perform harmful or unauthorised actions.

Virus

Malware that attaches to a host file or program and can spread when run.

Worm

Self-propagating malware which can spread between devices or networks.

Trojan

Malicious software disguised as something legitimate.

Ransomware

Malware that locks or encrypts access to data and demands payment.

Spyware

Software that secretly gathers information about activity or data.

KEY TERMINOLOGY

Key words for 1.4.1 · 2 of 2

Use the correct technical terms when explaining how each attack works and what it aims to achieve.

Social engineering

Manipulating people to reveal information or take unsafe actions.

Phishing

Deceptive messages or sites designed to steal information.

Brute force

Repeated systematic attempts to guess a password or secret.

DoS / DDoS

Overloading a service; distributed attacks use multiple sources.

Data interception

Capturing information during network transmission.

SQL injection

Untrusted input alters an application-generated SQL query.

WORKED EXAMPLES

Example 1 · A convincing school email

The central decision: does the attacker defeat software, or manipulate a person?

SCENARIO

A pupil receives a convincing message claiming to be from the school IT team. It asks them to confirm their password using a linked form.

STEP 1 · IDENTIFY THE ATTACK

This is phishing, an example of social engineering: the attacker impersonates a trusted sender.

STEP 2 · EXPLAIN HOW IT WORKS

The message may direct the pupil to a fake login page where credentials are entered and collected.

STEP 3 · STATE THE PURPOSE

The attacker wants account credentials and potentially unauthorised access to school data.

STEP 4 · COMPARE BRUTE FORCE

Brute force tries many possible passwords; phishing persuades a person to reveal one. Both may lead to compromised accounts.

WORKED EXAMPLES

Example 2 · A website under attack

The key skill: distinguish unavailability, intercepted data and a database query weakness.

SCENARIO

A school website suddenly stops responding. A separate report says confidential messages have been captured while travelling over the network.

DENIAL OF SERVICE

A flood of requests or traffic exhausts the service capacity, preventing legitimate access.

INTERCEPTION

Information is captured in transit. It may be read or stolen if it is not adequately protected.

SQL INJECTION

Unsafe database query construction allows input to alter the query, potentially exposing or modifying stored records.

LINKED EXAM ANSWER

Naming a threat is not enough. Explain how it is carried out, then give a linked purpose or effect. Do not describe these three as identical.

EXAM PRACTICE

Independent questions · 1 of 4

Answer before turning to the separate mark scheme. The original 12 questions remain unchanged; new coverage questions are clearly marked.

EXAM Q1 · 1 MARK

What is malware?

EXAM Q2 · 1 MARK

What is phishing?

EXAM Q3 · 1 MARK

What is a brute-force attack?

EXAM Q4 · 2 MARKS

State two examples of cyber attacks.

EXAM PRACTICE

Independent questions · 2 of 4

Use your own words. Explain the mechanism and purpose, not just the attack name.

EXAM Q5 · 2 MARKS

State two aims of a cyber attack.

EXAM Q6 · 3 MARKS

Explain how a brute- force attack works.

EXAM Q7 · 3 MARKS

Explain how phishing attacks work.

EXAM Q8 · 3 MARKS

Explain how a denial of service (DoS) attack affects a system.

EXAM PRACTICE

Independent questions · 3 of 4

Use your own words. Explain the mechanism and purpose, not just the attack name.

EXAM Q9 · 3 MARKS

Explain what is meant by data interception and theft.

EXAM Q10 · 4 MARKS

Describe two ways malware can harm a computer system.

EXAM Q11 · 4 MARKS

Describe two social engineering attacks.

EXAM Q12 · 6 MARKS

Explain why humans are often the weakest link in computer security.

EXAM PRACTICE

Independent questions · 4 of 4

Use your own words. Explain the mechanism and purpose, not just the attack name.

ADDED OCR COVERAGE · 3 MARKS

Explain the concept of SQL injection and state one possible purpose of the attack.

ADDED OCR COVERAGE · 3 MARKS

Explain the difference between a denial-of-service attack and data interception.

ANSWERS & MARK SCHEME

Model answers · 1 of 4

Indicative marking points. Match the meaning; exact wording is not always necessary.

EXAM Q1 · 1 MARK

What is malware?

Malware is software deliberately designed to carry out harmful or unauthorised actions on a system. (1)

EXAM Q2 · 1 MARK

What is phishing?

Phishing is a form of social engineering that uses deceptive messages or sites to trick a person into revealing information. (1)

EXAM Q3 · 1 MARK

What is a brute-force attack?

A brute-force attack repeatedly tries possible passwords or credentials to find the correct one. (1)

EXAM Q4 · 2 MARKS

State two examples of cyber attacks.

Any two: malware; phishing or social engineering; brute-force; denial of service; data interception and theft; SQL injection. (1 each, 2 maximum)

EXAM Q5 · 2 MARKS

State two aims of a cyber attack.

Any two: steal sensitive information; gain unauthorised access; disrupt a service; change/delete data; extort a victim. (1 each)

ANSWERS & MARK SCHEME

Model answers · 2 of 4

Indicative marking points. Match the meaning; exact wording is not always necessary.

EXAM Q6 · 3 MARKS

Explain how a brute- force attack works.

The attacker makes repeated guesses at a password (1), often using an automated program or systematic sequence (1), until a correct password is found and the account can be accessed (1).

EXAM Q7 · 3 MARKS

Explain how phishing attacks work.

An attacker sends a deceptive email or message that appears to come from a trusted person or organisation (1); it directs the person to a fake link/site or requests confidential information (1); the victim may reveal credentials or other private data (1).

EXAM Q8 · 3 MARKS

Explain how a denial of service (DoS) attack affects a system.

The attacker overwhelms a website, service or network with requests or traffic (1), using up its available capacity/resources (1), so legitimate users experience delays or cannot access it (1).

EXAM Q9 · 3 MARKS

Explain what is meant by data interception and theft.

Data is captured by an unauthorised party while it is being sent between devices (1); messages or packets can be read if not adequately protected (1); the captured confidential information may be stolen or misused (1).

ANSWERS & MARK SCHEME

Model answers · 3 of 4

Indicative marking points. Match the meaning; exact wording is not always necessary.

EXAM Q10 · 4 MARKS

Describe two ways malware can harm a computer system.

Any two harms, developed: malware can delete or corrupt important files, causing loss of work (up to 2); it can capture passwords or personal information, allowing identity theft or account compromise (up to 2). Award distinct valid mechanisms and effects to a maximum of 4.

EXAM Q11 · 4 MARKS

Describe two social engineering attacks.

Phishing: deceptive messages impersonate a trusted sender and entice a victim to reveal credentials or follow a false link (up to 2). Pharming: a victim is directed to a fraudulent website where information may be collected (up to 2). Award two distinct developed examples, such as pretexting or impersonation.

EXAM Q12 · 6 MARKS

Explain why humans are often the weakest link in computer security.

A developed response explains that attackers can impersonate trusted people or organisations (e.g. phishing); users may believe a convincing message, click a dangerous link or disclose credentials; social engineering bypasses technical barriers by exploiting human decisions; stolen credentials can enable account access or data theft; awareness and verification matter, but user errors are not the only source of vulnerabilities. Award up to 6 for relevant, linked points and clear explanation.

ANSWERS & MARK SCHEME

Model answers · 4 of 4

Indicative marking points. Match the meaning; exact wording is not always necessary.

ADDED QE1 · 3 MARKS

Explain the concept of SQL injection and state one possible purpose of the attack.

Untrusted text supplied to an application is improperly treated as part of a database query (1), changing the intended query behaviour (1), to access, change or delete data without authorisation (1).

ADDED QE2 · 3 MARKS

Explain the difference between a denial-of-service attack and data interception.

A denial-of-service attack floods or exhausts resources so legitimate users cannot access a service (1). Interception captures data during communication (1). Its purpose may be to discover confidential information, not necessarily to stop the service working (1).

THEORY

Seven ways to reduce cyber threats

Name the prevention method, explain what it does, and link it to the threat it limits.

THE EXAM RULE

Threat → Appropriate defence → How it works → Reduced risk. These seven common prevention methods are named by OCR. Some useful extra protections appear later but must not replace this list.

PENETRATION TESTING	An authorised tester simulates attacks to find weaknesses so they can be fixed.
ANTI-MALWARE SOFTWARE	Scans or monitors files/programs; can quarantine or remove identified malicious software.
FIREWALLS	Examines traffic against rules and permits or blocks selected network connections.
USER ACCESS LEVELS	Permissions restrict which users can access, change or delete particular resources.
PASSWORDS	Long, unique passwords make unauthorised account guessing more difficult.
ENCRYPTION	Transforms plaintext using a key so intercepted data is not readable without the key.
PHYSICAL SECURITY	Limits physical access to computers, cables, storage media and server rooms.

THEORY

Security defences and everyday analogies

Match each familiar security analogy to a specific defence and explain how it reduces risk.

Safety Tool	What it does (simple)	Like in real life
Anti-malware software	Finds and removes bad software	A **doctor** for your computer
User access levels	Only lets people do what they're allowed to do	Different **keys** for rooms
Strong passwords	Makes accounts hard to break into	A **strong lock**
Two-Factor Authentication (2FA)	Uses two checks to prove it's you	A lock **and** a code
User training	Teaches people how to stay safe	Learning **road safety**
Account lock-out policy	Locks accounts after too many wrong tries	A door that **locks itself**
Firewalls	Blocks bad traffic from the network	A **security guard**
Penetration testing	Finds weak spots before hackers do	A **friendly burglar**
Encryption	Scrambles messages so others can't read them	Writing in **secret code**
Input validation & sanitisation	Checks what people type to stop harm	A **safety checker**
Physical security	Protects computers in real life	**Locks, doors, CCTV**

STUDY THE COMPARISON TABLE

The analogy is a reminder, not a complete exam answer. A security guard suggests a firewall, but a firewall follows traffic rules rather than recognising every attack. Room keys suggest access levels, but those are software permissions. Secret code suggests encryption, which uses algorithms and keys.

SCOPE CHECK

OCR names seven core methods: penetration testing, anti-malware, firewalls, user access levels, passwords, encryption and physical security. Related measures include 2FA, training, account lockout and input checking.

THEORY

Four teaching analogies · access and detection

Picture → familiar comparison → exact action → the threat it limits.

PENETRATION TESTING

Teaching analogy: Friendly burglar

What it actually does: An authorised tester simulates attacks to find weaknesses so they can be fixed.

ANTI-MALWARE SOFTWARE

Teaching analogy: Doctor

What it actually does: Scans or monitors files/programs; can quarantine or remove identified malicious software.

FIREWALLS

Teaching analogy: Security guard

What it actually does: Examines traffic against rules and permits or blocks selected network connections.

USER ACCESS LEVELS

Teaching analogy: Different room keys

What it actually does: Permissions restrict which users can access, change or delete particular resources.

THEORY

Three teaching analogies · access, data, rooms

Explain the mechanism; avoid claiming any single defence guarantees security.

PASSWORDS

Teaching analogy: Strong lock

What it actually does: Long, unique passwords make unauthorised account guessing more difficult.

ENCRYPTION

Teaching analogy: Secret code

What it actually does: Transforms plaintext using a key so intercepted data is not readable without the key.

PHYSICAL SECURITY

Teaching analogy: Locked doors and CCTV

What it actually does: Limits physical access to computers, cables, storage media and server rooms.

USE SEVERAL DEFENCES

An encrypted school laptop can still be stolen. A locked server room does not prevent a phishing email. Defences cover different risks and work best in layers.

THEORY

Match each attack to a useful defence

An exam answer earns marks by naming the control and explaining how it limits the attack.

MALWARE

Anti-malware + access levels

Scan/quarantine malicious software; restrict installing or running unapproved programs.

PHISHING / SOCIAL ENGINEERING

Passwords + access levels

Unique secrets limit reuse; minimal permissions limit damage if credentials are stolen.

BRUTE FORCE

Strong passwords

Long, unpredictable passwords require far more guesses to find a valid login.

DENIAL OF SERVICE

Firewall rules + monitoring

Filter or restrict some unwanted traffic; a large-scale attack may still require other protections.

DATA INTERCEPTION

Encryption

Intercepted ciphertext is unintelligible without the appropriate key.

SQL INJECTION

Penetration testing + limited access

Testing can expose unsafe handling of database queries; low permissions limit damage. Parameterised queries fix the injection mechanism.

THEORY

Further security protections

These strengthen a security plan. They are supplementary to OCR's seven named prevention methods.

TWO-FACTOR AUTHENTICATION

Teaching analogy: A lock and a code

How it helps: Adds another verification factor when logging in.

USER TRAINING

Teaching analogy: Learning road safety

How it helps: Teaches users to notice suspicious messages and verify requests.

ACCOUNT LOCKOUT

Teaching analogy: A self-locking door

How it helps: Limits repeated failed login attempts.

INPUT HANDLING

Teaching analogy: Safety checker

How it helps: Validates input; safe parameterised queries prevent SQL query structure changes.

KEY TERMINOLOGY

Key words for 1.4.2 · 1 of 2

The yellow is the exact same translucent SVG highlighter used in classroom lessons.

Vulnerability

A weakness in a system that an attacker could exploit.

Penetration testing

An authorised attempt to expose system weaknesses using simulated attacks.

Anti-malware

Software that detects, quarantines or removes known malicious software.

Firewall

Software or hardware that controls traffic using security rules.

Access levels

Permissions determining the resources or actions a user can access.

Authentication

Process of checking a user or device is who it claims to be.

KEY TERMINOLOGY

Key words for 1.4.2 · 2 of 2

The yellow is the exact same translucent SVG highlighter used in classroom lessons.

Strong password

A long, unique secret difficult for an attacker to guess.

Encryption

Converting readable plaintext into ciphertext using a key and algorithm.

Physical security

Locks, access cards, surveillance and secure storage limiting physical access.

Two-factor authentication

Login checking two distinct types of evidence, e.g. password plus an authenticator.

Account lockout

Restricting login attempts after repeated failures to slow password guessing.

Defence in depth

Using several complementary protections so failure of one is less damaging.

WORKED EXAMPLES

Example 1 · A phishing message at school

Choose protections that address what is actually at risk.

SCENARIO

A teacher receives a fake IT-support email requesting account credentials. An attacker may then try to read confidential student files.

STEP 1 · IDENTIFY THE WEAKNESS

The attack tricks a person into revealing credentials. A stolen login might give access to confidential documents.

STEP 2 · CHOOSE AN OCR CONTROL

User access levels restrict which student records each account may open or change.

STEP 3 · EXPLAIN ITS EFFECT

Even if the teacher account is compromised, an attacker may be unable to view records outside that account’s permissions.

STEP 4 · ADD A COMPLEMENTARY CONTROL

Unique, strong passwords reduce guessing/reuse risks; staff training and 2FA provide useful additional layers.

WORKED EXAMPLES

Example 2 · A school exam portal

Write about the control’s mechanism rather than simply calling it secure.

SCENARIO

Exam data travels between staff laptops and a school server. The server also accepts uploads and database searches.

STEP 1 · INTERCEPTED DATA

Encryption protects the information in transit, so an interceptor without the key cannot read the contents.

STEP 2 · UNAUTHORISED NETWORK ACCESS

A firewall checks connections against rules and restricts unwanted network traffic. It does not stop every attack.

STEP 3 · SQL INJECTION RISK

Penetration testing can discover unsafe database handling. Parameterised queries prevent untrusted input changing SQL structure; this is a useful technical extension.

STEP 4 · LOST OR STOLEN DEVICES

Physical security limits access to machines; permissions and encryption help protect information if other controls fail.

EXAM PRACTICE

Practice questions · 1 of 4

Answer independently first; check the mark scheme afterwards.

EXAM Q1 · 1 MARK

What is a security vulnerability?

EXAM Q2 · 1 MARK

What is encryption used for?

EXAM Q3 · 1 MARK

What is a firewall?

EXAM Q4 · 2 MARKS

State two ways to prevent malware attacks.

EXAM PRACTICE

Practice questions · 2 of 4

Answer independently first; check the mark scheme afterwards.

EXAM Q5 · 2 MARKS

State two features of strong passwords.

EXAM Q6 · 3 MARKS

Explain how user access levels improve security.

EXAM Q7 · 3 MARKS

Explain how anti-malware software protects a system.

EXAM Q8 · 3 MARKS

Explain how account lockout policies prevent brute-force attacks.

EXAM PRACTICE

Practice questions · 3 of 4

Answer independently first; check the mark scheme afterwards.

EXAM Q9 · 3 MARKS

Explain how input validation helps prevent SQL injection.

EXAM Q10 · 4 MARKS

Describe two ways to prevent phishing attacks.

EXAM Q11 · 4 MARKS

Describe two ways firewalls help protect networks.

EXAM PRACTICE

EXAM Q12 + additional questions

Answer independently first; check the mark scheme afterwards.

EXAM Q12 · 6 MARKS

Explain how a range of security measures can be used to protect a computer system.

ADDED OCR COVERAGE E1 · 3 MARKS

Explain how an organisation could use penetration testing to reduce a security vulnerability.

ADDED OCR COVERAGE E2 · 3 MARKS

Explain how physical security and user access levels protect different weaknesses in a school.

ANSWERS & MARK SCHEME

Model answers · 1 of 4

Indicative points. Full credit depends on the quality of explanation, not memorising these exact sentences.

EXAM 1 · 1 MARK

What is a security vulnerability?

A vulnerability is a weakness in a computer, application or network that could be exploited by an attacker. (1)

EXAM 2 · 1 MARK

What is encryption used for?

Encryption converts readable data (plaintext) into unreadable ciphertext using an algorithm and key, protecting confidentiality if data is intercepted. (1)

EXAM 3 · 1 MARK

What is a firewall?

A firewall checks network traffic against security rules and allows or blocks connections. (1)

EXAM 4 · 2 MARKS

State two ways to prevent malware attacks.

Anti-malware software scans/detects potentially malicious programs (1), and software/security updates repair known weaknesses that malware might exploit (1).

ANSWERS & MARK SCHEME

Model answers · 2 of 4

Indicative points. Full credit depends on the quality of explanation, not memorising these exact sentences.

EXAM 5 · 2 MARKS

State two features of strong passwords.

Long enough to resist guessing (1) and unique/unpredictable rather than a common or reused password (1). Accept other justified features such as a suitable combination of character types.

EXAM 6 · 3 MARKS

Explain how user access levels improve security.

User permissions restrict who can see, change or delete information (1). A compromised account may have only limited access (1), reducing the possible harm to protected files or settings (1).

EXAM 7 · 3 MARKS

Explain how anti-malware software protects a system.

Anti-malware scans or monitors software for suspicious or known malicious patterns (1), identifies/quarantines threats to stop them running (1), and removes or isolates malicious software to reduce infection (1).

EXAM 8 · 3 MARKS

Explain how account lockout policies prevent brute-force attacks.

After a specified number of failed sign-ins, the system blocks/delays further attempts (1). This limits the speed of repeated guessing (1), reducing the effectiveness of brute-force attempts (1).

ANSWERS & MARK SCHEME

Model answers · 3 of 4

Indicative points. Full credit depends on the quality of explanation, not memorising these exact sentences.

EXAM 9 · 3 MARKS

Explain how input validation helps prevent SQL injection.

Validation checks data meets expected rules or format (1), rejecting or correcting unsuitable entries (1). However, validation alone is not a reliable SQL-injection defence: safely parameterised database queries keep untrusted data separate from SQL commands (1). Credit a clear supported explanation of the Practice question, not a false claim that all malicious input is detected.

EXAM 10 · 4 MARKS

Describe two ways to prevent phishing attacks.

Train users to spot suspicious messages and verify senders/links rather than disclosing credentials (up to 2). Two-factor authentication limits misuse of stolen passwords by requiring another independent factor (up to 2). Award two distinct developed approaches.

EXAM 11 · 4 MARKS

Describe two ways firewalls help protect networks.

A firewall can restrict incoming connections to allowed services and ports (up to 2). It can monitor/filter outgoing or incoming traffic against rules, blocking suspicious or unauthorised traffic (up to 2). Avoid the inaccurate idea that a firewall stops all attacks.

ANSWERS & MARK SCHEME

Model answers · 4 of 4

Indicative points. Full credit depends on the quality of explanation, not memorising these exact sentences.

EXAM 12 · 6 MARKS

Explain how a range of security measures can be used to protect a computer system.

Award up to six developed and linked points across several defences. For example, anti-malware identifies/quarantines infected files, firewalls restrict unauthorised traffic, access levels limit data access if an account is compromised, strong unique passwords reduce guessing, encryption protects intercepted data, and physical locks limit access to devices. Penetration testing identifies vulnerabilities so fixes can be applied. Credit explanations of how each method limits a named risk, not a mere list.

ADDED E1 · 3 MARKS

Explain how an organisation could use penetration testing to reduce a security vulnerability.

An authorised tester simulates realistic attacks (1), identifies weaknesses that could be exploited (1), and reports those weaknesses so the organisation can fix them before a real attacker exploits them (1).

ADDED E2 · 3 MARKS

Explain how physical security and user access levels protect different weaknesses in a school.

Physical security restricts who can reach devices or a server room (1). Access levels restrict what signed-in users can read or change (1). Together they address physical and account-based risks rather than relying on one barrier (1).

THEORY

Operating systems · five core jobs

The phone is a familiar example of a computer controlled by an operating system. The picture is a memory aid; use the exact technical explanation in exam answers.

WHAT IS AN OS?

An operating system is system software that manages the hardware and resources of a computer and provides an interface and services for applications and users.

1 · USER INTERFACE

Home-screen icons and swipes are how a person gives instructions. A GUI is one form of user interface.

2 · MEMORY + MULTITASKING

The OS allocates RAM to apps and schedules tasks for CPU time. An app switcher is a visible example, not the scheduling mechanism.

3 · PERIPHERALS + DRIVERS

The OS uses drivers to communicate with cameras, speakers, keyboards, printers and other devices, and manages data transfer.

4 · USER ACCOUNTS + ACCESS RIGHTS

The OS can authenticate users and control permissions to settings and data. An Apple ID is not exactly the same as a local OS account.

5 · FILE MANAGEMENT

The OS handles names, folders, saving, copying, moving and retrieving files on storage.

THEORY

Operating systems through an iPhone example

Use the phone examples to connect everyday actions with the OS functions that manage them.



TURN THE PICTURE INTO EXAM LANGUAGE

Select one labelled phone action. Name the operating-system function, describe how it works, then explain why it is useful.

THEORY

What each OS function really does

Keep the analogy, but describe the process rather than just naming a phone feature.

USER INTERFACE

Phone analogy: User taps a camera icon or selects from a menu.

Exam wording: The OS provides an interface for input and output; GUIs, text interfaces and touch gestures are examples.

MEMORY & MULTITASKING

Phone analogy: Phone changes between browser and photos.

Exam wording: The OS allocates RAM and schedules processor time, switching tasks so more than one program makes progress.

PERIPHERALS & DRIVERS

Phone analogy: Phone connects to a printer or processes camera input.

Exam wording: Drivers help the OS communicate with particular devices; the OS manages input/output data transfer.

USER MANAGEMENT

Phone analogy: Device checks a passcode or restricts an app permission.

Exam wording: Account authentication and access rights limit unauthorised use and access to protected resources.

FILE MANAGEMENT

Phone analogy: Phone saves a photo into a folder and moves it.

Exam wording: The OS gives files names, stores and organises them in folders, and enables saving, retrieval and movement.

THEORY

Operating-system functions

This table illustrates several OS functions, including memory, devices and interfaces. File management is explained below.

OS Function	What it does	Key features / examples	Why it matters (exam focus)
User Interface (UI)	Allows the user to interact with the computer	Windows, icons, menus, pointer, touchscreen	Makes the computer usable
Memory Management	Allocates RAM to applications and transfers data between memory and the CPU	Loads programs into memory, frees memory when closed	Prevents crashes and allows programs to run
Multitasking	Allows more than one program to run at the same time	CPU switches quickly between applications	Improves efficiency and user experience
Peripheral Management & Drivers	Controls input/output devices using drivers	Keyboard, mouse, printer, display	Allows hardware to communicate with software
Data Transfer Management	Manages data sent between devices and the processor	Input and output coordination	Prevents data loss or errors
User Management	Creates and manages user accounts	Login accounts, passwords, admin vs standard users	Improves security and controls access
Access Rights	Controls what users can do	Read/write permissions, restricted settings	Protects files and system settings
Security	Protects the system and user data	Passwords, permissions, account control	Prevents unauthorised access

FILE MANAGEMENT

File management organises folders and names, saves and retrieves data, and moves files. The OS manages resources but cannot guarantee that every application will run without errors.

EXAM SCOPE

OCR requires these operating-system functions and their purpose. Detailed implementation of paging and segmentation is not required.

THEORY

Phone-to-OS comparison

Your existing classroom table connects the phone to nine practical OS activities. Below it is the precise distinction to remember.

OS Function	What it does	📱 iPhone Analogy
User Interface (UI)	Allows the user to interact with the computer	The **iPhone home screen**, icons, swipe gestures, and touch
Memory Management	Allocates RAM to applications and transfers data between memory and the CPU	iOS deciding **how much memory each app gets**so apps don't crash
Multitasking	Allows more than one program to run at the same time	Switching between apps using the **app switcher**
Peripheral Management & Drivers	Controls input/output devices using drivers	iOS automatically recognising **AirPods, chargers, printers, keyboards**
Data Transfer Management	Manages data sent between devices and the processor	Touching the screen or taking a photo and iOS **sending that data to the processor**
User Management	Creates and manages user accounts	**Apple IDs** and user profiles on a device
Access Rights	Controls what users can do	**Parental controls** and app permissions (camera, microphone, location)
Security	Protects the system and user data	**Face ID, passcodes, secure boot, encryption**
File Management	Organises and stores data	The **Files app**, folders, saving and moving documents

Exam application – match the OS function: Match each task to its operating-system role: creating a new user profile; allocating RAM to running programs; organising a saved folder; sending data to a printer. Choose from user, memory, file and peripheral management, and explain one match.

WATCH OUT FOR TWO SHORTCUTS

The app switcher shows apps, but scheduling and RAM management are OS processes. An Apple ID manages access to Apple services; the OS also manages local authentication and device permissions. Use the five OCR functions when structuring an answer.

KEY TERMINOLOGY

Key terms for 1.5.1 · 1 of 2

Yellow highlighting indicates key terms; exam definitions explain what each function does.

Operating system

System software that manages hardware and resources, and provides services and an interface for users and applications.

User interface

How a person gives commands to and receives information from a computer (for example, icons, menus, touch or text commands).

Graphical UI (GUI)

A user interface using visual elements such as windows, icons, menus and pointers.

Memory management

The OS allocates RAM to processes and coordinates movement of programs and data through memory.

Multitasking

More than one task makes progress because the OS schedules CPU time and switches between tasks; multiple cores may also run tasks in parallel.

CPU scheduling

Selecting which ready task should run on the processor at a particular moment.

KEY TERMINOLOGY

Key terms for 1.5.1 · 2 of 2

Yellow highlighting indicates key terms; exam definitions explain what each function does.

Peripheral

Hardware used for input, output or storage, such as a keyboard, printer, camera or drive.

Device driver

Software that allows the OS to communicate with and control a particular hardware device.

Data transfer

Movement of data between devices, memory and processor that the operating system helps coordinate.

User account

An operating-system identity with credentials, settings and assigned permissions.

Access rights

Permissions controlling which files, resources and actions a user or program can use.

File management

Creating, naming, saving, organising, moving and deleting files and folders.

WORKED EXAMPLES

Example 1 · Switching between apps

A student records a voice note, opens a browser and returns to the recording.

SCENARIO

An iPhone has a notes app, camera and browser. The student wants to swap quickly between activities without losing their work.

STEP 1 · IDENTIFY TWO OS FUNCTIONS

Memory management allocates RAM to apps. Multitasking lets multiple programs make progress.

STEP 2 · EXPLAIN WHAT THE OS DOES

The OS allocates/reclaims memory and schedules CPU time. When a different app runs, the system switches active work.

STEP 3 · DISTINGUISH THE ANALOGY

The app-switcher screen is visible evidence of multiple apps, but not the mechanism that schedules CPU tasks.

STEP 4 · FORM AN EXAM ANSWER

"The OS allocates RAM and schedules processor time between running tasks, allowing a user to switch between applications."

WORKED EXAMPLES

Example 2 · A school computer

A pupil prints a document but must not edit the teacher-only folder.

SCENARIO

A school laptop connects to a printer. The pupil can print their work but cannot remove files from a protected shared folder.

STEP 1 · PERIPHERAL MANAGEMENT

The OS uses the printer driver to send data and commands to the printer in a format it can use.

STEP 2 · USER MANAGEMENT

The OS recognises/authenticates the pupil account and applies its assigned permissions.

STEP 3 · FILE MANAGEMENT

The pupil names and saves the assignment in a folder. The OS manages its location and retrieval.

STEP 4 · FORM AN EXAM ANSWER

"Drivers allow OS/device communication; access rights prevent the pupil deleting protected files; file management saves and organises the document."

EXAM PRACTICE

Questions · 1 of 4

Answer the questions first; the mark scheme follows later.

EXAM Q1 · 1 MARK

What is an operating system?

EXAM Q2 · 1 MARK

Name one function of an operating system.

EXAM Q3 · 1 MARK

What is a user interface (UI)?

EXAM Q4 · 2 MARKS

State two functions of an operating system.

EXAM PRACTICE

Questions · 2 of 4

Answer the questions first; the mark scheme follows later.

EXAM Q5 · 2 MARKS

State two reasons why an operating system is important.

EXAM Q6 · 3 MARKS

Explain how memory management helps a computer run efficiently.

EXAM Q7 · 3 MARKS

Explain how multitasking is handled by an operating system.

EXAM Q8 · 3 MARKS

Explain the role of device drivers in an operating system.

EXAM PRACTICE

Questions · 3 of 4

Answer the questions first; the mark scheme follows later.

EXAM Q9 · 3 MARKS

Explain how file management helps users.

EXAM Q10 · 4 MARKS

Describe the role of an operating system in managing users.

EXAM Q11 · 4 MARKS

Describe two security features provided by an operating system.

EXAM PRACTICE

Questions · 4 of 4

Answer the questions first; the mark scheme follows later.

EXAM Q12 · 6 MARKS

Explain the main functions of an operating system.

ADDED OCR PRACTICE E1 · 3 MARKS

A student switches between a notes app and the camera. Explain how the OS manages memory and processor time.

ADDED OCR PRACTICE E2 · 3 MARKS

A teacher installs a printer and gives a student account permission to print but not delete shared files. Explain the two OS functions involved.

ANSWERS & MARK SCHEME

Model answers · 1 of 4

These are indicative marking points, not a script that must be memorised word for word.

EXAM Q1 · 1 MARK

What is an operating system?

An operating system is system software that manages a computer's resources (hardware, memory, devices and files) and provides services/user interfaces for programs and people. (1)

EXAM Q2 · 1 MARK

Name one function of an operating system.

Any valid function, e.g. user interface, memory management, multitasking, peripheral/driver management, user management or file management. (1)

EXAM Q3 · 1 MARK

What is a user interface (UI)?

A user interface lets a person interact with or control a computer, providing input and viewing output. (1)

EXAM Q4 · 2 MARKS

State two functions of an operating system.

Any two distinct examples such as memory management (allocates RAM) and peripheral management (controls devices using drivers). (2)

ANSWERS & MARK SCHEME

Model answers · 2 of 4

These are indicative marking points, not a script that must be memorised word for word.

EXAM Q5 · 2 MARKS

State two reasons why an operating system is important.

Two justified reasons: enables users and applications to access hardware/resources (1); manages resources so tasks can run and system operation is coordinated/reliable (1). Avoid claiming crashes are impossible.

EXAM Q6 · 3 MARKS

Explain how memory management helps a computer run efficiently.

The OS allocates available RAM to running programs (1), loads/manages program data in memory (1), and reclaims or reallocates RAM when tasks no longer need it (1). Credit supported points on memory available for multitasking.

EXAM Q7 · 3 MARKS

Explain how multitasking is handled by an operating system.

The OS schedules time on the CPU between runnable tasks (1), saves/restores task state or rapidly switches CPU attention (1), so multiple applications can make progress or appear active simultaneously (1). Multiple cores can also perform tasks simultaneously.

EXAM Q8 · 3 MARKS

Explain the role of device drivers in an operating system.

A device driver provides a software interface between the OS and a particular hardware device (1), translating commands and data as needed (1), and enabling use/control of examples such as printers, keyboards or sound devices (1).

ANSWERS & MARK SCHEME

Model answers · 3 of 4

These are indicative marking points, not a script that must be memorised word for word.

EXAM Q9 · 3 MARKS

Explain how file management helps users.

File management lets users name and save files (1), organise them in folders and move/rename/delete them (1), and retrieve/open the correct stored work efficiently (1).

EXAM Q10 · 4 MARKS

Describe the role of an operating system in managing users.

The OS sets up and manages user accounts (1), authenticates users for access (1), assigns permissions such as read/write or administrator privileges (1), and restricts unauthorised use of resources (1).

EXAM Q11 · 4 MARKS

Describe two security features provided by an operating system.

Examples: authentication using a password/passcode protects access to an account (up to 2); access rights prevent a user reading/changing resources they are not allowed to use (up to 2). Other clearly OS-managed security features with proper explanations may be credited.

ANSWERS & MARK SCHEME

Model answers · 4 of 4

These are indicative marking points, not a script that must be memorised word for word.

EXAM Q12 · 6 MARKS

Explain the main functions of an operating system.

A developed six-mark answer should explain multiple functions: a GUI/CLI lets users interact with the system; memory management allocates RAM and loads data; multitasking schedules processor time; peripheral management uses drivers to control input/output; user management authenticates users and restricts access via permissions; file management names, stores and organises data. Award distinct explained functions rather than a mere list. Do not require paging or segmentation.

ADDED OCR PRACTICE E1 · 3 MARKS

A student switches between a notes app and the camera. Explain how the OS manages memory and processor time.

The OS allocates memory to the active programs (1), schedules processor time between runnable tasks (1), and manages data/state as apps become active or inactive (1). A visible app switcher alone is not the scheduling algorithm.

ADDED OCR PRACTICE E2 · 3 MARKS

A teacher installs a printer and gives a student account permission to print but not delete shared files. Explain the two OS functions involved.

Peripheral management and the printer driver enable the OS to control/send data to the printer (1). User management assigns a student account (1) and access rights restrict deletion while allowing authorised tasks (1).

THEORY

Utility software · extra housekeeping

The operating system manages core computer resources; utilities carry out additional housekeeping and protection tasks, sometimes integrated with the OS.

THE CORE IDEA

Utility software helps maintain, protect or manage a system. OCR J277 focuses on three purposes: encryption, defragmentation and data compression.

ENCRYPTION SOFTWARE

Makes information unreadable to someone without the correct key, protecting confidentiality if a file or device falls into the wrong hands.

DEFRAGMENTATION

Rearranges scattered file blocks on a magnetic HDD to reduce the distance the moving disk head travels and potentially improve access speed.

DATA COMPRESSION

Represents information using fewer bits. It can save storage space and reduce the amount of data that needs transferring.

OS OR UTILITY?

The OS provides core services; utility software handles particular maintenance tasks. Some utilities are built into the OS or run only when requested.

THEORY

Utility software on a mobile phone

Use this comparison to distinguish what the operating system does from the purposes of utility software.

Utility Software	What it does (simple)	Why it is needed	📱 Mobile phone analogy
Encryption software	Scrambles data so only authorised users can read it	Protects personal and sensitive data if a device is lost, stolen or hacked	Locking your phone with **Face ID / fingerprint** so no one else can read your messages
Defragmentation	Reorganises data on a hard disk so files are stored together	Makes files load faster and improves performance on HDDs	Tidying apps and storage so your phone doesn't feel slow
Data compression	Reduces file size by removing unnecessary data	Saves storage space and speeds up file transfer	Your phone **shrinking photos or videos** to save space
Utility software (overall)	Performs extra housekeeping tasks	Keeps the system fast, secure and organised	Background tools that **quietly keep your phone running smoothly**

IMPORTANT: THE ANALOGY IS A MEMORY AID

The Face ID example is authentication (who may access the device), not the encryption algorithm. Tidying phone apps is not the same as HDD defragmentation. Use the following pages for technically accurate descriptions.

THEORY

Three analogies · three real processes

Keep the familiar phone illustration, but distinguish the everyday example from the computing action.

1 · LOCKING A PHONE

Picture/analogy: Face ID or a passcode checks who you are.

Exam explanation: Encryption transforms the underlying information into ciphertext. The device may unlock access to the key after authentication.

2 · TIDYING A DEVICE

Picture/analogy: Rearranging apps on a phone screen makes the interface neat.

Exam explanation: HDD defragmentation relocates separate blocks of file data on a rotating magnetic disk; that is a different process.

3 · SHRINKING A VIDEO

Picture/analogy: The phone can reduce a photo or video file so it takes up less storage.

Exam explanation: Compression uses fewer bits to represent content. Lossless methods preserve data exactly; lossy methods may remove some detail.

THEORY

Encryption protects confidentiality

Use accurate encryption language: the key is the distinction between readable data and data that cannot be

BEFORE · PLAINTEXT

A school stores a document containing student information. In its original form the information can be read by an authorised program.

ENCRYPT · METHOD AND KEY

Encryption software changes the plaintext into ciphertext using an encryption method and key.

IF THE FILE IS INTERCEPTED

An attacker may still obtain or copy the data, but cannot normally understand the protected ciphertext without the appropriate decryption key.

AUTHORISATION AND AUTHENTICATION

A device uses a passcode or biometric check to control access. This can protect access to an encryption key, but the identity check itself is not encryption.

THEORY

Defragmentation · on magnetic HDDs

A simple model: fragments of a single file may be physically separated across the spinning disk.

THE SITUATION

A large file can be stored in separated blocks on an HDD. The moving read/write head may travel to different positions to gather all the data.

BEFORE · FRAGMENTED



A–D are file blocks; X/Y/Z are unrelated blocks. This is a simplified HDD model, not a picture of an SSD.

AFTER · DEFRAGMENTED



A–D are file blocks; X/Y/Z are unrelated blocks. This is a simplified HDD model, not a picture of an SSD.

WHY IT CAN HELP

More contiguous blocks can reduce head movement/seek delays on a rotating HDD. An SSD has no moving read head, so HDD-style defragmentation does not bring the same benefit.

THEORY

Data compression · smaller files

See J277 1.2.5 for lossy/lossless details; here focus on why compression is useful as utility software.

PROBLEM

A large video or backup takes up storage space and may take longer to transfer.

UTILITY ACTION

A data-compression tool encodes the file using fewer bits. Depending on the technique, it might be restored exactly (lossless) or with some detail removed (lossy).

ADVANTAGES

Uses less storage and can transfer faster if fewer bits are sent over a fixed-speed connection. The saving depends on the content, method and settings.

KEY DIFFERENCE

Compression changes file size, not access permission. Encryption changes readability/confidentiality, not necessarily how much storage a file uses.

THEORY

Operating system vs utility software

Study the comparison, then use the technical explanation to prepare an exam answer.

Operating System	Utility Software
Makes the device work	Looks after the device
Manages hardware, memory, users and files	Performs extra maintenance jobs
Always running	Often hidden or runs in the background

WHEN STUDENTS COMPARE THEM

Operating system: provides core services such as memory allocation, device communication, file management and user interface. Utility software: carries out maintenance or protection jobs such as encryption, compression and defragmentation. Utilities may be part of an OS; neither category is defined solely by whether a program runs in the background.

TWO MARKS? GIVE TWO CLEAR DIFFERENCES

Name a relevant OS function (e.g. allocating RAM) and a utility role (e.g. compressing a file). If asked for two differences, develop contrasting roles or examples rather than simply saying “one works and the other cleans”.

KEY TERMINOLOGY

Exam vocabulary · 1 of 2

Yellow highlighting identifies the key terms to use in exam answers.

Utility software

System software that performs additional housekeeping, security or maintenance tasks.

Encryption

Transforming readable plaintext into ciphertext using an encryption method and key.

Plaintext

Data that can be read in its original form.

Ciphertext

Encrypted data that cannot normally be understood without the correct key.

Decryption key

Information needed to convert encrypted data back into readable form with an appropriate method.

Authentication

Checking the identity of a user, such as by a passcode, fingerprint or face recognition.

KEY TERMINOLOGY

Exam vocabulary · 2 of 2

Yellow highlighting identifies the key terms to use in exam answers.

Defragmentation

Rearranging fragmented file blocks on a magnetic hard disk so related blocks are closer together.

Fragmentation

File data stored in separate non-contiguous blocks on a drive.

Hard disk (HDD)

Magnetic storage device whose moving read/write head can incur delays between separated blocks.

Solid-state drive (SSD)

Flash-based storage without a moving read head; routine HDD-style defragmentation is not normally useful.

Data compression

Reducing the number of bits required to represent data, according to a chosen method.

Lossless / lossy

Lossless preserves the exact original when restored; lossy deliberately discards information to reduce size.

WORKED EXAMPLES

Example 1 · The stolen laptop

A school laptop containing pupil data has been lost. Explain how encryption may help.

SCENARIO

The laptop contains a student progress spreadsheet. A thief may possess the laptop but the school has enabled device encryption.

STEP 1 · IDENTIFY THE THREAT

The thief may try to access confidential information stored on the laptop.

STEP 2 · NAME THE UTILITY

Encryption software transforms the protected spreadsheet into ciphertext using a key.

STEP 3 · EXPLAIN THE BENEFIT

Without the correct key, the thief cannot normally interpret the information even if they access a copy of the data.

STEP 4 · AVOID A COMMON ERROR

Face ID/passcodes are authentication controls, not the actual method of scrambling the stored data.

WORKED EXAMPLES

Example 2 · Different utilities for different jobs

Choose the right software for a storage-performance or file-transfer scenario.

SCENARIO A · SLOW MAGNETIC HDD

A school desktop uses a spinning hard disk. Large files have become fragmented across the disk surface.

STEP-BY-STEP HDD ANSWER

1. Choose defragmentation software. 2. It moves separate pieces of a file closer together on a magnetic drive. 3. The read/write head may move less when accessing the file, which can reduce access time. 4. Do not suggest this provides extra disk capacity or is the same as cleaning phone apps.

SCENARIO B · SHARING A LARGE VIDEO

A student needs to send a file over a slow internet connection. The video is larger than necessary.

STEP-BY-STEP FILE TRANSFER ANSWER

1. Choose data compression. 2. It represents the file with fewer bits. 3. A smaller file needs less storage and usually takes less time to send at the same transfer rate. 4. Select lossless if all information must be preserved, or a suitable lossy method where some detail loss is acceptable.

EXAM PRACTICE

Independent questions · 1 of 3

Practice questions and mark values are unchanged. Attempt without looking at the answers on later pages.

EXAM Q1 · 1 MARK

What is utility software?

EXAM Q2 · 1 MARK

Name one type of utility software.

EXAM Q3 · 1 MARK

What is encryption software used for?

EXAM Q4 · 2 MARKS

State two tasks carried out by utility software.

EXAM Q5 · 2 MARKS

State two differences between an operating system and utility software.

EXAM PRACTICE

Independent questions · 2 of 3

Practice questions and mark values are unchanged. Attempt without looking at the answers on later pages.

EXAM Q6 · 3 MARKS

Explain how defragmentation improves performance.

EXAM Q7 · 3 MARKS

Explain why encryption software is important.

EXAM Q8 · 3 MARKS

Explain how data compression benefits users.

EXAM Q9 · 4 MARKS

Describe two examples of utility software.

EXAM PRACTICE

Independent questions · 3 of 3

Practice questions and mark values are unchanged. Attempt without looking at the answers on later pages.

EXAM Q10 · 4 MARKS

Describe the difference between utility software and an operating system.

EXAM Q11 · 6 MARKS

Explain the role of utility software in maintaining a computer system.

ADDITIONAL OCR PRACTICE E1 · 3 MARKS

A student says Face ID is the same as encryption. Explain why this is not quite correct and how the two can work together.

ADDITIONAL OCR PRACTICE E2 · 3 MARKS

A school replaces a magnetic hard disk with a solid-state drive. Explain why running a regular HDD defragmentation utility is no longer an appropriate way to improve speed.

ANSWERS & MARK SCHEME

Indicative model answers · 1 of 3

These are exam-accurate marking points. Equivalent technically correct explanations can receive credit.

EXAM Q1 · 1 MARK

What is utility software?

Utility software performs additional maintenance or housekeeping tasks to help protect or manage a computer. (1)

EXAM Q2 · 1 MARK

Name one type of utility software.

One of: encryption, defragmentation, or data compression. (1)

EXAM Q3 · 1 MARK

What is encryption software used for?

It encrypts readable data into ciphertext using an algorithm/key so someone without the correct key cannot readily understand it. (1)

EXAM Q4 · 2 MARKS

State two tasks carried out by utility software.

Two different tasks, e.g. encrypting information to help keep it confidential (1) and compressing files to reduce the size for storage/transfer (1).

EXAM Q5 · 2 MARKS

State two differences between an operating system and utility software.

OS: manages everyday computer resources and provides core interfaces/services, e.g. managing RAM or hardware (1). Utility software: provides an additional housekeeping or security task, e.g. encryption or compression (1). Other clear differences may receive credit.

ANSWERS & MARK SCHEME

Indicative model answers · 2 of 3

These are exam-accurate marking points. Equivalent technically correct explanations can receive credit.

EXAM Q6 · 3 MARKS

Explain how defragmentation improves performance.

On a magnetic hard disk, fragments of one file may be stored far apart (1). Defragmentation groups those file blocks more closely (1), reducing how far the read/write head moves and potentially reducing access time (1). Do not generalise this benefit to SSDs.

EXAM Q7 · 3 MARKS

Explain why encryption software is important.

Encryption converts readable information into ciphertext (1); if a device is lost, stolen, or data is intercepted (1), someone without the correct decryption key cannot easily read the protected information (1). A phone lock is authentication, not the encryption process itself.

EXAM Q8 · 3 MARKS

Explain how data compression benefits users.

Compression reduces how many bits are needed to represent a file (1), so less storage space is required (1) and smaller files can usually be transferred more quickly at the same network speed (1).

EXAM Q9 · 4 MARKS

Describe two examples of utility software.

Two developed examples: encryption protects confidentiality by turning information into ciphertext that needs a key (up to 2); defragmentation moves separated blocks of a file closer together on an HDD, potentially reducing seek time (up to 2). Compression can also be credited if explained.

ANSWERS & MARK SCHEME

Indicative model answers · 3 of 3

These are exam-accurate marking points. Equivalent technically correct explanations can receive credit.

EXAM Q10 · 4 MARKS

Describe the difference between utility software and an operating system.

The OS provides core resource management and services, such as allocating memory, running tasks or controlling hardware (up to 2). Utility programs perform additional maintenance and security tasks, such as encryption, HDD defragmentation, and compression (up to 2). Some tools are integrated into an OS, so do not define them solely by whether they run in the background.

EXAM Q11 · 6 MARKS

Explain the role of utility software in maintaining a computer system.

A developed answer covers: encryption changes plaintext to ciphertext using a key so untrusted readers cannot understand intercepted or stolen data (up to 2); defragmentation reorganises file blocks on a magnetic hard disk and can reduce read-head movement/access time (up to 2); compression reduces file size and therefore uses less space and transfers more quickly (up to 2). Explain the purpose and usefulness of each rather than simply listing their names.

ADDITIONAL OCR PRACTICE E1 · 3 MARKS

A student says Face ID is the same as encryption. Explain why this is not quite correct and how the two can work together.

Face ID checks the user's identity (authentication) (1). Encryption turns readable data into ciphertext protected by a key (1). A device can use successful authentication to permit access to an encryption key or protected data (1).

ADDITIONAL OCR PRACTICE E2 · 3 MARKS

A school replaces a magnetic hard disk with a solid-state drive. Explain why running a regular HDD defragmentation utility is no longer an appropriate way to improve speed.

An HDD has moving read/write heads whose seek time is affected by separated file blocks (1). An SSD does not have a moving read head (1), so bringing the file blocks together does not provide the same seek-time benefit and creates unnecessary writes (1).

THEORY

Technology affects real people

Use the five-issue comparison to identify each impact, then explain its effects on people or society.

Issue	What it means (simple)	Example of digital technology	Impact on society
Ethical	What is right or wrong when using technology	AI facial recognition	Could reduce crime but may unfairly target groups
Legal	Laws that control how technology is used	Copyright on YouTube	People can be fined for illegal downloads
Cultural	How technology changes behaviour and lifestyle	Social media	Changes how people communicate
Environmental	How technology affects the planet	Data centres	Uses lots of electricity
Privacy	How personal data is collected and used	Smartphones tracking location	People lose control of personal information

FROM EXAMPLE TO EXPLANATION

Ethical: facial-recognition bias may unfairly affect groups. Legal: copying content may break copyright rules. Cultural: social media may change communication. Environmental: data centres consume energy and resources. Privacy: location tracking can expose personal information.

EXAM HABIT

For longer questions, connect the technology to an issue, explain an effect, and reach a reasoned result. Avoid answers made only of a list of issue names.

THEORY · TIER

Use the TIER writing framework

TIER stands for Technology, Issues, Effects and Result. Use these four stages to structure an extended answer.

Students lose marks in longer questions because they **list** instead of **discuss** always use **TIER** to avoid **tears**
T–I–E–R (Technology → Issues → Effects → Result)

Think in TIER. Write in paragraphs. Never list. If the question says **discuss**, aim for **6 marks** by using more than one (> 1) issue and explaining the > 1 effects.

TIER STEPS

TIER Steps	What it means	Helpful sentence starter	Example (Social Media)
T – Technology	The digital technology you are talking about	This technology is...	Social media
I – Issues	The problems the technology causes (more than one)	This causes problems because... and...	People share too much personal information and behaviour online changes
E – Effects	What happens to people or society because of the issues	This can lead to... and...	People lose privacy and mental health or behaviour can be affected
R – Result	What happens next because of the effects	As a result...	Privacy settings, rules and online safety awareness are used

HOW TIER EARNS MARKS

T: identify the technology. I: name the issues requested by the question. E: explain consequences for real people or organisations. R: give a reasoned result or judgement. In an extended answer, linked explanations and balance earn marks; the acronym alone does not.

THEORY · WORKED MODEL

How TIER becomes an answer

Study this model of a social-media answer, with the thinking structure beside a written student answer.

Question (6 marks)	TIER (Thinking Structure)	Student exam answer
Discuss the ethical, cultural and privacy issues caused by the use of social media.	**T** – Social media lets people talk and share things online. **I** – This causes problems because: personal data is collected (privacy issue) AND online behaviour changes (cultural issue). **E** – This can lead to: people losing privacy AND people sharing too much information and spending less time talking face to face, which can affect mental health. **R** – As a result, privacy settings, rules and online safety awareness are used.	Social media causes privacy and cultural issues because personal data is collected and online behaviour changes. This can lead to people losing privacy and sharing too much information, as well as spending less time talking face to face, which can affect mental health. As a result, privacy settings and online safety rules are used.

WHAT TO NOTICE

The worked table contains an issue (privacy and changed social behaviour), an effect (loss of control over data and changed communication), then a result (better choices and safeguards). The final answer makes these links in prose.

IMPROVE THE EXAM ARGUMENT

Add balance when asked to discuss: online communication can connect distant families, but data collection and pressure to post also have drawbacks. Explain both rather than repeating only disadvantages.

THEORY · MORE ANALOGIES

Worked examples using TIER

The guide applies TIER to everyday technology and legal/ethical issues, not just social media.

Discuss the legal and ethical issues caused by online shopping websites.	**T** – Online shopping lets people buy items on the internet. **I** – This causes problems because:• customer rights must be protected by law (legal issue)AND• personal data is collected and used (ethical issue). **E** – This can lead to:• customers being tricked or treated unfairlyAND• personal data being misused. **R** – As a result, consumer laws and secure payment systems are used.	Online shopping causes legal and ethical issues because customer rights must be protected and personal data is collected. This can lead to customers being tricked and personal data being misused. As a result, consumer protection laws and secure payment systems are used.
Discuss the ethical and privacy issues caused by artificial intelligence in everyday life.	**T** – Artificial intelligence is used to make choices and analyse data. **I** – This causes problems because:• AI decisions may be unfair (ethical issue)AND• large amounts of personal data are collected (privacy issue). **E** – This can lead to:• people being treated unfairlyAND• people losing control of their personal information. **R** – As a result, rules and controls are introduced to manage AI use.	Artificial intelligence causes ethical and privacy issues because decisions may be unfair and lots of personal data is collected. This can lead to people being treated unfairly and losing control of their information. As a result, rules and controls are introduced to manage how AI is used.

TECHNICAL ACCURACY CHECK

For online shopping, the issue is not simply that collecting personal data is illegal. Businesses can process it when they have a valid lawful basis and satisfy other data-protection duties.

THEORY

Explain the impact on different groups

A strong response considers who benefits, who may be disadvantaged, and what follows.

AI IN RECRUITMENT

Benefit: may process many applications consistently. Risk: biased training data may discriminate unfairly. Judgement: human review and checks can reduce some risks.

SMARTPHONES & LOCATION

Benefit: maps and emergency services can use location to help. Risk: excessive location tracking exposes sensitive habits and movements.

DATA CENTRES

Benefit: online services and cloud storage support school and work. Risk: electricity, water and equipment manufacture carry environmental costs.

SOCIAL MEDIA

Benefit: families and communities can stay connected. Risk: oversharing, harmful content or intense usage can affect privacy and wellbeing.

RIGHT TO REPAIR

Benefit: longer-lived products can save money and reduce e-waste. Trade-off: repairs may be difficult or costly for some devices.

THEORY

Three laws you must recognise

The OCR specification names these laws. Explain the purpose, what actions they allow or restrict, and the impact on technology users.

DATA PROTECTION ACT 2018

Alongside UK GDPR, sets rules for processing personal data. Organisations need a valid lawful basis, appropriate security and respect for people’s rights. Consent is not automatically required if another lawful basis applies.

COMPUTER MISUSE ACT 1990

Makes specified unauthorised access to computer material and other harmful unauthorised computer acts offences. Authorised access does not give unlimited permission to alter or disrupt data.

COPYRIGHT, DESIGNS AND PATENTS ACT 1988

Protects rights in creative works, including software. Authors and rights holders can control copying and distribution, grant licences and challenge infringement.

WHAT OCR DOES NOT REQUIRE

OCR focuses on the purpose and the effects of these acts in computing scenarios—not memorising every section number, court procedure, or complete UK GDPR lawful-basis list. Licensing will receive its own full revised reference section.

THEORY · LEGAL COMPARISON

Computing legislation

Use this legislation comparison and the TIER example to link legal requirements to real effects.

Legislation	Purpose	Allows	Prohibits
Data Protection Act 2018	Protect personal data	Lawful data processing	Misuse or insecure storage of data
Computer Misuse Act 1990	Protect systems from cybercrime	Authorised access	Hacking, malware, DoS attacks
Copyright Designs and Patents Act 1988	Protect creative work	Licensed use of content	Piracy and plagiarism

IMPORTANT DATA-PROTECTION CLARIFICATION

Consent is one possible lawful basis for processing personal data, but not the only one. Organisations still need to handle data lawfully, fairly, transparently and securely.

Exam Question	T – Technology	I – Issue (Law)	E – Effect (Allows / Prohibits)	R – Result (Why it matters)
Discuss the impact of the Data Protection Act 2018 on the use of online databases by organisations.	Online databases store personal data such as names, addresses, and dates of birth.	The DPA2018 applies because this is personal data.	Allows lawful data processing and secure storage but prohibits sharing data without consent or insecure storage.	Organisations must protect data properly, reducing privacy breaches and misuse.

TIER + THE LAW

Use TIER to connect a law to what it allows or restricts, how people are affected and a justified conclusion.

KEY TERMINOLOGY

Core exam vocabulary · 1 of 2

The translucent yellow strokes are the same SVG paths and colours used in the classroom lesson pages.

Ethical issue

A question of right, wrong, fairness or responsibility when developing or using a technology.

Legal issue

Whether use of a technology complies with relevant laws and obligations.

Cultural issue

An effect on how people communicate, live, work, behave or share beliefs.

Environmental impact

An effect on energy use, resources, emissions, waste or natural surroundings.

Privacy

Control over access to personal information and how it is collected, used or shared.

Personal data

Information relating to a living person who can be identified directly or indirectly.

KEY TERMINOLOGY

Core exam vocabulary · 2 of 2

The translucent yellow strokes are the same SVG paths and colours used in the classroom lesson pages.

Data Protection Act 2018

UK legislation governing personal-data handling alongside the UK GDPR, protecting rights and setting rules.

Lawful basis

A valid legal reason for processing personal data; consent is only one possible lawful basis.

Computer Misuse Act 1990

UK law creating offences for specified unauthorised computer access and harmful unauthorised acts.

Copyright, Designs and Patents Act 1988

UK law protecting authors’ and creators’ rights over works, including computer software.

Stakeholder

A person or group affected by a decision, such as users, businesses, communities or the environment.

Balanced discussion

An extended answer explaining different effects, relevant evidence and a reasoned judgement.

WORKED EXAMPLES

Example 1 · Social-media discussion

Apply TIER to a six-mark issue question, then convert the notes into connected paragraphs.

QUESTION

Discuss the ethical, cultural and privacy issues caused by social media. (6 marks)

T — TECHNOLOGY

People post messages, images and opinions; content reaches friends and wider audiences.

I — ISSUES

Apps may collect personal data (privacy); posts may influence behaviour, expectations and relationships (cultural/ethical).

E — EFFECTS

Connecting people is beneficial, but oversharing may expose private details; online pressure may affect relationships and wellbeing.

R — RESULT

Privacy controls and responsible behaviour help, but users and platform operators share responsibility. Weigh the benefits against risks.

EXAM-WORTHY SENTENCE

“Social media can help distant communities communicate, but data collection and public posts can reduce users’ control over personal information; balanced rules and sensible privacy choices help manage those risks.”

WORKED EXAMPLES

Example 2 · A school online database

Use TIER when a question connects a named law to a realistic technology.

QUESTION

Discuss the impact of the Data Protection Act 2018 on an organisation using an online database. (6 marks)

T — TECHNOLOGY

A school database stores pupils’ names and contact information to run school services.

I — LEGAL ISSUE

The school must use personal data lawfully and fairly and apply appropriate protection; the DPA 2018 operates alongside UK GDPR.

E — EFFECT

The school may need safeguards, staff training and controls on access. Weak security can harm pupils and lead to investigation.

R — RESULT

The rules help protect pupils’ rights and trust, while allowing necessary data use when there is a valid lawful basis.

EXAM WARNING

Never say every lawful use of personal data needs consent: other lawful bases exist.

WORKED EXAMPLES

Example 3 · A reasoned judgement

Use this phone-battery dilemma to evaluate different stakeholders.

EXTENDED-RESPONSE SCENARIO

A phone company makes replacement batteries difficult to obtain but offers cheaper new phones. Discuss benefits and drawbacks for the company, customers and the environment. (8 marks, additional source exercise)

T — TECHNOLOGY

New phone models are offered cheaply, while replacement batteries are harder to obtain.

I — ISSUES

The company may earn more from upgrades; customers lose some repair choices; manufacturing raises environmental concerns.

E — EFFECTS

Customers may benefit from cheaper devices but pay more over time. Extra production uses energy/materials and increases e-waste.

R — JUDGEMENT

A repairable, longer-lasting phone may be preferable environmentally, although affordability and technical limitations matter. Explain why.

EXAM PRACTICE

Practice questions · 1 of 4

Attempt these independently; no model answers appear until the later mark-scheme section.

EXAM Q1 · 1 MARK

What is meant by an ethical issue in computing?

EXAM Q2 · 1 MARK

What is meant by privacy?

EXAM Q3 · 1 MARK

Name one law related to computer use.

EXAM Q4 · 2 MARKS

State two social or cultural issues caused by digital technology.

EXAM PRACTICE

Practice questions · 2 of 4

Attempt these independently; no model answers appear until the later mark-scheme section.

EXAM Q5 · 2 MARKS

State two environmental issues caused by computer systems.

EXAM Q6 · 3 MARKS

Explain one privacy issue caused by smartphones.

EXAM Q7 · 3 MARKS

Explain one ethical issue caused by artificial intelligence.

EXAM Q8 · 3 MARKS

Explain one cultural issue caused by social media.

EXAM PRACTICE

Practice questions · 3 of 4

Attempt these independently; no model answers appear until the later mark-scheme section.

EXAM Q9 · 4 MARKS

Describe the purpose of the Data Protection Act 2018.

EXAM Q10 · 4 MARKS

Describe what the Computer Misuse Act 1990 protects against.

EXAM Q11 · 4 MARKS

Describe what the Copyright, Designs and Patents Act 1988 protects.

EXAM PRACTICE

Practice questions · 4 of 4

Attempt these independently; no model answers appear until the later mark-scheme section.

EXAM Q12 · 6 MARKS

Discuss the ethical, cultural and privacy issues caused by the use of social media.

EXAM Q13 · 6 MARKS

Discuss the impact of the Data Protection Act 2018 on organisations using online databases.

EXTENSION QUESTION · 8 MARKS

A phone company makes replacement batteries difficult to obtain but offers cheaper new phones. Discuss benefits and drawbacks for the company, customers and the environment.

ANSWERS & MARK SCHEME

Indicative marking · 1 of 4

Match the marked explanation to the Practice question; these are marking points, not required memorised scripts.

EXAM Q1 · 1 MARK

What is meant by an ethical issue in computing?

An ethical issue concerns whether using a digital technology is right, fair or responsible, including effects on people. (1)

EXAM Q2 · 1 MARK

What is meant by privacy?

Privacy is a person’s ability or right to control access to and use of information about them; it includes what organisations collect, store and share. (1)

EXAM Q3 · 1 MARK

Name one law related to computer use.

One valid law: Data Protection Act 2018, Computer Misuse Act 1990, or Copyright, Designs and Patents Act 1988. (1)

EXAM Q4 · 2 MARKS

State two social or cultural issues caused by digital technology.

Any two developed cultural/social issues, such as: online messaging can replace some face-to-face contact (1); social media can change attitudes and expected behaviour/communication (1). Accept other distinct effects. (2)

ANSWERS & MARK SCHEME

Indicative marking · 2 of 4

Match the marked explanation to the Practice question; these are marking points, not required memorised scripts.

EXAM Q5 · 2 MARKS

State two environmental issues caused by computer systems.

Any two distinct environmental impacts, e.g.: data centres consume significant electricity (1); manufacturing/disposal creates electronic waste and uses scarce materials (1). Other supported points may include heat, water consumption or transport emissions. (2)

EXAM Q6 · 3 MARKS

Explain one privacy issue caused by smartphones.

Phones/apps may collect or infer a user’s location (1); that information can be stored or shared and used to track activities (1), so the person may lose control over who knows where they have been (1). (3)

EXAM Q7 · 3 MARKS

Explain one ethical issue caused by artificial intelligence.

An AI system trained on biased or unrepresentative data may make unfair decisions (1), such as unequal screening of job candidates (1); this could discriminate against individuals or groups (1). (3)

EXAM Q8 · 3 MARKS

Explain one cultural issue caused by social media.

Social media can shift communication away from face-to-face interactions (1), promote trends and influence language or social norms (1), and affect how people behave or maintain relationships (1). (3)

ANSWERS & MARK SCHEME

Indicative marking · 3 of 4

Match the marked explanation to the Practice question; these are marking points, not required memorised scripts.

EXAM Q9 · 4 MARKS

Describe the purpose of the Data Protection Act 2018.

The Data Protection Act 2018 sits alongside UK data-protection law and regulates processing of personal data (1). Organisations must process information lawfully and fairly (1), protect it with appropriate security (1), and respect relevant individual rights such as access or correction (1). (4)

EXAM Q10 · 4 MARKS

Describe what the Computer Misuse Act 1990 protects against.

Computer Misuse Act 1990 makes specified unauthorised access to computer material an offence (1), including access with intent to commit further offences (1); it also covers unauthorised actions intended to impair computer operation/data (1). This protects users and organisations from attacks such as hacking (1). (4)

EXAM Q11 · 4 MARKS

Describe what the Copyright, Designs and Patents Act 1988 protects.

The Copyright, Designs and Patents Act 1988 protects original creative works including software (1); it gives rights holders control over copying/distribution (1), allows legal licensing and permitted use (1), and can make unauthorised reproduction or sharing an infringement (1). (4)

ANSWERS & MARK SCHEME

Indicative marking · 4 of 4

Match the marked explanation to the Practice question; these are marking points, not required memorised scripts.

EXAM Q12 · 6 MARKS

Discuss the ethical, cultural and privacy issues caused by the use of social media.

Use TIER and cover several issues: T: social media allows rapid communication and sharing. I: platforms may collect personal data, while online norms, trends or overuse change how people communicate. E: positives include maintaining distant relationships, but risks include loss of privacy and reduced face-to-face interaction or pressure on wellbeing. R: explain safeguards such as controls on personal-data sharing and responsible use, with a reasoned conclusion. For six marks, explain linked impacts, not simply name categories. (6)

EXAM Q13 · 6 MARKS

Discuss the impact of the Data Protection Act 2018 on organisations using online databases.

T: organisations use online databases to store customers' personal information. I: the Data Protection Act 2018 and UK data-protection rules require lawful, fair handling of that data. E: organisations need appropriate security, transparency and a valid lawful basis (consent is not automatically required), and must respect data rights. Unlawful processing or weak security may lead to investigations, fines and loss of trust. R: compliance helps protect individuals and enables responsible use of databases. Explain both organisational obligations and consequences. (6)

EXTENSION QUESTION · 8 MARKS

A phone company makes replacement batteries difficult to obtain but offers cheaper new phones. Discuss benefits and drawbacks for the company, customers and the environment.

A developed, balanced discussion should use TIER. Technology: cheaper new smartphones, with replacement batteries difficult to obtain. Issues: customer cost/convenience, business sales/profit, product longevity, resources and waste. Effects: customers may benefit from cheaper newer models but pay more often or lose repair choice; the company may increase sales but suffer reputational damage; new manufacture uses materials/energy and creates electronic waste. Result: discuss repair access, right-to-repair approaches or trade-in/refurbishment and end with a justified judgement balancing stakeholders. This is a source extended task, not one of the original 13 table questions.

THEORY · VISUAL ANALOGY

Software licences

The recipe analogy helps explain permissions to use, change and share software.

THE RECIPE ANALOGY



SHARED RECIPE

Open source: you may read and adapt the recipe if you follow the sharing rules.



SEALED RECIPE

Proprietary: you can use the finished meal, but cannot assume permission to change the secret recipe.

Exam connection: The “recipe” represents the source code. A licence defines the permissions and restrictions. Unlike a recipe analogy, the legal terms matter: some open-source licences have conditions for sharing changes, while proprietary licences may allow use without sharing source code.

Definition: A software licence sets out the permissions and conditions for using, modifying or sharing software.

THEORY · COMPARE

Open source versus proprietary

Compare source-code permissions, not just prices or brand names.

Feature	Open source	Proprietary
Source code	Available to examine	Usually not provided
Modification	Permitted by its licence	Usually not permitted
Cost	May be free or paid	May be free or paid
Redistribution	Allowed with conditions	Often restricted
Support	Community or paid services	Vendor or external services
Examples	School coding tool / library	Commercial closed-source app

Important: Open source does not mean “always free”. Proprietary does not mean “always paid”, “always more reliable” or “always professionally supported”. Those details depend on the product and licence.

EXAM TECHNIQUE

For a recommendation, connect the licence to the user’s needs: access to source code → permitted changes → benefit to that user. Then give a realistic drawback or trade-off.

THEORY · RULES AND RIGHTS

What a software licence controls

Compare what each type of software licence permits.

Use

Who may install or run the software, and on how many devices or accounts.

Copy and share

Whether the program can be redistributed to somebody else.

Modify

Whether the user can inspect and change the source code.

Attribution and conditions

Whether credit, a copy of the licence or other conditions must accompany redistribution.

Support and updates

May be included or purchased separately; not guaranteed by licence type alone.

Remember: ?To prevent illegal copying? is only one possible purpose. A licence also tells users what they are allowed to do. Copyright still applies to both open-source and proprietary software.

THEORY · OCR COVERAGE

What the examination needs

Software licences are included in OCR J277 specification 1.6.1.

- Explain why software needs a licence and what a licence is for.
- Identify the two types: open source and proprietary.
- Compare access to source code and permission to change it.
- Recommend the suitable type of licence for a given scenario, including a benefit and a drawback.

LINK TO TIER

Software licensing can raise legal and ethical issues. Use TIER to structure an extended discussion:

Technology: software used by the school or business
Issues: licence restrictions, cost, maintenance, access
Effects: who can modify, share, support or afford it
Result: balanced judgement linked to the scenario

Study the two licence types, complete the worked recommendation, then attempt the 14 practice questions before checking the answers.

KEY TERMINOLOGY

Key terminology · 1 of 2

Learn the definitions; the yellow stroke is the same as in lessons.

Software licence

Legal terms and conditions specifying how software may be used or shared.

Source code

Program instructions in a readable form that a developer can edit.

Open source

Software whose source code is accessible and may be modified under its licence.

Proprietary

Software usually distributed without public access or permission to modify its source.

Copyright

Legal protection for an original work, including computer software.

Modification

Making changes to the original program’s source code.

KEY TERMINOLOGY

Key terminology · 2 of 2

Learn the definitions; the yellow stroke is the same as in lessons.

Redistribution

Giving copies of software or its modified versions to others.

Off-the-shelf

Ready-made software designed for a broad range of users.

Customisation

Altering a program to meet particular user or organisation needs.

Licence conditions

Rules that must be followed when installing, sharing or changing software.

Technical support

Help with installing, maintaining or troubleshooting software.

Vendor lock-in

Dependence on a supplier that makes it difficult or expensive to switch.

WORKED EXAMPLES · MODEL 1

A school wants editable code

Plan your recommendation before reading the model answer.

Six-mark scenario: A school wants students to learn how programs work and change the code. Recommend a type of software licence and justify your answer.

1 Identify the requirement

Students need to view and modify source code.

2 Choose a matching licence

Open source, because the licence allows source access and changes.

3 Explain the benefit

Pupils can investigate real algorithms, experiment and learn through editing.

4 Weigh a limitation

The school must check compatibility, support availability and licence conditions.

5 Reach a judgement

Open source is most appropriate where learning through source-code editing is the priority.

EXAM ANSWER

Choose open source. Students can inspect and modify the program code, helping them understand how it works. The school may also save on licence fees, although this is not guaranteed. It should check support and compatibility. Overall it best meets the requirement for editable code.

WORKED EXAMPLES · MODEL 2

A business needs supported software

An exam recommendation must justify a real scenario, not repeat stereotypes.

Six-mark scenario: A business needs reliable word-processing software with technical support. Recommend a type of software licence and justify your choice.

Requirement

Reliable word processing, predictable updates and accessible technical help.

Likely choice

Proprietary may be appropriate if the selected package includes a support contract.

Benefit

A named provider may offer a service agreement and compatible document formats.

Trade-off

Licence or subscription costs and reliance on the supplier can increase costs.

Important caution

Open-source applications can also have paid support, so test the actual product.

BALANCED CONCLUSION

A proprietary package may be a good choice where the vendor offers verified technical support, updates and format compatibility. The business should weigh those benefits against recurring costs and dependence on a supplier. The best choice depends on the actual products available.

EXAM PRACTICE · Practice questions Practice · Questions 1?5

Attempt every question before turning to the Answers & Mark Scheme section.

Q1 · Define · 1 mark

What is a software licence?

Q2 · Define · 1 mark

State one purpose of a software licence.

Q3 · Define · 1 mark

Name one type of software licence.

Q4 · Explain · 2 marks

State two purposes of a software licence.

Q5 · Explain · 2 marks

State two features of open source software.

EXAM PRACTICE · Practice questions Practice · Questions 6?9

Attempt every question before turning to the Answers & Mark Scheme section.

Q6 · Explain · 2 marks

State two features of proprietary software.

Q7 · Explain · 3 marks

Explain why open source software allows users to modify the program.

Q8 · Explain · 3 marks

Explain why proprietary software is often sold as off-the-shelf software.

Q9 · Explain · 3 marks

Explain one benefit of using open source software.

EXAM PRACTICE · Practice questions Practice · Questions 10?12

Attempt every question before turning to the Answers & Mark Scheme section.

Q10 · Describe · 4 marks

Describe the difference between open source and proprietary software.

Q11 · Describe · 4 marks

Describe two drawbacks of proprietary software.

Q12 · Recommend · 6 marks

A school wants students to learn how programs work and change the code. Recommend a type of software licence and justify your answer.

EXAM PRACTICE · Practice questions Practice · Questions 13?14

Attempt every question before turning to the Answers & Mark Scheme section.

Q13 · Recommend · 6 marks

A business needs reliable word-processing software with technical support. Recommend a type of software licence and justify your choice.

Q14 · Discuss · 6 marks

Discuss the benefits and drawbacks of using open source software.

ANSWERS & MARK SCHEME

Answers · Questions 1–4

Use these indicative marking points to improve your answer.

Q1 · 1 mark

What is a software licence?

A licence sets out permitted use, installation, copying and distribution of software.

Q2 · 1 mark

State one purpose of a software licence.

To grant permission for use under stated conditions; it can protect the copyright holder’s rights.

Q3 · 1 mark

Name one type of software licence.

Open source or proprietary.

Q4 · 2 marks

State two purposes of a software licence.

Specify conditions of permitted use and protect the creator’s rights or restrict unauthorised copying/distribution.

ANSWERS & MARK SCHEME

Answers · Questions 5–8

Use these indicative marking points to improve your answer.

Q5 · 2 marks

State two features of open source software.

Users can access its source code; the licence permits modification (and usually redistribution under conditions).

Q6 · 2 marks

State two features of proprietary software.

The source code is normally unavailable to users; the licence restricts modification and redistribution.

Q7 · 3 marks

Explain why open source software allows users to modify the program.

Its licence gives access to source code and permission to change it, enabling users to customise or fix features.

Q8 · 3 marks

Explain why proprietary software is often sold as off-the-shelf software.

It may be developed for a broad market, supplied as a ready-to-use package and sold or licensed without source-code modification rights.

ANSWERS & MARK SCHEME

Answers · Questions 9–11

Use these indicative marking points to improve your answer.

Q9 · 3 marks

Explain one benefit of using open source software.

It can be adapted to specific needs because developers can inspect and modify source code; licensing charges may also be lower.

Q10 · 4 marks

Describe the difference between open source and proprietary software.

Open source grants access and modification rights; proprietary usually keeps source code closed and places tighter limits on copying and modification.

Q11 · 4 marks

Describe two drawbacks of proprietary software.

Licence fees or recurring subscription costs may apply; lack of source access restricts how users can customise software.

ANSWERS & MARK SCHEME

Answers · Questions 12–14

Use these indicative marking points to improve your answer.

Q12 · 6 marks

A school wants students to learn how programs work and change the code. Recommend a type of software licence and justify your answer.

Open source: students can view and modify source code, learn by experimentation and potentially share improvements under the licence. Some packages may need extra training or support; evaluate suitability before concluding.

Marking guidance: For full credit: a justified choice, relevant benefits, an explained limitation, and a conclusion tied to the scenario.

Q13 · 6 marks

A business needs reliable word-processing software with technical support. Recommend a type of software licence and justify your choice.

Proprietary is defensible if a suitable product includes a support agreement, updates and compatibility; compare total cost and vendor dependence. Open-source software can also be supported professionally, so reliability and support must be evidenced for the selected product.

Marking guidance: For full credit: a justified choice, relevant benefits, an explained limitation, and a conclusion tied to the scenario.

Q14 · 6 marks

Discuss the benefits and drawbacks of using open source software.

Benefits: source-code access, adaptability, community review and potential licensing savings. Drawbacks: availability of specialist support, maintenance effort, compatibility and training can vary. A balanced answer reaches a context-dependent judgement.

Marking guidance: For full credit: balance several explained benefits and drawbacks and make a clear contextual judgement.

Resources

Exam Paper 1 2026 by Topic

Question 1 – CPU, Architecture and Performance

1(a) Which component controls the operation of the CPU? **(1)**

Select one option.

☐ ALU
☐ Cache
☐ Control Unit
☐ RAM

1(b) State two registers used in the fetch–decode–execute cycle. **(2)**

1(c) Explain why registers are located inside the CPU rather than in main memory. **(3)**

1(d) Complete the table showing what each register stores: Program Counter, MAR, MDR and Accumulator. **(4)**

Program Counter: _____

MAR: _____

MDR: _____

Accumulator: _____

1(e) Explain how clock speed, cache size and number of cores together affect CPU performance. **(6)**

Question 2 – Data Representation

2(a) Binary is a number system that uses: **(1)**

Select one option.

☐ 0–9
☐ 0 and 1
☐ A–F
☐ 1–10

2(b) Explain why computers use binary to store data. **(3)**

2(c) Convert the hexadecimal number **3F** into denary.

Show your working. **(4)**

2(d) An image is 800 × 600 pixels with a colour depth of 16 bits.

Calculate the file size in MB. **(4)**

Question 3 – Memory, Storage and Embedded Systems

3(a) Match each memory type to its description: RAM, ROM and Cache. **(2)**

RAM → _____

ROM → _____

Cache → _____

3(b) Explain why embedded systems usually use ROM to store their programs. **(3)**

3(c) Explain why SSDs are often preferred to HDDs in portable devices. **(3)**

3(d) Compare RAM and ROM. **(4)**

Question 4 – Networks and Security

4(a) Which device forwards data packets between different networks? **(1)**

Select one option.

☐ Switch
☐ Router
☐ Hub
☐ NIC

4(b) Explain the role of a switch in a school network. **(3)**

4(c) Describe how encryption helps protect data when using Wi-Fi. **(4)**

4(d) Explain how data is transmitted securely from a school computer to a web server. **(6)**

Question 5 – Systems Software and Utilities

5(a) System software is software that: **(1)**

Select one option.

☐ Manages hardware and resources
☐ Creates documents
☐ Plays media
☐ Edits images

5(b) Explain one role of an operating system related to memory or process management. **(3)**

5(c) State two examples of utility software. **(2)**

5(d) Explain why utility software is important for maintaining computer performance. **(4)**

Question 6 – Ethical, Legal and Environmental Issues

6(a) State two ethical issues related to digital technology. **(2)**

6(b) Name one law related to computer use and state its purpose. **(2)**

6(c) Explain how the Data Protection Act 2018 affects how organisations store personal data. **(4)**

6(d) Discuss the impact of digital technology on individuals or society. **(8)**

End of Paper

Answers and Misconceptions by topic

Question	Right answer (student-friendly)	Common misconceptions
1(a) CPU control	**Control Unit** controls the operation of the CPU.	Choosing ALU (does calculations) or RAM (not part of CPU).
1(b) CPU registers	Any two of: **PC, MAR, MDR, Accumulator**.	Naming cache/RAM; inventing registers.
1(c) Registers location	Registers are **very fast, small storage inside the CPU** for data/addresses in use now.	Saying they store lots of data or are slower than RAM.
1(d) Register roles	**PC:** address of next instruction. **MAR:** address being accessed. **MDR:**data/instruction fetched. **Accumulator:** results of calculations.	Mixing MAR/MDR; saying PC stores data.
1(e) CPU performance	**Clock speed:** cycles per second. **Cache:** faster access to frequent data. **Cores:** parallel processing/multitasking.	Saying higher clock speed = more cores; cache replaces RAM.
2(a) Binary	Uses **0 and 1**.	Choosing A–F or 0–9.
2(b) Why binary	Hardware has **two states (on/off)** so binary is reliable.	"Computers prefer binary" with no hardware link.
2(c) Hex → denary	**3F = (3×16) + 15 = 63**.	Forgetting F=15; not showing working.
2(d) Image file size	Size = **pixels × colour depth**, then convert bits → bytes → MB.	Missing unit conversion; confusing bits/bytes.
3(a) Memory matching	**RAM:** running programs. **ROM:** start-up instructions. **Cache:** frequent data near CPU.	Saying cache is permanent; swapping RAM/ROM.
3(b) ROM in embedded	**Non-volatile**, reliable start-up every time.	Saying ROM is faster than RAM.
3(c) SSD vs HDD	**SSDs** are faster, durable, low power, no moving parts.	Saying SSDs are magnetic or have moving parts.
3(d) Compare RAM/ROM	**RAM:** volatile, used while running. **ROM:** non-volatile, stores start-up code.	Listing facts without comparison words.
4(a) Network device	**Router** forwards packets between networks.	Choosing switch (within a LAN).
4(b) Switch role	Sends frames to correct device using **MAC addresses**.	Confusing switch with router.
4(c) Encryption	**Scrambles data** so intercepted data is unreadable without a key.	Saying it speeds up Wi-Fi.
4(d) Secure transmission	Data is **encrypted**, sent via network devices, **decrypted** at destination.	Listing tools without explaining stages.
5(a) System software	**Manages hardware and resources**.	Choosing apps like documents/media.
5(b) OS role	**Memory/process management** (allocates RAM, schedules tasks).	Vague answers like "controls the computer".
5(c) Utilities	Examples: **antivirus, backup, compression, defragmentation**.	Naming apps (Word, Chrome).
5(d) Utilities importance	**Maintain**	Saying they create files for users.

	performance/security (clean, protect, organise data).	
6(a) Ethical issues	Examples: **privacy**, **addiction**, **job loss**, **surveillance**.	Naming laws instead of issues.
6(b) Law & purpose	**Data Protection Act 2018** protects personal data.	Naming a law with no purpose.
6(c) DPA impact	Data must be **secure, lawful, limited, accurate**.	Listing principles with no explanation.
6(d) Impact discussion	**Balanced**: benefits (communication, efficiency) and drawbacks (privacy, environment).	One-sided lists; no discussion.

Synoptic practice set B

Question 1

1(a) Which number system is base 2? **(1)**

Select one option.

☐ Denary
☐ Binary
☐ Hexadecimal
☐ ASCII

1(b) Convert the denary number **25** into an **8-bit binary** number. **(2)**

1(c) A binary number is shifted **right by one place**.

Explain the effect this has on its denary value. **(3)**

1(d) Convert the hexadecimal number **7A** into denary.

Show your working. **(4)**

Question 2

2(a) Which register stores the address of the next instruction to be fetched? **(1)**

Select one option.

☐ MAR
☐ MDR
☐ Program Counter
☐ Accumulator

2(b) State two registers used during the fetch–decode–execute cycle. **(2)**

2(c) Explain why the Control Unit is essential in the fetch–decode–execute cycle. **(3)**

2(d) Complete the table to show what each register stores. **(4)**

MAR:	
MDR:	
Accumulator:	
Program Counter:	

Question 3

3(a) Which address uniquely identifies a device on a local network? **(1)**

Select one option.

☐ IP address
☐ MAC address
☐ URL
☐ Domain name

3(b) Give one example of a valid IPv4 address. **(1)**

3(c) State two features of a MAC address. **(2)**

3(d) Describe how a switch sends data to the correct device on a local area network. **(4)**

Question 4

4(a) Define each memory type to its correct description. **(2)**

RAM → _____

ROM → _____

Cache → _____

4(b) Explain why ROM is suitable for storing the program in an embedded system such as a washing machine. **(3)**

4(c) State two differences between RAM and ROM. **(2)**

4(d) A portable device uses an SSD rather than an HDD.

Explain two reasons why an SSD is more suitable. **(4)**

Question 5

5(a) Which operating system task controls how processor time is shared between programs? **(1)**

Select one option.

☐ File management
☐ Scheduling
☐ Compression
☐ Backup

5(b) Explain what is meant by *peripheral management* in an operating system. **(3)**

5(c) Explain why device drivers are required for peripheral devices. **(3)**

5(d) State one example of utility software and describe its purpose. **(3)**

WORKING SPACE

Use this page for workings or longer responses to the preceding questions.

Question 6

6(a) Which register stores the result of calculations carried out by the ALU? **(1)**

Select one option.

☐ MAR
☐ MDR
☐ Accumulator
☐ Program Counter

6(b) Add the following binary numbers.
Show your working. **(4)**

```
00110110
00011011
```

6(c) Explain how clock speed, cache memory and number of cores *together* affect CPU performance. **(6)**

Question 7

7(a) A hospital uses a wired network for desktop computers and wireless connections for tablets.

Explain one reason why **wired** connections are suitable for desktop computers in this hospital. **(3)**

7(b) Explain one reason why **wireless** connections are suitable for tablets in this hospital. **(3)**

7(c) Explain how encryption protects data when it is transmitted over a public Wi-Fi network. **(4)**

Question 8

8(a) State two ethical issues related to the use of digital technology. **(2)**

8(b) Name one law related to computer use and state its purpose. **(2)**

8(c) A student is releasing a game online.

Explain one benefit and one drawback of using an **open source licence** for this game. **(4)**

End of Paper

Answers and Misconceptions - likely OCR style exam

Question	Right answer (student-friendly)	Common misconceptions
1(a) Base 2	**Binary** uses base 2.	Choosing hexadecimal or thinking ASCII is a number system.
1(b) 25 → binary	**00011001** (16 + 8 + 1).	Missing leading zeros; adding bits instead of place values.
1(c) Right shift	Shifting right **divides the number by 2** each shift.	Saying it "moves bits" with no effect; saying it multiplies.
1(d) 7A → denary	**122** (7×16 + 10).	Forgetting A = 10; writing 7 + 10.
2(a) PC register	**Program Counter** stores the next instruction's address.	Choosing MAR or saying it stores data.
2(b) FDE registers	Any two of: **PC, MAR, MDR, Accumulator**.	Naming cache or RAM; inventing registers.
2(c) Control Unit	The CU **decodes instructions and tells components what to do**.	Saying "it controls the CPU" with no explanation.
2(d) Register roles	**MAR = address**, **MDR = data**, **PC = next instruction**, **Accumulator = results**.	Swapping MAR/MDR; saying PC stores data.
3(a) Network ID	**MAC address** uniquely identifies a device on a LAN.	Choosing IP address or URL.
3(b) IPv4 example	Any valid format e.g. **192.168.1.1**.	Numbers over 255; missing dots.
3(c) MAC features	Uses **hexadecimal**, usually **48 bits**, fixed to device.	Saying it changes location; confusing with IP.
3(d) Switch	Uses a **MAC address table** to send data only to the correct device.	Saying it broadcasts like a hub; calling it a router.
4(a) Memory match	**RAM = running programs**, **ROM = start-up**, **Cache = frequent data near CPU**.	Saying cache is permanent; swapping RAM/ROM.
4(b) ROM in embedded	ROM is **non-volatile** so the program is always there on power-up.	Saying ROM is faster than RAM.
4(c) RAM vs ROM	RAM = **volatile / temporary**, ROM = **non-volatile / permanent**.	Just listing facts with no comparison.
4(d) SSD vs HDD	SSDs are **faster, more durable, no moving parts**.	Saying SSDs are magnetic or cheaper.
5(a) Scheduling	**Scheduling** shares CPU time between programs.	Choosing file management or backup.
5(b) Peripheral management	The OS **controls peripherals using drivers**.	Writing "manages peripherals" with no detail.
5(c) Drivers	Drivers **let the OS communicate with hardware**.	Saying drivers store files or control the CPU.
5(d) Utility	e.g. **antivirus protects against malware**.	Naming apps like Word or Chrome.
6(a) Accumulator	Stores the **result of ALU calculations**.	Choosing MDR or PC.
6(b) Binary addition	Correct binary sum with **working shown**.	Adding like denary; missing carry bits.
6(c) CPU performance	**Clock speed = cycles**, **cache = faster access**, **cores = parallel processing**.	Saying "faster CPU" repeatedly; mixing factors up.
7(a) Wired (hospital)	**Reliable, fast, secure** for fixed desktop use.	Giving generic advantages with no hospital link.
7(b) Wireless (tablets)	**Mobility and flexibility** for moving staff.	Saying "it's better" with no reason.
7(c) Encryption	Data is **scrambled** so intercepted data is	Saying encryption blocks

	unreadable without the key.	hackers directly.
8(a) Ethical issues	e.g. **privacy, addiction, surveillance, job loss**.	Naming laws instead of issues.
8(b) Law & purpose	e.g. **Data Protection Act – protects personal data**.	Naming a law with no purpose.
8(c) Open source licence	**Benefit:** can edit/share code. **Drawback:** less control/profit.	Only listing benefits; not linking to game scenario.